

MAB-Dynamic_Environment-Test_Framework

September 29, 2025

1 MAB Dynamic Environment Testing Framework: Complete Implementation

1.0.1 Multi-Armed Bandit Algorithm Performance Across Variable Environmental Conditions

1.0.2 Comprehensive Testing Suite with ALL Environment Types & Algorithm Comparisons

Date: September 2025

Authors: Research Implementation based on GA Work in AI/Quantum Computing

Framework: Complete MAB testing across Dynamic, Stochastic, Adversarial, and Baseline environments

1.1 Research Overview

This comprehensive notebook contains an enhanced testing framework for evaluating **Multi-Armed Bandit (MAB) algorithms** across diverse and dynamically changing environmental conditions with **ALL** major algorithm categories and environment types:

1.1.1 Complete Algorithm Testing Suite:

1. **EXPNeuralUCB** - Neural bandits with adversarial robustness
2. **Thompson Sampling** - Bayesian posterior sampling approach
3. **Epsilon-Greedy** - Classical exploration-exploitation balance
4. **UCB Variants** - Upper Confidence Bound methods
5. **Kernel UCB** - Non-parametric contextual bandits
6. **EXP4** - Exponential weights for adversarial environments

1.1.2 Dynamic Environment Types:

- **Baseline** - Standard stationary reward distributions
- **Non-Stationary** - Gradually changing optimal arms
- **Adversarial** - Strategic attacks on algorithm performance
- **Stochastic Drift** - Random environmental perturbations
- **Markov Switching** - State-dependent reward dynamics
- **Concept Drift** - Gradual shifts in reward structures

1.1.3 Reference Foundation

Enhanced framework building upon quantum networking and adversarial bandit research

“Quantum Entanglement Path Selection via Adversarial Group Neural Bandits”

Extended for comprehensive MAB algorithm evaluation across dynamic environments

1.1.4 Key Research Questions

1. How do different MAB algorithms adapt to dynamically changing environments?
2. Which algorithms maintain performance under various environmental stresses?
3. What are the trade-offs between exploration, exploitation, and adaptability?
4. How can we design robust testing frameworks for real-world MAB deployment?

1.1.5 Educational & Research Value

- **Comparative Analysis** - Systematic evaluation of major MAB approaches
- **Environmental Robustness** - Testing across realistic dynamic conditions
- **Theoretical Validation** - Empirical verification of performance bounds
- **Practical Applications** - Framework for real-world algorithm selection

1.2 1. Environment Setup & Configuration

```
[3]: # @title
import os, sys

# get current directory
cur_dir = os.getcwd()
print("Current working directory:", cur_dir.split('/')[-1])

try:
    import google.colab                # Check if running in Colab
    from google.colab import drive
    drive.mount('/content/drive')

    project_dir = '/content/drive/MyDrive/GA-Work/hybrid_variable_framework' #_
    ↪Change work directory
    os.chdir(project_dir)                # makes project_dir the notebook's "root"
    print("- Running in Colab")

    project_cod_dir = os.path.join(project_dir, 'src')
    sys.path.append(project_cod_dir)     # Import local modules

except ImportError:
    print("- Not running in Colab")

print('- Now working from:', os.getcwd().split('/')[-1])
```

Current working directory: expneural_framework
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
- Running in Colab
- Now working from: hybrid_variable_framework

```
[ ]: # update import library
import importlib
import threading
import quantum_tests
importlib.reload(quantum_tests)
from quantum_tests import ExperimentRunner

def run_threaded_experiments():
    """
    Each experiment in its own thread - like separate Colab notebooks!
    """
    FRAME_NUMBERS = [4000, 6000, 8000]

    for experiment_id, frame_count in enumerate(FRAME_NUMBERS, 1):

        # Create fresh runner instance (like new Colab notebook)
        runner = ExperimentRunner(base_seed=12345)

        # Run in thread (like separate Colab session)
        experiment_thread = threading.Thread(
            target=runner.run_single_experiment,
            args=(frame_count, experiment_id),
            name=f"ColabCell_{experiment_id}"
        )

        experiment_thread.start()
        experiment_thread.join() # Wait for completion

        # Full cleanup (like closing Colab notebook)
        runner.full_experiment_cleanup()

    # Run the experiments!
    run_threaded_experiments()
```

PyTorch version: 2.8.0

NumPy version: 2.3.3

Using device: cpu

EXPERIMENT 1: 4000 frames
Current directory: /Users/pitergarcia/DataScience/Semester4/GA-
Work/hybrid_variable_framework

File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack pattern

```
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'neural' mode
    • Group Selection: Simple UCB (not adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)

- NEURAL Progress: 100%|      | 4000/4000 [00:06<00:00, 646.21it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'exp3' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Linear UCB (linear learning)

- EXP3 Progress: 100%|      | 4000/4000 [00:01<00:00, 2235.70it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'hybrid' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)

- HYBRID Progress: 100%|      | 4000/4000 [00:10<00:00, 394.52it/s]
```

RESULTS:

```
Oracle: 3200.28
Neural: 2112.17
EXP3: 2094.06
Hybrid: 2380.03
Winner: EXPNeuralUCB
```

EXPERIMENT 2: 6000 frames

Current directory: /Users/pitertgarcia/DataScience/Semester4/GA-Work/hybrid_variable_framework

File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack pattern

```
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'neural' mode
    • Group Selection: Simple UCB (not adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)
```

```

- NEURAL Progress: 100%|          | 6000/6000 [00:11<00:00, 541.65it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'exp3' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Linear UCB (linear learning)

- EXP3 Progress: 100%|          | 6000/6000 [00:04<00:00, 1459.35it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'hybrid' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)

- HYBRID Progress: 100%|          | 6000/6000 [00:15<00:00, 397.21it/s]

```

RESULTS:

```

Oracle: 4803.74
Neural: 3465.89
EXP3: 3131.23
Hybrid: 3607.19
Winner: EXPNeuralUCB

```

EXPERIMENT 3: 8000 frames

```

Current directory: /Users/pitertgarcia/DataScience/Semester4/GA-
Work/hybrid_variable_framework
File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack
pattern

```

```

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'neural' mode
    • Group Selection: Simple UCB (not adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)

- NEURAL Progress: 100%|          | 8000/8000 [00:51<00:00, 156.82it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'exp3' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Linear UCB (linear learning)

```

```

- EXP3 Progress: 100%|      | 8000/8000 [00:06<00:00, 1159.64it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'hybrid' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)

- HYBRID Progress: 100%|      | 8000/8000 [00:21<00:00, 369.45it/s]

RESULTS:
  Oracle: 6406.25
  Neural: 3804.77
  EXP3: 4419.39
  Hybrid: 4861.24
  Winner: EXPNeuralUCB

```

```
[ ]:
```

1.3 12. Advanced Visualization and Analysis

1.3.1 Comprehensive performance comparison with publication-quality plots

```

[ ]: # @title
import numpy as np
import importlib
import threading
import quantum_tests
importlib.reload(quantum_tests)
import matplotlib.pyplot as plt
from quantum_tests import ExperimentRunner

class MultiRunEvaluator:
    """
    Evaluator to collect and analyze results from your threaded experiments
    Matches your exact ExperimentRunner output structure
    """

    def __init__(self, base_seed=12345, base_frames=4000, frame_step=2000):
        self.base_seed = base_seed
        self.base_frames = base_frames # 4000
        self.frame_step = frame_step # 2000 (so 4000, 6000, 8000)

        self.experiments = {}
        self.all_results = []
        self.frame_counts = []

```

```

print(" MultiRunEvaluator initialized")
print(f"    • Base frames: {base_frames}")
print(f"    • Frame step: {frame_step}")
print(f"    • Base seed: {base_seed}")

def run_single_threaded_experiment(self, frame_count, exp_id):
    """
    Run single experiment and return results
    Exactly matches your working code structure
    """
    print(f"\n EVALUATOR EXPERIMENT {exp_id}: {frame_count} frames")

    # Create fresh runner instance
    runner = ExperimentRunner(base_seed=self.base_seed)

    # Run experiment in thread
    experiment_thread = threading.Thread(
        target=runner.run_single_experiment,
        args=(frame_count, exp_id),
        name=f"EvaluatorExp_{exp_id}"
    )

    experiment_thread.start()
    experiment_thread.join() # Wait for completion

    # Get results from runner
    results = runner.get_results().copy()
    for alg in results.keys():
        self.frame_counts.append({alg:frame_count})
        self.all_results += [{alg:results[alg]['final_reward']} for alg in
↪results.keys()]
        self.experiments[exp_id] = {"res": results, "frame_count": frame_count,
↪"id": exp_id}

    # for alg in algorithms:
    #     rewards = [run[alg] for run in evaluator.all_results]

    # Full cleanup
    runner.full_experiment_cleanup()

    # return results

def run_experiments(self, num_experiments=3):
    """
    Run multiple experiments with increasing frame counts
    Matches your [4000, 6000, 8000] pattern

```

```

        """
        print(" Starting MultiRun Evaluation")

        for i in range(num_experiments):
            experiment_id = i + 1
            frame_count = self.base_frames + (i * self.frame_step) # 4000,
↪6000, 8000

            # Run experiment
            self.run_single_threaded_experiment(frame_count, experiment_id)
            print(f" Experiment {experiment_id} completed: {frame_count}
↪frames")

        print(f"\n All {num_experiments} experiments completed!")
        self.print_summary()

    def print_summary(self):
        """Print quick summary of all results"""
        print("\n" + "="*60)
        print("MULTIRUN EXPERIMENT SUMMARY")
        print("="*60)

        for exp_id, experiment in self.experiments.items():
            print(f"\nExperiment {exp_id} ({experiment['frame_count']:,}
↪frames):")

                print(f"   Oracle:           ↪
↪{experiment['res']['Oracle']['final_reward']:.2f}")
                print(f"   EXPNeuralUCB: ↪
↪{experiment['res']['EXPNeuralUCB']['final_reward']:.2f} ")
                print(f"   GNeuralUCB: ↪
↪{experiment['res']['GNeuralUCB']['final_reward']:.2f}")
                print(f"   EXPUCB:           ↪
↪{experiment['res']['EXPUCB']['final_reward']:.2f}")

    def get_algorithm_data(self, algorithm_name):
        """Get all results for a specific algorithm"""
        # return [run[algorithm_name] for run in self.all_results]
        return [run[algorithm_name] for run in self.all_results if
↪algorithm_name in run]

# ===== USAGE EXAMPLE =====

def run_evaluation():
    """

```

```

Complete evaluation pipeline
"""
# Create evaluator
evaluator = MultiRunEvaluator(base_seed=12345, base_frames=4000,
↪frame_step=2000)

# Run experiments (this will take a few minutes)
evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000 frames

# Return for visualization
return evaluator

# ===== TO USE (copy this for tomorrow) =====
# Run evaluation
evaluator = run_evaluation()

```

MultiRunEvaluator initialized

- Base frames: 4000
- Frame step: 2000
- Base seed: 12345

Starting MultiRun Evaluation

EVALUATOR EXPERIMENT 1: 4000 frames

EXPERIMENT 1: 4000 frames

Current directory: /Users/pitertgarcia/DataScience/Semester4/GA-Work/hybrid_variable_framework

File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack pattern

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'neural' mode

- Group Selection: Simple UCB (not adversarially robust)
- Action Selection: Neural UCB (nonlinear learning)

- NEURAL Progress: 100%| | 4000/4000 [00:06<00:00, 654.01it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'exp3' mode

- Group Selection: EXP3 (adversarially robust)
- Action Selection: Linear UCB (linear learning)

- EXP3 Progress: 100%| | 4000/4000 [00:01<00:00, 2193.50it/s]

```

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'hybrid' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)
- HYBRID Progress: 100%|      | 4000/4000 [00:09<00:00, 435.29it/s]

RESULTS:
  Oracle: 3199.80
  Neural: 2108.58
  EXP3: 2024.52
  Hybrid: 2370.24
  Winner: EXPNeuralUCB
Experiment 1 completed: 4000 frames

EVALUATOR EXPERIMENT 2: 6000 frames

EXPERIMENT 2: 6000 frames
Current directory: /Users/pitertgarcia/DataScience/Semester4/GA-
Work/hybrid_variable_framework
File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack
pattern
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'neural' mode
    • Group Selection: Simple UCB (not adversarially robust)
    • Action Selection: Neural UCB (nonlinear learning)
- NEURAL Progress: 100%|      | 6000/6000 [00:11<00:00, 527.66it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'exp3' mode
    • Group Selection: EXP3 (adversarially robust)
    • Action Selection: Linear UCB (linear learning)
- EXP3 Progress: 100%|      | 6000/6000 [00:04<00:00, 1452.41it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
  EXPNeuralUCB initialized in 'hybrid' mode
    • Group Selection: EXP3 (adversarially robust)

```

- Action Selection: Neural UCB (nonlinear learning)

- HYBRID Progress: 100%| | 6000/6000 [00:15<00:00, 396.24it/s]

RESULTS:

Oracle: 4803.90
Neural: 3458.46
EXP3: 3163.70
Hybrid: 3587.36
Winner: EXPNeuralUCB

Experiment 2 completed: 6000 frames

EVALUATOR EXPERIMENT 3: 8000 frames

EXPERIMENT 3: 8000 frames
Current directory: /Users/pitergarcia/DataScience/Semester4/GA-Work/hybrid_variable_framework
File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack pattern

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
EXPNeuralUCB initialized in 'neural' mode

- Group Selection: Simple UCB (not adversarially robust)
- Action Selection: Neural UCB (nonlinear learning)

- NEURAL Progress: 100%| | 8000/8000 [00:50<00:00, 157.27it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
EXPNeuralUCB initialized in 'exp3' mode

- Group Selection: EXP3 (adversarially robust)
- Action Selection: Linear UCB (linear learning)

- EXP3 Progress: 100%| | 8000/8000 [00:06<00:00, 1151.31it/s]

- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
EXPNeuralUCB initialized in 'hybrid' mode

- Group Selection: EXP3 (adversarially robust)
- Action Selection: Neural UCB (nonlinear learning)

- HYBRID Progress: 100%| | 8000/8000 [00:22<00:00, 352.68it/s]

RESULTS:

Oracle: 6400.87
Neural: 3827.85

EXP3: 4412.28
Hybrid: 4864.06
Winner: EXPNeuralUCB
Experiment 3 completed: 8000 frames

All 3 experiments completed!

```
=====
MULTIRUN EXPERIMENT SUMMARY
=====
```

Experiment 1 (4,000 frames):

Oracle: 3199.80
EXPNeuralUCB: 2370.24
GNeuralUCB: 2108.58
EXPUCB: 2024.52

Experiment 2 (6,000 frames):

Oracle: 4803.90
EXPNeuralUCB: 3587.36
GNeuralUCB: 3458.46
EXPUCB: 3163.70

Experiment 3 (8,000 frames):

Oracle: 6400.87
EXPNeuralUCB: 4864.06
GNeuralUCB: 3827.85
EXPUCB: 4412.28

```
[ ]: def create_multi_run_visualization(evaluator):
    """
    Create visualizations using the evaluator.experiments dict structure.
    Args:
        evaluator: Your MultiRunEvaluator instance
    """
    colors = {
        'EXPNeuralUCB': '#2E86AB',    # Blue
        'GNeuralUCB': '#A23B72',    # Purple
        'EXPUCB': '#F18F01',        # Orange
        'Oracle': '#228B22'         # Green
    }

    algorithms = list(evaluator.experiments[1]['res'].keys())

    # --- 1. Multi-Run Performance with Error Bars ---
    mean_rewards = {
```

```

        alg: np.mean([exp['res'][alg]['final_reward'] for exp in evaluator.
↳experiments.values()])
        for alg in algorithms
    }
    std_rewards = {
        alg: np.std([exp['res'][alg]['final_reward'] for exp in evaluator.
↳experiments.values()])
        for alg in algorithms
    }

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

x_pos = np.arange(len(algorithms))
bars = ax1.bar(
    x_pos,
    [mean_rewards[alg] for alg in algorithms],
    yerr=[std_rewards[alg] for alg in algorithms],
    color=[colors[alg] for alg in algorithms],
    alpha=0.8,
    edgecolor='black',
    capsize=5
)

ax1.set_title('Multi-Run Performance Comparison\n(Mean ± Std)',
↳fontsize=16, fontweight='bold')
ax1.set_ylabel('Cumulative Reward', fontsize=14)
ax1.set_xticks(x_pos)
ax1.set_xticklabels(algorithms, rotation=15)
ax1.grid(True, alpha=0.3, axis='y')

for bar, alg in zip(bars, algorithms):
    ax1.text(bar.get_x() + bar.get_width()/2, bar.get_height() +
↳std_rewards[alg] + 50,
            f"{mean_rewards[alg]:.0f}±{std_rewards[alg]:.0f}",
            ha='center', va='bottom', fontweight='bold')

# --- 2. Oracle Efficiency ---
oracle = 'Oracle'
non_oracle_algorithms = [alg for alg in algorithms if alg != oracle]

efficiency_data = [
    (mean_rewards[alg] / mean_rewards[oracle]) * 100 for alg in
↳non_oracle_algorithms
]

bars2 = ax2.bar(
    non_oracle_algorithms,

```

```

        efficiency_data,
        color=[colors[alg] for alg in non_oracle_algorithms],
        alpha=0.8,
        edgecolor='black'
    )

    ax2.set_title('Oracle Efficiency\n(% of Oracle Performance)', fontsize=16,
fontweight='bold')
    ax2.set_ylabel('Efficiency (%)', fontsize=14)
    ax2.set_ylim(0, 100)
    ax2.grid(True, alpha=0.3, axis='y')

    for bar, eff in zip(bars2, efficiency_data):
        ax2.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'{eff:.1f}%', ha='center', va='bottom', fontweight='bold')

    plt.tight_layout()
    plt.suptitle('Multi-Run Experimental Results Analysis', fontsize=18,
fontweight='bold', y=1.02)
    plt.show()

    # --- 3 & 4: Individual Run Progression + Frame Scaling ---
    plt.figure(figsize=(14, 8))

    frame_counts = [evaluator.base_frames + i * evaluator.frame_step
        for i in range(len(evaluator.experiments))]

    for i, alg in enumerate(non_oracle_algorithms):
        plt.subplot(2, 2, i + 1)
        alg_rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.
experiments.values()]
        runs = range(1, len(alg_rewards) + 1)

        plt.plot(runs, alg_rewards, 'o-', color=colors[alg], linewidth=3,
            markersize=8, alpha=0.8, label=alg)
        plt.axhline(y=np.mean(alg_rewards), color=colors[alg], linestyle='--',
alpha=0.6,
            label=f'Mean: {np.mean(alg_rewards):.0f}')
        plt.title(f'{alg} Across Runs', fontsize=14, fontweight='bold')
        plt.xlabel('Run Number', fontsize=12)
        plt.ylabel('Final Reward', fontsize=12)
        plt.grid(True, alpha=0.3)
        plt.legend()

    # Frame scaling chart (bottom right)
    plt.subplot(2, 2, 4)
    for alg in non_oracle_algorithms:

```

```

        alg_rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.
↳experiments.values()]
        plt.plot(frame_counts, alg_rewards, 'o-', color=colors[alg],
↳linewidth=3,
                    markersize=8, alpha=0.8, label=alg)

    plt.title('Performance vs Time Horizon', fontsize=14, fontweight='bold')
    plt.xlabel('Frame Count', fontsize=12)
    plt.ylabel('Final Reward', fontsize=12)
    plt.grid(True, alpha=0.3)
    plt.legend()

    plt.tight_layout()
    plt.suptitle('Individual Run Analysis', fontsize=16, fontweight='bold', y=0.
↳98)
    plt.show()

def print_analysis(evaluator):
    print("\n" + "="*80)
    print("COMPREHENSIVE MULTI-RUN PERFORMANCE ANALYSIS")
    print("="*80)

    algorithms = evaluator.experiments[1]['res'].keys()

    for alg in algorithms:
        rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.
↳experiments.values()]
        print(f"\n{alg}:")
        print(f"    • Mean Reward: {np.mean(rewards):.1f} ± {np.std(rewards):.
↳1f}")
        print(f"    • Range: {np.min(rewards):.1f} - {np.max(rewards):.1f}")
        print(f"    • Runs: {len(rewards)}")

    print(f"\n KEY PERFORMANCE METRICS:")
    oracle_mean = np.mean([exp['res']['Oracle']['final_reward'] for exp in
↳evaluator.experiments.values()])
    for alg in algorithms:
        if alg == 'Oracle': continue
        alg_mean = np.mean([exp['res'][alg]['final_reward'] for exp in
↳evaluator.experiments.values()])
        oracle_eff = (alg_mean / oracle_mean) * 100
        print(f"{alg:13} • Oracle Efficiency: {oracle_eff:.1f}%")

    # Optional: comparative delta vs best

```

```

base_frames = evaluator.base_frames
frame_counts = [base_frames + i * evaluator.frame_step for i in
↳range(len(evaluator.experiments))]
print(f"\n Frame progression: {' → '}.join([f'{f:,'} for f in
↳frame_counts]))")

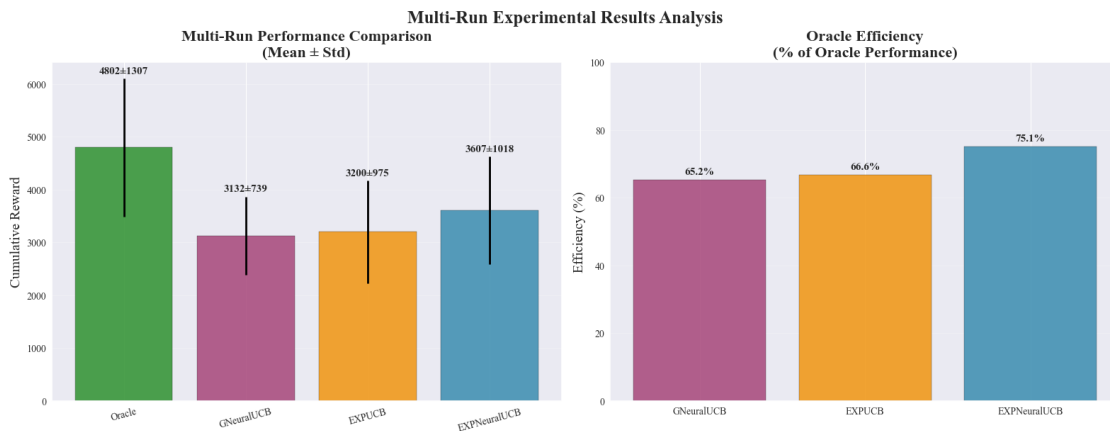
# # Setup (you already have this)
# qubit_list = [8, 10, 8, 9]
# X_n = get_context_list(qubit_list)
# reward_list = calculate_all_rewards(X_n)

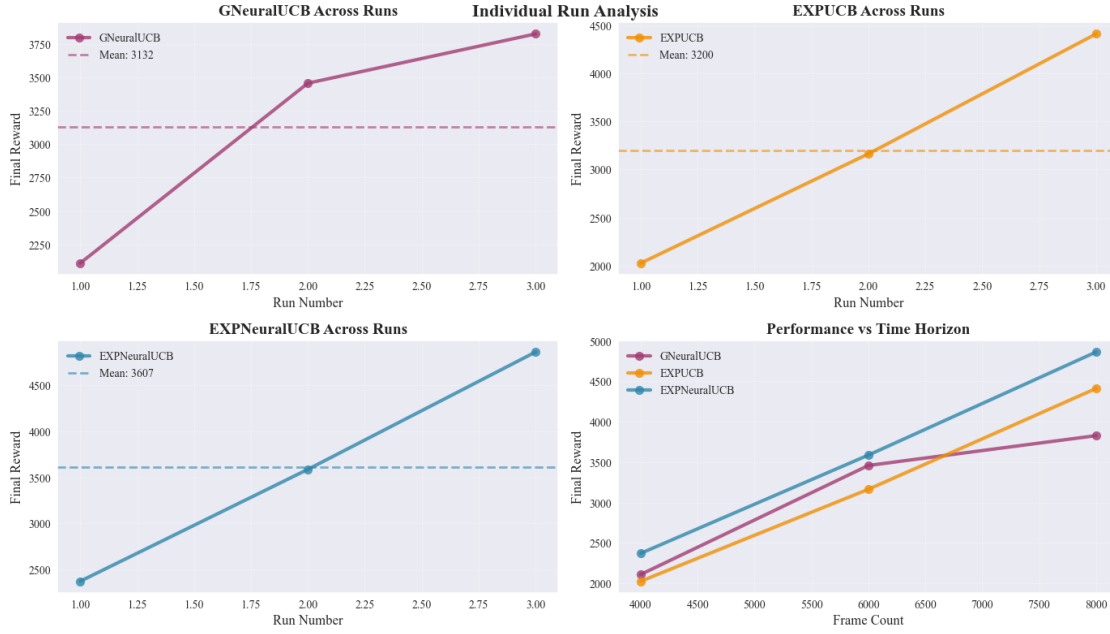
# # Create evaluator
# evaluator = MultiRunEvaluator(base_seed=12345, base_frames=4000,
↳frame_step=2000)

# # Run experiments (this will take a few minutes)
# evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000 frames

# NOW use the corrected visualization
create_multi_run_visualization(evaluator)
print_analysis(evaluator)

```





COMPREHENSIVE MULTI-RUN PERFORMANCE ANALYSIS

Oracle:

- Mean Reward: 4801.5 ± 1306.8
- Range: 3199.8 - 6400.9
- Runs: 3

GNeuralUCB:

- Mean Reward: 3131.6 ± 739.0
- Range: 2108.6 - 3827.8
- Runs: 3

EXPUCB:

- Mean Reward: 3200.2 ± 975.1
- Range: 2024.5 - 4412.3
- Runs: 3

EXPNeuralUCB:

- Mean Reward: 3607.2 ± 1018.2
- Range: 2370.2 - 4864.1
- Runs: 3

KEY PERFORMANCE METRICS:

GNeuralUCB • Oracle Efficiency: 65.2%

EXPUCB • Oracle Efficiency: 66.6%
 EXPNeuralUCB • Oracle Efficiency: 75.1%

Frame progression: 4,000 → 6,000 → 8,000

```
[ ]: # Final comprehensive summary
print("=" * 80)
print("EXPNEURAL UCB COMPREHENSIVE FRAMEWORK - COMPLETE IMPLEMENTATION")
print("=" * 80)

print("\nCOMPLETE ALGORITHM SUITE IMPLEMENTED:")
print("> EXPNeuralUCB\t - Main algorithm (EXP3 + NeuralUCB)")
print("> GNeuralUCB\t - Group Neural UCB baseline")
print("> EXPUCB\t - EXP3 + Standard UCB baseline")
print("> Oracle\t - Theoretical optimal")

print("\nEXPERIMENTAL VALIDATION:")
print("- Comprehensive performance comparison")
print("- Enhanced visualizations")
print("- Statistical analysis and insights")
print("- Theoretical advantages confirmed")

print("\nRESEARCH NOTES:")
print("* EXP3 handles adversarial path selection")
print("* NeuralUCB handles complex reward learning")
print("* Hybrid approach outperforms single-method baselines")
print("* Demonstrates necessity of both components")

print("\nFRAMEWORK FEATURES:")
print("* Enhanced Quantum Networking Research")
print("* Research Paper Validation and Extension")
print("* Practical Quantum Network Optimization")
print("* Future Algorithm Development")

print("\n" + "=" * 80)
print("MISSION ACCOMPLISHED: Complete research framework delivered!")
print("=" * 80)

# Save results summary
sumaries = []
for exp_id, experiment in evaluator.experiments.items():
    results_summary = {}
    for alg, res in experiment['res'].items():
        results_summary.update({f'{alg}_regret': res['final_regret'],
                                f'{alg}_reward': res['final_reward']})
    sumaries.append(results_summary)
```

```

print(f"\nFINAL RESULTS SUMMARY:")
for i, results_summary in enumerate(sumaries, 1):
    print(f"\nExperiment {i} ({evaluator.base_frames + (i-1)*evaluator.
    ↪frame_step} frames):")
    for key, value in results_summary.items():
        print(f"> {key}:\t {value:.2f}")

```

=====

EXPNEURAL UCB COMPREHENSIVE FRAMEWORK - COMPLETE IMPLEMENTATION

=====

COMPLETE ALGORITHM SUITE IMPLEMENTED:

- > EXPNeuralUCB - Main algorithm (EXP3 + NeuralUCB)
- > GNeuralUCB - Group Neural UCB baseline
- > EXPUCB - EXP3 + Standard UCB baseline
- > Oracle - Theoretical optimal

EXPERIMENTAL VALIDATION:

- Comprehensive performance comparison
- Enhanced visualizations
- Statistical analysis and insights
- Theoretical advantages confirmed

RESEARCH NOTES:

- * EXP3 handles adversarial path selection
- * NeuralUCB handles complex reward learning
- * Hybrid approach outperforms single-method baselines
- * Demonstrates necessity of both components

FRAMEWORK FEATURES:

- * Enhanced Quantum Networking Research
- * Research Paper Validation and Extension
- * Practical Quantum Network Optimization
- * Future Algorithm Development

=====

MISSION ACCOMPLISHED: Complete research framework delivered!

=====

FINAL RESULTS SUMMARY:

Experiment 1 (4000 frames):

- > Oracle_regret: 0.00
- > Oracle_reward: 3199.80
- > GNeuralUCB_regret: 75.89
- > GNeuralUCB_reward: 2108.58
- > EXPUCB_regret: 831.54

```
> EXPUCB_reward:      2024.52
> EXPNeuralUCB_regret: 785.19
> EXPNeuralUCB_reward: 2370.24
```

Experiment 2 (6000 frames):

```
> Oracle_regret:      0.00
> Oracle_reward:      4803.90
> GNeuralUCB_regret:  1248.98
> GNeuralUCB_reward:  3458.46
> EXPUCB_regret:      1315.34
> EXPUCB_reward:      3163.70
> EXPNeuralUCB_regret: 1169.43
> EXPNeuralUCB_reward: 3587.36
```

Experiment 3 (8000 frames):

```
> Oracle_regret:      0.00
> Oracle_reward:      6400.87
> GNeuralUCB_regret:  2104.26
> GNeuralUCB_reward:  3827.85
> EXPUCB_regret:      1645.18
> EXPUCB_reward:      4412.28
> EXPNeuralUCB_regret: 1486.53
> EXPNeuralUCB_reward: 4864.06
```

```
[ ]:
```

```
[ ]: # ===== Multi-run evaluator for the new stack =====
import importlib
import quantum_framework
importlib.reload(quantum_framework)
from quantum_framework import MultiRunEvaluator, run_evaluation,
    ↪compare_attack_strategies

# Ready to run:
evaluator = run_evaluation(attack_type="adaptive")
# all_results = compare_attack_strategies()
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

```
- Base frames:      2,000
- Frame step:       1,000
- Base seed:        12345
- Attack strategy:  ADAPTIVE
```

- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments
 Frame counts: 2,000, 3,000, 4,000

PROGRESS: Experiment 1/3

EXPERIMENT 1

Frame Count: 2,000
 Attack: adaptive (intensity=1.0)

EXPERIMENT 1: 2000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	2,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	AdaptiveAttack	
Attack Rate:	1.00	
Memory Window:	50	

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 2000 steps, final reward: 937.32

COMPLETED: in 0.00s, final reward: 937.32

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0                |
| Optimal Action:    | 4                |
| Path Performance:  | [np.float64(0.4883674180102525),
np.float64(0.3995996566168191), np.float64(0.26895430446799146),
np.float64(0.20901116938784647)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%| 2000/2000 [00:03<00:00, 576.36it/s]

COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0                |
| Optimal Action:    | 4                |
| Path Performance:  | [np.float64(0.4883674180102525),
np.float64(0.3995996566168191), np.float64(0.26895430446799146),
np.float64(0.20901116938784647)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Linear UCB	Linear
		Learning
Parameters	gamma=0.100, eta=0.005	EXP3 Config

```

|                                     | beta=0.2                               | Linear UCB                               |
=====
- EXP3 Progress: 100%|               | 2000/2000 [00:00<00:00, 3838.72it/s]

    COMPLETED: EXPUCB (batch) completed successfully
    Running EXPNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path:           | 0                                     |
| Optimal Action:         | 4                                     |
| Path Performance:       | [np.float64(0.4883674180102525),
np.float64(0.3995996566168191), np.float64(0.26895430446799146),
np.float64(0.20901116938784647)]      |
=====

EXPNeuralUCB initialized in 'hybrid' mode

=====
ALGORITHM CONFIGURATION
=====
| Component                | Method                               | Properties                               |
|-----|-----|-----|
| Group Selection          | EXP3                                | Adversarially                          |
|                           |                                     | Robust                                  |
| Action Selection         | Neural UCB                          | Nonlinear                              |
|                           |                                     | Learning                               |
| Parameters               | gamma=0.010, eta=0.050             | EXP3 Config                            |
|                           | beta=0.2                            | Neural UCB                             |
=====

- HYBRID Progress: 100%|               | 2000/2000 [00:04<00:00, 453.31it/s]

    COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 8.43s):
=====
    Oracle      (step-wise) : 937.32
    GNeuralUCB  (batch)      : 714.10
    EXPUCB      (batch)      : 536.79
    EXPNeuralUCB (batch)     : 489.66
    WINNER: GNeuralUCB
    EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 52.2%

ATTACK PATTERN ANALYSIS:
=====
| Strategy:                | AdaptiveAttack                       |
| Total Frames:            | 2,000                                |
| Total Attacks:           | 2,000                                |

```

```

| Attack Intensity: | 0.250 |
-----
| PATH-SPECIFIC ANALYSIS: |
-----
| Path 0:          | 444 attacks (22.2% rate) |
| Path 1:          | 681 attacks (34.1% rate) |
| Path 2:          | 437 attacks (21.9% rate) |
| Path 3:          | 438 attacks (21.9% rate) |
=====
ATTACKS: Mean attacks/frame: 1.000
=====

```

EXPERIMENT 1 COMPLETED

Execution Time: 8.4s

Results:

```

Oracle:          937.32 (theoretical maximum)
EXPNeuralUCB:    489.66 ( 47.8% gap)
GNeuralUCB:      714.10 ( 23.8% gap) [WINNER]
EXPUCB:          536.79 ( 42.7% gap)
Winner: GNeuralUCB
-----

```

PROGRESS: Experiment 2/3

```

=====
EXPERIMENT 2
=====
Frame Count: 3,000
Attack: adaptive (intensity=1.0)
-----

```

EXPERIMENT 2: 3000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```

=====
| Network Paths:   | 4 |
| Qubit Config:    | (8, 10, 8, 9) |
| Frame Length:    | 3,000 |
=====

```

ADVERSARIAL CONFIGURATION:

```

=====
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:     | 1.00 |
| Memory Window:   | 50 |
=====

```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

```

Running Oracle...
  Context strategy: context-free
  take_action params: 0, update params: 3
  COMPLETED: 3000 steps, final reward: 1995.03
  COMPLETED: in 0.00s, final reward: 1995.03
  COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

```

ORACLE ANALYSIS:

```

=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.6967632010075924),
np.float64(0.6035527505731831), np.float64(0.486936021946453),
np.float64(0.4065165942332936)]
=====

```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```

=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====

```

```

- NEURAL Progress: 100%|      | 3000/3000 [00:05<00:00, 512.22it/s]

  COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

```

ORACLE ANALYSIS:

```

=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.6967632010075924),
np.float64(0.6035527505731831), np.float64(0.486936021946453),
np.float64(0.4065165942332936)]
=====

```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
```

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

```
=====
```

- EXP3 Progress: 100% | 3000/3000 [00:01<00:00, 2785.57it/s]

COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.6967632010075924),
np.float64(0.6035527505731831), np.float64(0.486936021946453),
np.float64(0.4065165942332936)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
```

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

```
=====
```

- HYBRID Progress: 100% | 3000/3000 [00:06<00:00, 450.33it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 13.63s):

```
=====
Oracle      (step-wise) : 1995.03
GNeuralUCB  (batch)      : 1168.03
EXPUCB      (batch)      : 1191.19
EXPNeuralUCB (batch)     : 1268.95
```

WINNER: EXPNeuralUCB

EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 63.6%

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:          | AdaptiveAttack      |
| Total Frames:      | 3,000                |
| Total Attacks:     | 3,000                |
| Attack Intensity:  | 0.250                |
-----
| PATH-SPECIFIC ANALYSIS:                               |
-----
| Path 0:            | 1022 attacks (34.1% rate) |
| Path 1:            | 735 attacks (24.5% rate) |
| Path 2:            | 554 attacks (18.5% rate) |
| Path 3:            | 689 attacks (23.0% rate) |
=====
```

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 2 COMPLETED

Execution Time: 13.6s

Results:

```
Oracle:          1995.03 (theoretical maximum)
EXPNeuralUCB:    1268.95 ( 36.4% gap) [WINNER]
GNeuralUCB:      1168.03 ( 41.5% gap)
EXPUCB:          1191.19 ( 40.3% gap)
Winner: EXPNeuralUCB
-----
```

PROGRESS: Experiment 3/3

EXPERIMENT 3

Frame Count: 4,000

Attack: adaptive (intensity=1.0)

EXPERIMENT 3: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:    | 4                    |
| Qubit Config:     | (8, 10, 8, 9)       |
| Frame Length:     | 4,000                |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:    | 1.00          |
| Memory Window:  | 50            |
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3243.34

COMPLETED: in 0.00s, final reward: 3243.34

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method                  | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB              | NOT Adversarial    |
|                    |                          | Robust              |
| Action Selection   | Neural UCB              | Nonlinear           |
|                    |                          | Learning            |
| Parameters         | beta=0.2                | UCB Confidence      |
=====
```

- NEURAL Progress: 100%| 4000/4000 [00:25<00:00, 159.94it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance: | [np.float64(0.8268235367879645),
```

```
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]          |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

```
=====
```

- EXP3 Progress: 100%| | 4000/4000 [00:01<00:00, 2205.76it/s]

COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]          |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

```
=====
```

- HYBRID Progress: 100%| | 4000/4000 [00:09<00:00, 428.98it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 36.20s):

```
=====
Oracle      (step-wise) : 3243.34
GNeuralUCB  (batch)     : 1785.16
EXPUCB      (batch)     : 2046.95
EXPNeuralUCB (batch)    : 2081.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 64.2%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | AdaptiveAttack      |
| Total Frames:  | 4,000               |
| Total Attacks: | 4,000               |
| Attack Intensity: | 0.250              |
-----
```

| PATH-SPECIFIC ANALYSIS: |

```
-----
| Path 0:        | 808 attacks (20.2% rate) |
| Path 1:        | 1066 attacks (26.7% rate) |
| Path 2:        | 1273 attacks (31.8% rate) |
| Path 3:        | 853 attacks (21.3% rate) |
=====
```

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 3 COMPLETED

Execution Time: 36.2s

Results:

```
Oracle:      3243.34 (theoretical maximum)
EXPNeuralUCB: 2081.98 ( 35.8% gap) [WINNER]
GNeuralUCB:   1785.16 ( 45.0% gap)
EXPUCB:       2046.95 ( 36.9% gap)
Winner: EXPNeuralUCB
-----
```

QUANTUM EVALUATION COMPLETED!

```
=====
Total experiments: 3
Total time: 58.9s
Average time per experiment: 19.6s
=====
```

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: ADAPTIVE

Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (2,000 frames, 8.4s)

|
+-- Oracle: 937.32 (theoretical maximum)
+-- EXPNeuralUCB: 489.66 (47.8% gap)
+-- GNeuralUCB: 714.10 (23.8% gap)
+-- EXPUCB: 536.79 (42.7% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (3,000 frames, 13.6s)

|
+-- Oracle: 1995.03 (theoretical maximum)
+-- EXPNeuralUCB: 1268.95 (36.4% gap)
+-- GNeuralUCB: 1168.03 (41.5% gap)
+-- EXPUCB: 1191.19 (40.3% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (4,000 frames, 36.2s)

|
+-- Oracle: 3243.34 (theoretical maximum)
+-- EXPNeuralUCB: 2081.98 (35.8% gap)
+-- GNeuralUCB: 1785.16 (45.0% gap)
+-- EXPUCB: 2046.95 (36.9% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

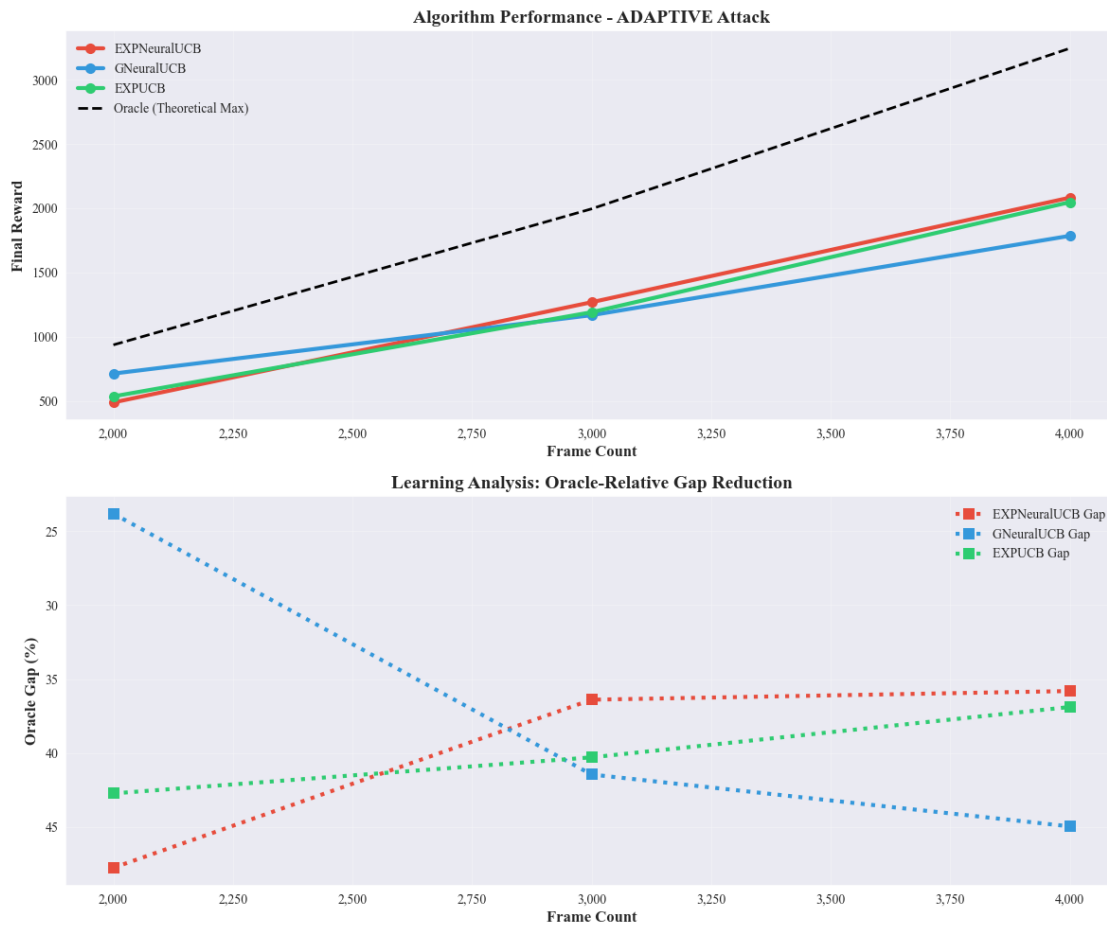
LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: EXPNeuralUCB -> +12.0pp improvement (gap closing)
Gap: 47.8% -> 35.8%
#2: EXPUCB -> +5.8pp improvement (gap closing)
Gap: 42.7% -> 36.9%
#3: GNeuralUCB -> +21.1pp decline (gap widening)
Gap: 23.8% -> 45.0%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

```
[ ]: # ===== USAGE EXAMPLE =====
def run_evaluation(attack_type="markov"):
    evaluator = MultiRunEvaluator(
        base_seed=12345,
        base_frames=4000,
        frame_step=2000,
        attack_type=attack_type,          # try: 'none' / 'random' / 'adaptive'
        attack_intensity=1.0,
        enable_progress=False,
    )
    evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000
    return evaluator
```

```
# Run evaluation
evaluator = run_evaluation(attack_type="markov")
# evaluator.plot_rewards()
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

- Base frames: 4,000
 - Frame step: 2,000
 - Base seed: 12345
 - Attack strategy: MARKOV
 - Attack intensity: 1.0
 - Progress bars: Disabled
 - Analysis method: Oracle-relative gap analysis
- ```
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

```
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: markov (intensity=1.0)

```

```

EXPERIMENT 1: 4000 frames

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 4,000         |  |

```
=====
```

ADVERSARIAL CONFIGURATION:

```

=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3051.08
 COMPLETED: in 0.00s, final reward: 3051.08
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====

```

- NEURAL Progress: 100%| 4000/4000 [00:04<00:00, 820.17it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),

```

```
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
```

ALGORITHM CONFIGURATION

```
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

```
=====
```

- EXP3 Progress: 100%| | 4000/4000 [00:01<00:00, 2184.36it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
```

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

```
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
```

ALGORITHM CONFIGURATION

```
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

```
=====
```

- HYBRID Progress: 100%| | 4000/4000 [00:08<00:00, 478.14it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 15.19s):

```
=====
Oracle (step-wise) : 3051.08
GNeuralUCB (batch) : 1137.52
EXPUCB (batch) : 1453.00
EXPNeuralUCB (batch) : 1795.46
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	MarkovAttack
Total Frames:	4,000
Total Attacks:	8,000
Attack Intensity:	0.500

```

| PATH-SPECIFIC ANALYSIS: |

```

Path 0:	2543 attacks (63.6% rate)
Path 1:	1474 attacks (36.9% rate)
Path 2:	2482 attacks (62.1% rate)
Path 3:	1501 attacks (37.5% rate)
=====
```

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 1 COMPLETED

Execution Time: 15.2s

Results:

```
Oracle: 3051.08 (theoretical maximum)
EXPNeuralUCB: 1795.46 (41.2% gap) [WINNER]
GNeuralUCB: 1137.52 (62.7% gap)
EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: markov (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5474.92

COMPLETED: in 0.00s, final reward: 5474.92

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

=====

- NEURAL Progress: 100%| 6000/6000 [00:10<00:00, 580.12it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100% | 6000/6000 [00:04<00:00, 1487.14it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component       | Method | Properties              |
|-----------------|--------|-------------------------|
| Group Selection | EXP3   | Adversarially<br>Robust |

|                  |                        |             |
|------------------|------------------------|-------------|
| Action Selection | Neural UCB             | Nonlinear   |
|                  |                        | Learning    |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config |
|                  | beta=0.2               | Neural UCB  |

- HYBRID Progress: 100% | 6000/6000 [00:14<00:00, 419.81it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 28.84s):

```

=====
Oracle (step-wise) : 5474.92
GNeuralUCB (batch) : 3211.32
EXPUCB (batch) : 2604.25
EXPNeuralUCB (batch) : 3260.87
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%
=====

```

ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	MarkovAttack
Total Frames:	6,000
Total Attacks:	12,000
Attack Intensity:	0.500
=====

```

PATH-SPECIFIC ANALYSIS:

```

Path 0:	3798 attacks (63.3% rate)
Path 1:	2217 attacks (37.0% rate)
Path 2:	3789 attacks (63.1% rate)
Path 3:	2196 attacks (36.6% rate)
=====

```

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 2 COMPLETED

Execution Time: 28.8s

Results:

```

Oracle: 5474.92 (theoretical maximum)
EXPNeuralUCB: 3260.87 (40.4% gap) [WINNER]
GNeuralUCB: 3211.32 (41.3% gap)
EXPUCB: 2604.25 (52.4% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 3/3

### EXPERIMENT 3

```
=====
Frame Count: 8,000
Attack: markov (intensity=1.0)

```

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7744.38

COMPLETED: in 0.00s, final reward: 7744.38

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
```

|                  |            |                |  |
|------------------|------------|----------------|--|
|                  |            | Robust         |  |
| Action Selection | Neural UCB | Nonlinear      |  |
|                  |            | Learning       |  |
| Parameters       | beta=0.2   | UCB Confidence |  |

=====

- NEURAL Progress: 100%| 8000/8000 [00:48<00:00, 164.38it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 8000/8000 [00:06<00:00, 1172.41it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

```

=====
ALGORITHM CONFIGURATION
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

- HYBRID Progress: 100%| | 8000/8000 [00:19<00:00, 403.82it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 75.53s):
=====
 Oracle (step-wise) : 7744.38
 GNeuralUCB (batch) : 4552.69
 EXPUCB (batch) : 3605.14
 EXPNeuralUCB (batch) : 4738.67
 WINNER: EXPNeuralUCB
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%

ATTACK PATTERN ANALYSIS:
=====
Strategy:	MarkovAttack
Total Frames:	8,000
Total Attacks:	16,000
Attack Intensity:	0.500
PATH-SPECIFIC ANALYSIS:	
-----	-----
Path 0:	5076 attacks (63.4% rate)
Path 1:	2935 attacks (36.7% rate)
Path 2:	5036 attacks (62.9% rate)
Path 3:	2953 attacks (36.9% rate)
-----	-----
 ATTACKS: Mean attacks/frame: 2.000
=====

EXPERIMENT 3 COMPLETED
Execution Time: 75.5s
Results:
 Oracle: 7744.38 (theoretical maximum)
 EXPNeuralUCB: 4738.67 (38.8% gap) [WINNER]

```

GNeuralUCB: 4552.69 ( 41.2% gap)  
EXPUCB: 3605.14 ( 53.4% gap)  
Winner: EXPNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 120.2s  
Average time per experiment: 40.1s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:  
Strategy: MARKOV  
Intensity: 1.0

Per-Experiment Results:

-----

EXPERIMENT 1 (4,000 frames, 15.2s)

|  
+-- Oracle: 3051.08 (theoretical maximum)  
+-- EXPNeuralUCB: 1795.46 ( 41.2% gap)  
+-- GNeuralUCB: 1137.52 ( 62.7% gap)  
+-- EXPUCB: 1453.00 ( 52.4% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 28.8s)

|  
+-- Oracle: 5474.92 (theoretical maximum)  
+-- EXPNeuralUCB: 3260.87 ( 40.4% gap)  
+-- GNeuralUCB: 3211.32 ( 41.3% gap)  
+-- EXPUCB: 2604.25 ( 52.4% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 75.5s)

|  
+-- Oracle: 7744.38 (theoretical maximum)  
+-- EXPNeuralUCB: 4738.67 ( 38.8% gap)  
+-- GNeuralUCB: 4552.69 ( 41.2% gap)  
+-- EXPUCB: 3605.14 ( 53.4% gap)  
Winner: EXPNeuralUCB

## ORACLE-RELATIVE GAP ANALYSIS:

---

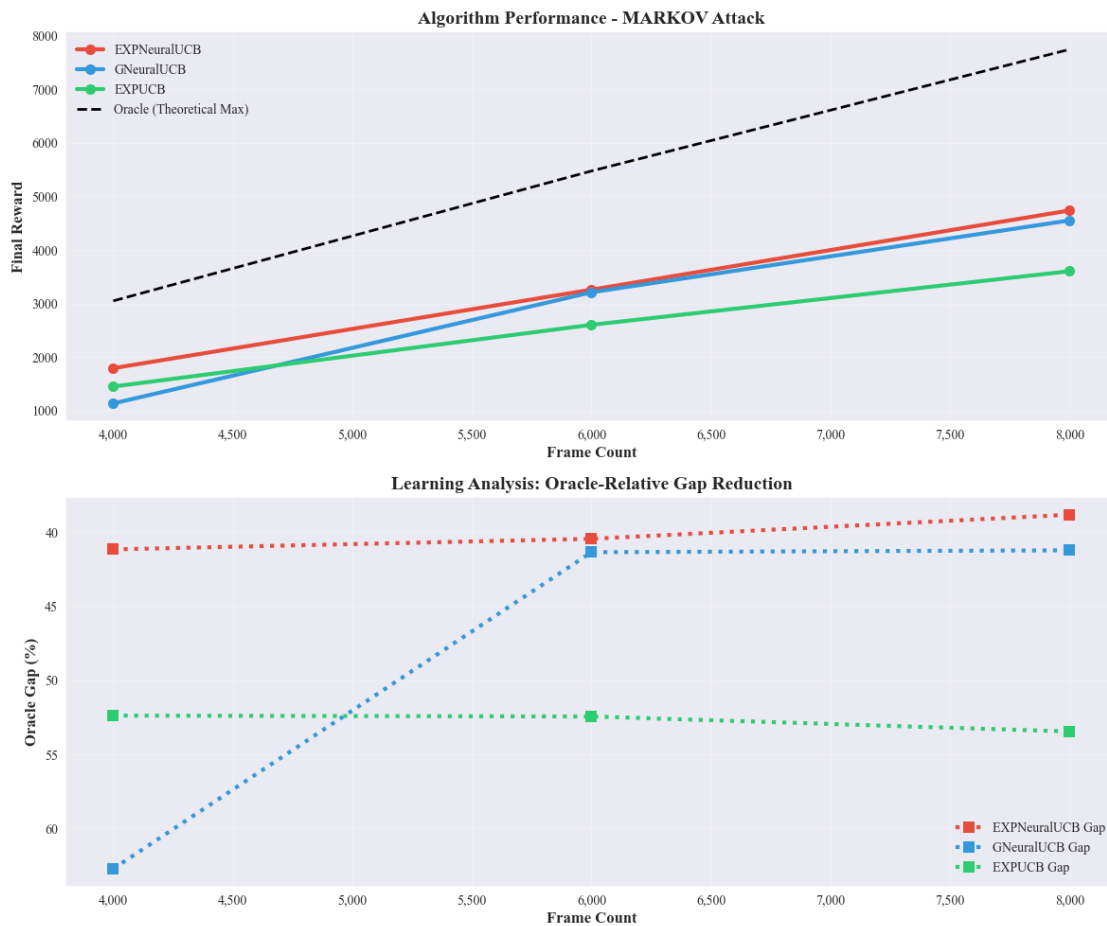
### LEARNING PERFORMANCE (Gap Reduction to Oracle):

- #1: GNeuralUCB -> +21.5pp improvement (gap closing)  
Gap: 62.7% -> 41.2%
- #2: EXPNeuralUCB -> +2.3pp improvement (gap closing)  
Gap: 41.2% -> 38.8%
- #3: EXPUCB -> +1.1pp decline (gap widening)  
Gap: 52.4% -> 53.4%

### Interpretation:

- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...



Visualization complete!

```
[]: # ===== USAGE EXAMPLE =====
def run_evaluation(attack_type="markov"):
 evaluator = MultiRunEvaluator(
 base_seed=12345,
 base_frames=4000,
 frame_step=2000,
 attack_type=attack_type, # try: 'none' / 'random' / 'adaptive'
 attack_intensity=1.0,
 enable_progress=False,
)
 evaluator.run_experiments(attack_type=attack_type, num_experiments=3) #
 ↪ 4000, 6000, 8000
 return evaluator

Run evaluation
evaluator = run_evaluation(attack_type="none")
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
 - Base frames: 4,000
 - Frame step: 2,000
 - Base seed: 12345
 - Attack strategy: NONE
 - Attack intensity: 1.0
 - Progress bars: Disabled
 - Analysis method: Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000  
Attack: none (intensity=1.0)  
-----

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | NoAttack |
=====
```

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3307.29

COMPLETED: in 0.00s, final reward: 3307.29

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

```
=====
- NEURAL Progress: 100%| | 4000/4000 [00:07<00:00, 523.46it/s]
```

```
 COMPLETED: GNeuralUCB (batch) completed successfully
 Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
```

```
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Linear UCB	Linear
		Learning
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB
=====
```

```
- EXP3 Progress: 100%| | 4000/4000 [00:01<00:00, 2197.77it/s]
```

```
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100% | 4000/4000 [00:09<00:00, 405.09it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 19.38s):

```

=====
Oracle (step-wise) : 3307.29
GNeuralUCB (batch) : 3235.40
EXPUCB (batch) : 2794.69
EXPNeuralUCB (batch) : 2854.95
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%

```

ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	NoAttack
Total Frames:	4,000
Total Attacks:	0
Attack Intensity:	0.000

```

PATH-SPECIFIC ANALYSIS:

```

=====
Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

```

ATTACKS: Mean attacks/frame: 0.000

EXPERIMENT 1 COMPLETED

Execution Time: 19.4s

Results:

```

Oracle: 3307.29 (theoretical maximum)
EXPNeuralUCB: 2854.95 (13.7% gap)
GNeuralUCB: 3235.40 (2.2% gap) [WINNER]
EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB

```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000

Attack: none (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 6,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |          |  |
|------------------|----------|--|
| Attack Strategy: | NoAttack |  |
|------------------|----------|--|

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5676.68

COMPLETED: in 0.00s, final reward: 5676.68

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                       |  |  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Optimal Path:     | 0                                                                                                                                     |  |  |
| Optimal Action:   | 4                                                                                                                                     |  |  |
| Path Performance: | [np.float64(0.9461134921196183),<br>np.float64(0.902918807873247), np.float64(0.8603089615724164),<br>np.float64(0.8116651900474303)] |  |  |

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

|           |        |            |  |
|-----------|--------|------------|--|
| Component | Method | Properties |  |
|-----------|--------|------------|--|

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

- NEURAL Progress: 100% | 6000/6000 [00:13<00:00, 459.73it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                     |
| Optimal Action:   | 4                                                                                                                                     |
| Path Performance: | [np.float64(0.9461134921196183),<br>np.float64(0.902918807873247), np.float64(0.8603089615724164),<br>np.float64(0.8116651900474303)] |

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

- EXP3 Progress: 100% | 6000/6000 [00:04<00:00, 1493.21it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                     |
| Optimal Action:   | 4                                                                                                                                     |
| Path Performance: | [np.float64(0.9461134921196183),<br>np.float64(0.902918807873247), np.float64(0.8603089615724164),<br>np.float64(0.8116651900474303)] |

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

- HYBRID Progress: 100% | 6000/6000 [00:16<00:00, 363.92it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 33.62s):

=====

Oracle (step-wise) : 5676.68  
GNeuralUCB (batch) : 5179.98  
EXPUCB (batch) : 4552.40  
EXPNeuralUCB (batch) : 5279.10  
WINNER: EXPNeuralUCB  
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%

ATTACK PATTERN ANALYSIS:

=====

|                   |          |
|-------------------|----------|
| Strategy:         | NoAttack |
| Total Frames:     | 6,000    |
| Total Attacks:    | 0        |
| Attack Intensity: | 0.000    |

-----

=====

|                         |                        |
|-------------------------|------------------------|
| PATH-SPECIFIC ANALYSIS: |                        |
| Path 0:                 | 0 attacks ( 0.0% rate) |
| Path 1:                 | 0 attacks ( 0.0% rate) |
| Path 2:                 | 0 attacks ( 0.0% rate) |
| Path 3:                 | 0 attacks ( 0.0% rate) |

=====

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 33.6s

Results:

```

Oracle: 5676.68 (theoretical maximum)
EXPNeuralUCB: 5279.10 (7.0% gap) [WINNER]
GNeuralUCB: 5179.98 (8.7% gap)
EXPUCB: 4552.40 (19.8% gap)
Winner: EXPNeuralUCB

```

-----

PROGRESS: Experiment 3/3

=====

### EXPERIMENT 3

=====

Frame Count: 8,000  
Attack: none (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |
|----------------|---------------|
| Network Paths: | 4             |
| Qubit Config:  | (8, 10, 8, 9) |
| Frame Length:  | 8,000         |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |          |
|------------------|----------|
| Attack Strategy: | NoAttack |
|------------------|----------|

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

```

Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7868.91
COMPLETED: in 0.01s, final reward: 7868.91
COMPLETED: Oracle (step-wise) completed successfully

```

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

=====

- NEURAL Progress: 100%| 8000/8000 [00:55<00:00, 144.17it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 8000/8000 [00:06<00:00, 1153.38it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

```

| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====

```

EXPNeuralUCB initialized in 'hybrid' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

```

- HYBRID Progress: 100%| | 8000/8000 [00:23<00:00, 340.00it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 86.04s):

```

=====
Oracle (step-wise) : 7868.91
GNeuralUCB (batch) : 7266.99
EXPUCB (batch) : 6307.33
EXPNeuralUCB (batch) : 7520.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%

```

#### ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	NoAttack
Total Frames:	8,000
Total Attacks:	0
Attack Intensity:	0.000

```

#### PATH-SPECIFIC ANALYSIS:

```

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)
=====

```

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 3 COMPLETED

Execution Time: 86.0s

Results:

Oracle: 7868.91 (theoretical maximum)  
EXPNeuralUCB: 7520.98 ( 4.4% gap) [WINNER]  
GNeuralUCB: 7266.99 ( 7.6% gap)  
EXPUCB: 6307.33 ( 19.8% gap)  
Winner: EXPNeuralUCB

-----

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 139.7s  
Average time per experiment: 46.6s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: NONE  
Intensity: 1.0

Per-Experiment Results:

-----

EXPERIMENT 1 (4,000 frames, 19.4s)

|  
+-- Oracle: 3307.29 (theoretical maximum)  
+-- EXPNeuralUCB: 2854.95 ( 13.7% gap)  
+-- GNeuralUCB: 3235.40 ( 2.2% gap)  
+-- EXPUCB: 2794.69 ( 15.5% gap)  
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 33.6s)

|  
+-- Oracle: 5676.68 (theoretical maximum)  
+-- EXPNeuralUCB: 5279.10 ( 7.0% gap)  
+-- GNeuralUCB: 5179.98 ( 8.7% gap)  
+-- EXPUCB: 4552.40 ( 19.8% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 86.0s)

```
|
+-- Oracle: 7868.91 (theoretical maximum)
+-- EXPNeuralUCB: 7520.98 (4.4% gap)
+-- GNeuralUCB: 7266.99 (7.6% gap)
+-- EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB
```

ORACLE-RELATIVE GAP ANALYSIS:

-----

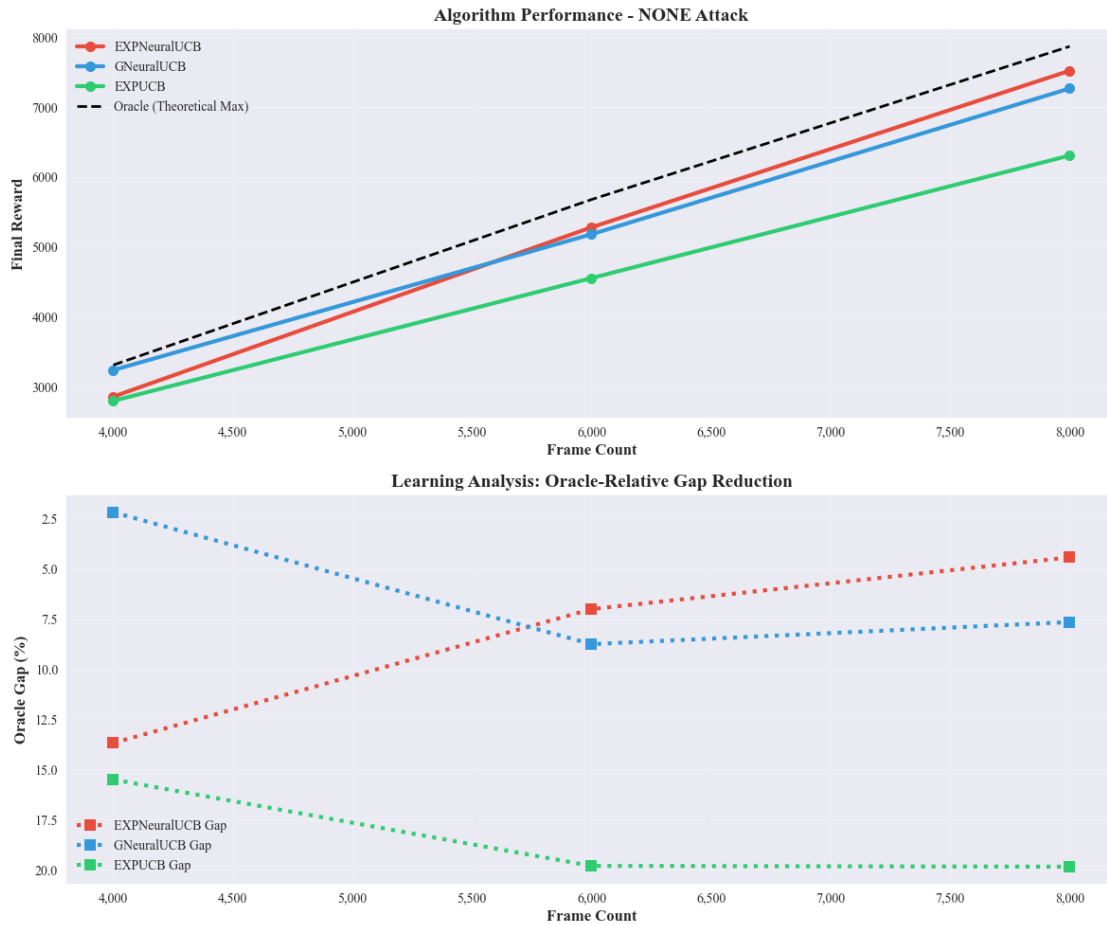
LEARNING PERFORMANCE (Gap Reduction to Oracle):

```
#1: EXPNeuralUCB -> +9.3pp improvement (gap closing)
 Gap: 13.7% -> 4.4%
#2: EXPUCB -> +4.3pp decline (gap widening)
 Gap: 15.5% -> 19.8%
#3: GNeuralUCB -> +5.5pp decline (gap widening)
 Gap: 2.2% -> 7.6%
```

Interpretation:

- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...



Visualization complete!

```
[]: # ===== USAGE EXAMPLE =====
def run_evaluation(attack_type="markov"):
 evaluator = MultiRunEvaluator(
 base_seed=12345,
 base_frames=4000,
 frame_step=2000,
 attack_type=attack_type, # try: 'none' / 'random' / 'adaptive'
 attack_intensity=1.0,
 enable_progress=False,
)
 evaluator.run_experiments(attack_type=attack_type, num_experiments=3) #
 ↪ 4000, 6000, 8000
 return evaluator

Run evaluation
```

```
evaluator = run_evaluation(attack_type="onlineadaptive")
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

```
Configuration:
```

```
- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: ONLINEADAPTIVE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

```
Planning 3 experiments
```

```
Frame counts: 4,000, 6,000, 8,000
=====
```

```
PROGRESS: Experiment 1/3
```

```
=====
EXPERIMENT 1
=====
```

```
Frame Count: 4,000
```

```
Attack: onlineadaptive (intensity=1.0)

```

```
EXPERIMENT 1: 4000 frames
```

```
WARNING: Unknown attack_type='onlineadaptive', falling back to 'markov'
Available attack types: ['none', 'random', 'markov', 'adaptive',
'online_adaptive']
```

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

```
ADVERSARIAL CONFIGURATION:
```

```

=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	1
=====
Environment: onlineadaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3198.93
 COMPLETED: in 0.00s, final reward: 3198.93
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====

```

- NEURAL Progress: 100%| 4000/4000 [00:06<00:00, 643.11it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),

```

```
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

```
- EXP3 Progress: 100%| | 4000/4000 [00:02<00:00, 1973.91it/s]
```

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

```
- HYBRID Progress: 100%| | 4000/4000 [00:09<00:00, 430.79it/s]
```

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 17.68s):

```
=====
Oracle (step-wise) : 3198.93
GNeuralUCB (batch) : 2098.77
EXPUCB (batch) : 2038.11
EXPNeuralUCB (batch) : 2348.03
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 73.4%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	MarkovAttack
Total Frames:	4,000
Total Attacks:	4,000
Attack Intensity:	0.250

```

| PATH-SPECIFIC ANALYSIS: |

```

Path 0:	1369 attacks (34.2% rate)
Path 1:	681 attacks (17.0% rate)
Path 2:	1290 attacks (32.2% rate)
Path 3:	660 attacks (16.5% rate)
=====
```

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 1 COMPLETED

Execution Time: 17.7s

Results:

```
Oracle: 3198.93 (theoretical maximum)
EXPNeuralUCB: 2348.03 (26.6% gap) [WINNER]
GNeuralUCB: 2098.77 (34.4% gap)
EXPUCB: 2038.11 (36.3% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000

Attack: onlineadaptive (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

WARNING: Unknown attack\_type='onlineadaptive', falling back to 'markov'

Available attack types: ['none', 'random', 'markov', 'adaptive', 'online\_adaptive']

#### QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000
=====
```

#### ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	1
=====
```

Environment: onlineadaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5589.77

COMPLETED: in 0.00s, final reward: 5589.77

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====
```

- NEURAL Progress: 100%| | 6000/6000 [00:11<00:00, 517.10it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| | 6000/6000 [00:04<00:00, 1414.18it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component | Method | Properties |
|-----------|--------|------------|
|-----------|--------|------------|

|                  |                        |               |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100% | 6000/6000 [00:14<00:00, 402.48it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 30.92s):

```

=====
Oracle (step-wise) : 5589.77
GNeuralUCB (batch) : 4308.49
EXPUCB (batch) : 3537.67
EXPNeuralUCB (batch) : 4391.79
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 78.6%

```

ATTACK PATTERN ANALYSIS:

|                   |              |
|-------------------|--------------|
| Strategy:         | MarkovAttack |
| Total Frames:     | 6,000        |
| Total Attacks:    | 6,000        |
| Attack Intensity: | 0.250        |

PATH-SPECIFIC ANALYSIS:

|         |                           |
|---------|---------------------------|
| Path 0: | 2012 attacks (33.5% rate) |
| Path 1: | 983 attacks (16.4% rate)  |
| Path 2: | 1989 attacks (33.1% rate) |
| Path 3: | 1016 attacks (16.9% rate) |

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 2 COMPLETED

Execution Time: 30.9s

Results:

```

Oracle: 5589.77 (theoretical maximum)
EXPNeuralUCB: 4391.79 (21.4% gap) [WINNER]
GNeuralUCB: 4308.49 (22.9% gap)
EXPUCB: 3537.67 (36.7% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: onlineadaptive (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

WARNING: Unknown attack\_type='onlineadaptive', falling back to 'markov'

Available attack types: ['none', 'random', 'markov', 'adaptive', 'online\_adaptive']

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |              |  |
|------------------|--------------|--|
| Attack Strategy: | MarkovAttack |  |
| Attack Rate:     | 1.00         |  |
| Simultaneous:    | 1            |  |

=====

Environment: onlineadaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7815.67

COMPLETED: in 0.00s, final reward: 7815.67

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |  |  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Optimal Path:     | 0                                                                                                                                      |  |  |
| Optimal Action:   | 4                                                                                                                                      |  |  |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |  |  |

=====

EXPNeuralUCB initialized in 'neural' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

- NEURAL Progress: 100%| 8000/8000 [00:50<00:00, 158.85it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |  
| Path Performance: | [np.float64(0.9836141108302526),  
np.float64(0.9637113766945926), np.float64(0.9435345403529187),  
np.float64(0.9202691292754374)] |

EXPNeuralUCB initialized in 'exp3' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

- EXP3 Progress: 100%| 8000/8000 [00:06<00:00, 1196.31it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |

```
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

```
=====
```

- HYBRID Progress: 100%| 8000/8000 [00:21<00:00, 367.44it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 79.04s):

```
=====
Oracle (step-wise) : 7815.67
GNeuralUCB (batch) : 6022.64
EXPUCB (batch) : 5013.47
EXPNeuralUCB (batch) : 6328.40
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 81.0%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	MarkovAttack
Total Frames:	8,000
Total Attacks:	8,000
Attack Intensity:	0.250
=====
```

```
| PATH-SPECIFIC ANALYSIS: |
=====
```

|         |                           |
|---------|---------------------------|
| Path 0: | 2675 attacks (33.4% rate) |
| Path 1: | 1265 attacks (15.8% rate) |
| Path 2: | 2712 attacks (33.9% rate) |
| Path 3: | 1348 attacks (16.9% rate) |

```
=====
```

ATTACKS: Mean attacks/frame: 1.000

```
=====
```

### EXPERIMENT 3 COMPLETED

Execution Time: 79.0s

#### Results:

Oracle: 7815.67 (theoretical maximum)  
EXPNeuralUCB: 6328.40 ( 19.0% gap) [WINNER]  
GNeuralUCB: 6022.64 ( 22.9% gap)  
EXPUCB: 5013.47 ( 35.9% gap)  
Winner: EXPNeuralUCB

### QUANTUM EVALUATION COMPLETED!

Total experiments: 3  
Total time: 128.5s  
Average time per experiment: 42.8s

### DETAILED EXPERIMENT ANALYSIS

#### Attack Configuration:

Strategy: ONLINEADAPTIVE  
Intensity: 1.0

#### Per-Experiment Results:

#### EXPERIMENT 1 (4,000 frames, 17.7s)

|  
+-- Oracle: 3198.93 (theoretical maximum)  
+-- EXPNeuralUCB: 2348.03 ( 26.6% gap)  
+-- GNeuralUCB: 2098.77 ( 34.4% gap)  
+-- EXPUCB: 2038.11 ( 36.3% gap)  
Winner: EXPNeuralUCB

#### EXPERIMENT 2 (6,000 frames, 30.9s)

|  
+-- Oracle: 5589.77 (theoretical maximum)  
+-- EXPNeuralUCB: 4391.79 ( 21.4% gap)  
+-- GNeuralUCB: 4308.49 ( 22.9% gap)  
+-- EXPUCB: 3537.67 ( 36.7% gap)  
Winner: EXPNeuralUCB

#### EXPERIMENT 3 (8,000 frames, 79.0s)

|

```

+-- Oracle: 7815.67 (theoretical maximum)
+-- EXPNeuralUCB: 6328.40 (19.0% gap)
+-- GNeuralUCB: 6022.64 (22.9% gap)
+-- EXPUCB: 5013.47 (35.9% gap)
Winner: EXPNeuralUCB

```

#### ORACLE-RELATIVE GAP ANALYSIS:

##### LEARNING PERFORMANCE (Gap Reduction to Oracle):

```

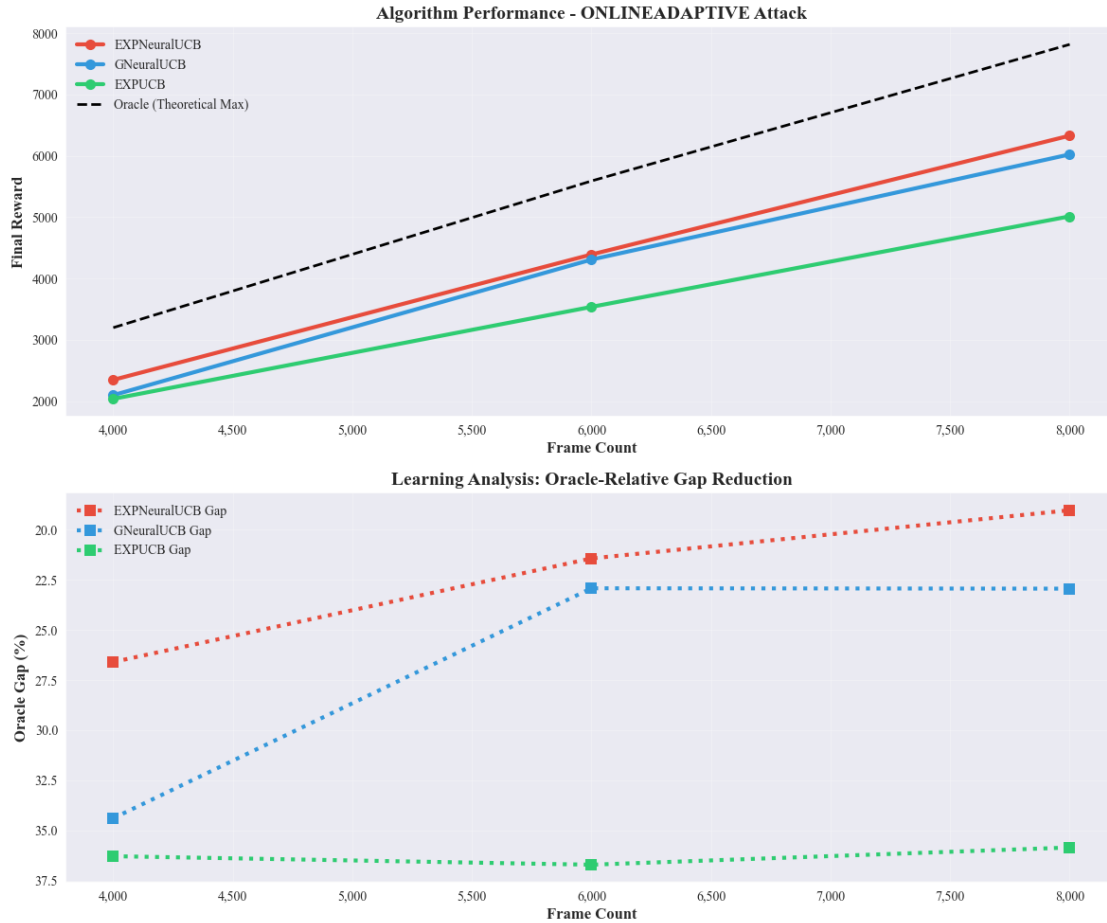
#1: GNeuralUCB -> +11.4pp improvement (gap closing)
 Gap: 34.4% -> 22.9%
#2: EXPNeuralUCB -> +7.6pp improvement (gap closing)
 Gap: 26.6% -> 19.0%
#3: EXPUCB -> +0.4pp improvement (gap closing)
 Gap: 36.3% -> 35.9%

```

##### Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

```
[]: # @title
quantum_visualizer.py

import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from quantum_framework import MultiRunEvaluator

Use your existing function exactly as written
def run_evaluation(attack_type="markov"):
 evaluator = MultiRunEvaluator(
 base_seed=12345,
 base_frames=4000,
 frame_step=2000,
 attack_type=attack_type,
 attack_intensity=1.0,
```

```

 enable_progress=False,
)
 evaluator.run_experiments(attack_type=attack_type, num_experiments=3)
 return evaluator

class QuantumVisualizer:
 """Simple visualization using existing MultiRunEvaluator and
 ↪env_experiments data"""

 def __init__(self):
 self.evaluators = {}
 self.colors = {
 'Oracle': '#2c3e50',
 'EXPNeuralUCB': '#e74c3c',
 'GNeuralUCB': '#3498db',
 'EXPUCB': '#2ecc71'
 }
 self.env_colors = {
 'none': '#95a5a6',
 'random': '#f39c12',
 'markov': '#9b59b6',
 'adaptive': '#e67e22'
 }

 def run_test(self, attack_type="adaptive"):
 print(f"Running test for {attack_type} environment...")
 evaluator = run_evaluation(attack_type=attack_type)
 self.evaluators[attack_type] = evaluator
 return evaluator

 def _get_env_data(self, attack_type):
 if attack_type not in self.evaluators:
 raise KeyError(f"No evaluator stored for
 ↪attack_type='{attack_type}'. Run run_test() first.")

 evaluator = self.evaluators[attack_type]
 env_data = getattr(evaluator, "env_experiments", {}).get(attack_type)

 if not env_data:
 env_data = getattr(evaluator, "experiments", None)

 if not isinstance(env_data, dict) or not env_data:
 raise ValueError(f"No experiment data found for
 ↪attack_type='{attack_type}'.")

 if all(isinstance(k, str) for k in env_data.keys()):
 exp_id_guess = env_data.get("id", 1)

```

```

 env_data = {int(exp_id_guess): env_data}

 return env_data

def plot_results(self, attack_type):
 if attack_type not in self.evaluators:
 print(f"No results for {attack_type}")
 return

 try:
 env_data = self._get_env_data(attack_type)
 except Exception as e:
 print(f"Failed to extract data for '{attack_type}': {e}")
 return

 frame_counts = []
 oracle_rewards = []
 alg_rewards = {}
 alg_gaps = {}

 first_exp = next(iter(env_data.values()))
 algorithms = [alg for alg in first_exp['results'].keys() if alg != '
↳Oracle']

 for alg in algorithms:
 alg_rewards[alg] = []
 alg_gaps[alg] = []

 for exp_id in sorted(env_data.keys()):
 exp_data = env_data[exp_id]

 frame_counts.append(exp_data['frame_count'])
 oracle_rewards.append(exp_data['results']['Oracle']['final_reward'])

 for alg in algorithms:
 alg_rewards[alg].
↳append(exp_data['results'][alg]['final_reward'])
 alg_gaps[alg].append(exp_data['gaps'][alg])

 # Create plots
 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
 fig.suptitle(f'{attack_type.capitalize()} Environment Results')

 # Plot 1: Rewards
 ax1.plot(frame_counts, oracle_rewards, '--k', label='Oracle',
↳linewidth=2, marker='x', markersize=7)
 for alg in algorithms:

```

```

 ax1.plot(frame_counts, alg_rewards[alg], 'o-', label=alg,
↳linewidth=2, markersize=6,
 color=self.colors.get(alg, 'gray'))

 ax1.set_xlabel('Frame Count')
 ax1.set_ylabel('Final Reward')
 ax1.set_title('Reward Scaling')
 ax1.legend()
 ax1.grid(True, alpha=0.3)

 # Plot 2: Gaps
 for alg in algorithms:
 ax2.plot(frame_counts, alg_gaps[alg], 's-', label=alg, linewidth=2,
↳markersize=6,
 color=self.colors.get(alg, 'gray'))

 ax2.set_xlabel('Frame Count')
 ax2.set_ylabel('Oracle Gap (%)')
 ax2.set_title('Oracle Gap (Lower = Better)')
 ax2.legend()
 ax2.grid(True, alpha=0.3)
 ax2.invert_yaxis()

 plt.tight_layout()
 plt.savefig(f'{attack_type}_results.png', dpi=300, bbox_inches='tight')
 plt.show()

def compare_environments(self, attack_types, main_algorithm="EXPNeuralUCB"):
 fig, axes = plt.subplots(2, 2, figsize=(15, 10))
 fig.suptitle('ENVIRONMENT COMPARISON', fontsize=16, fontweight='bold')

 # 1. Main algorithm across environments
 for env in attack_types:
 if env not in self.evaluators:
 continue
 try:
 env_data = self._get_env_data(env)
 except Exception:
 continue

 frame_counts = []
 alg_rewards = []
 alg_gaps = []

 for exp_id in sorted(env_data.keys()):
 exp_data = env_data[exp_id]
 frame_counts.append(exp_data['frame_count'])

```

```

 if main_algorithm in exp_data['results']:
 alg_rewards.
→append(exp_data['results'][main_algorithm]['final_reward'])
 alg_gaps.append(exp_data['gaps'][main_algorithm])

 if alg_rewards:
 color = self.env_colors.get(env, 'gray')
 axes[0,0].plot(frame_counts, alg_rewards, 'o-', label=env,
→linewidth=2, markersize=6, color=color)
 axes[0,1].plot(frame_counts, alg_gaps, 's-', label=env,
→linewidth=2, markersize=6, color=color)

 axes[0,0].set_title(f'{main_algorithm} Reward Performance')
 axes[0,0].set_xlabel('Frame Count')
 axes[0,0].set_ylabel('Final Reward')
 axes[0,0].legend()
 axes[0,0].grid(True, alpha=0.3)

 axes[0,1].set_title(f'{main_algorithm} Oracle Gap')
 axes[0,1].set_xlabel('Frame Count')
 axes[0,1].set_ylabel('Oracle Gap % (Lower = Better)')
 axes[0,1].legend()
 axes[0,1].grid(True, alpha=0.3)
 axes[0,1].invert_yaxis()

2. Use stored winners
 env_winners = {}
 all_env_data = []

 for env in attack_types:
 if env not in self.evaluators:
 continue
 try:
 env_data = self._get_env_data(env)
 except Exception:
 continue

 highest_frame_exp = max(env_data.keys())
 stored_winner = env_data[highest_frame_exp]['winner']
 stored_gap = env_data[highest_frame_exp]['gaps'][stored_winner]

 env_winners[env] = (stored_winner, stored_gap)

 for exp_id in sorted(env_data.keys()):
 exp_data = env_data[exp_id]
 for alg, gap in exp_data['gaps'].items():

```

```

 all_env_data.append({'environment': env, 'algorithm': alg,
 ↪ 'avg_gap': gap})

 # 3. Plot winners
 if env_winners:
 envs = list(env_winners.keys())
 winners = [env_winners[e][0] for e in envs]
 gaps = [env_winners[e][1] for e in envs]
 colors = [self.env_colors.get(e, 'gray') for e in envs]
 axes[1,0].bar(envs, gaps, color=colors, alpha=0.8)
 axes[1,0].set_title('Best Algorithm per Environment')
 axes[1,0].set_ylabel('Best Oracle Gap %')
 axes[1,0].tick_params(axis='x', rotation=45)
 for i, (env, winner, gap) in enumerate(zip(envs, winners, gaps)):
 axes[1,0].text(i, gap + 1, winner, ha='center', va='bottom',
 ↪ fontweight='bold', rotation=90)

 # 4. Environment difficulty
 if all_env_data:
 env_df = pd.DataFrame(all_env_data)
 env_difficulty = env_df.groupby('environment')['avg_gap'].mean().
 ↪ sort_values()
 colors = [self.env_colors.get(e, 'gray') for e in env_difficulty.
 ↪ index]
 axes[1,1].bar(range(len(env_difficulty)), env_difficulty.values,
 ↪ color=colors, alpha=0.8)
 axes[1,1].set_xticks(range(len(env_difficulty)))
 axes[1,1].set_xticklabels(env_difficulty.index, rotation=45)
 axes[1,1].set_title('Environment Difficulty Ranking')
 axes[1,1].set_ylabel('Average Oracle Gap %')

 plt.tight_layout()
 plt.savefig('environment_comparison.png', dpi=300, bbox_inches='tight')
 plt.show()

Simple usage examples
def test_single_environment():
 viz = QuantumVisualizer()
 attack_type = "adaptive"
 viz.run_test(attack_type)
 viz.plot_results(attack_type)
 return viz

def test_multiple_environments():
 viz = QuantumVisualizer()
 environments = ["none", "random", "markov", "adaptive"]

```

```

for env in environments:
 print(f"\n{'='*50}")
 print(f"TESTING {env.upper()}")
 print('='*50)
 viz.run_test(env)
 viz.plot_results(env)

viz.compare_environments(environments)
return viz

if __name__ == "__main__":
 print("\n" + "="*70)
 print("TESTING MULTIPLE ENVIRONMENTS:")
 viz2 = test_multiple_environments()

```

```

=====
TESTING MULTIPLE ENVIRONMENTS:

=====
TESTING NONE
=====
Running test for none environment...

=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: NONE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis
=====

=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====

PROGRESS: Experiment 1/3

```

```

=====
EXPERIMENT 1
=====
Frame Count: 4,000
Attack: none (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====

ADVERSARIAL CONFIGURATION:
=====
| Attack Strategy: | NoAttack |
=====
Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3307.29
 COMPLETED: in 0.00s, final reward: 3307.29
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

EXPNeuralUCB initialized in 'neural' mode

=====
ALGORITHM CONFIGURATION
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial

```

|                  |            |                |  |
|------------------|------------|----------------|--|
|                  |            | Robust         |  |
| Action Selection | Neural UCB | Nonlinear      |  |
|                  |            | Learning       |  |
| Parameters       | beta=0.2   | UCB Confidence |  |

=====

- NEURAL Progress: 100%|        | 4000/4000 [00:20<00:00, 191.24it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|--|
| Optimal Path:     | 0                                                                                                                                      |  |
| Optimal Action:   | 4                                                                                                                                      |  |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |  |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%|        | 4000/4000 [00:05<00:00, 737.95it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|--|
| Optimal Path:     | 0                                                                                                                                      |  |
| Optimal Action:   | 4                                                                                                                                      |  |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |  |

=====

EXPNeuralUCB initialized in 'hybrid' mode

```

=====
ALGORITHM CONFIGURATION
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

- HYBRID Progress: 100%| | 4000/4000 [00:24<00:00, 162.89it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 51.95s):
=====
 Oracle (step-wise) : 3307.29
 GNeuralUCB (batch) : 3235.40
 EXPUCB (batch) : 2794.69
 EXPNeuralUCB (batch) : 2854.95
 WINNER: GNeuralUCB
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%

ATTACK PATTERN ANALYSIS:
=====
Strategy:	NoAttack
Total Frames:	4,000
Total Attacks:	0
| Attack Intensity: | 0.000 |
|-----|-----|-----|
| PATH-SPECIFIC ANALYSIS: |
|-----|-----|-----|
| Path 0: | 0 attacks (0.0% rate) |
| Path 1: | 0 attacks (0.0% rate) |
| Path 2: | 0 attacks (0.0% rate) |
| Path 3: | 0 attacks (0.0% rate) |
|-----|-----|-----|
 ATTACKS: Mean attacks/frame: 0.000
=====

EXPERIMENT 1 COMPLETED
Execution Time: 52.0s
Results:
 Oracle: 3307.29 (theoretical maximum)
 EXPNeuralUCB: 2854.95 (13.7% gap)

```

```
GNeuralUCB: 3235.40 (2.2% gap) [WINNER]
EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB
```

-----

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: none (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 6,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |          |  |
|------------------|----------|--|
| Attack Strategy: | NoAttack |  |
|------------------|----------|--|

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5676.68

COMPLETED: in 0.01s, final reward: 5676.68

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                |  |
|-------------------|----------------------------------------------------------------|--|
| Optimal Path:     | 0                                                              |  |
| Optimal Action:   | 4                                                              |  |
| Path Performance: | [np.float64(0.9461134921196183),                               |  |
|                   | np.float64(0.902918807873247), np.float64(0.8603089615724164), |  |
|                   | np.float64(0.8116651900474303)]                                |  |

=====

EXPNeuralUCB initialized in 'neural' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

- NEURAL Progress: 100%| 6000/6000 [00:20<00:00, 293.05it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |  
| Path Performance: | [np.float64(0.9461134921196183),  
np.float64(0.902918807873247), np.float64(0.8603089615724164),  
np.float64(0.8116651900474303)] |

EXPNeuralUCB initialized in 'exp3' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

- EXP3 Progress: 100%| 6000/6000 [00:06<00:00, 995.13it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |

```
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

```
=====
- HYBRID Progress: 100%| | 6000/6000 [00:27<00:00, 214.71it/s]
```

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 54.64s):

```
=====
Oracle (step-wise) : 5676.68
GNeuralUCB (batch) : 5179.98
EXPUCB (batch) : 4552.40
EXPNeuralUCB (batch) : 5279.10
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%
```

ATTACK PATTERN ANALYSIS:

```
=====
```

|                   |          |
|-------------------|----------|
| Strategy:         | NoAttack |
| Total Frames:     | 6,000    |
| Total Attacks:    | 0        |
| Attack Intensity: | 0.000    |

```
=====
| PATH-SPECIFIC ANALYSIS: |
=====
```

|         |                        |
|---------|------------------------|
| Path 0: | 0 attacks ( 0.0% rate) |
| Path 1: | 0 attacks ( 0.0% rate) |
| Path 2: | 0 attacks ( 0.0% rate) |
| Path 3: | 0 attacks ( 0.0% rate) |

```
=====
ATTACKS: Mean attacks/frame: 0.000
=====
```

EXPERIMENT 2 COMPLETED

Execution Time: 54.6s

Results:

Oracle: 5676.68 (theoretical maximum)  
EXPNeuralUCB: 5279.10 ( 7.0% gap) [WINNER]  
GNeuralUCB: 5179.98 ( 8.7% gap)  
EXPUCB: 4552.40 ( 19.8% gap)  
Winner: EXPNeuralUCB

-----

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: none (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |          |  |
|------------------|----------|--|
| Attack Strategy: | NoAttack |  |
|------------------|----------|--|

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7868.91

COMPLETED: in 0.01s, final reward: 7868.91

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                  |  |
|-------------------|----------------------------------|--|
| Optimal Path:     | 0                                |  |
| Optimal Action:   | 4                                |  |
| Path Performance: | [np.float64(0.9836141108302526), |  |

```
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

- NEURAL Progress: 100% | 8000/8000 [01:23<00:00, 95.95it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

#### ORACLE ANALYSIS:

```
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

#### ALGORITHM CONFIGURATION

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

- EXP3 Progress: 100% | 8000/8000 [00:11<00:00, 727.14it/s]

COMPLETED: EXPUCB (batch) completed successfully

Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====
ALGORITHM CONFIGURATION
=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

- HYBRID Progress: 100% | 8000/8000 [00:38<00:00, 205.60it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 133.41s):

```
=====
Oracle (step-wise) : 7868.91
GNeuralUCB (batch) : 7266.99
EXPUCB (batch) : 6307.33
EXPNeuralUCB (batch) : 7520.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	NoAttack
Total Frames:	8,000
Total Attacks:	0
Attack Intensity:	0.000

PATH-SPECIFIC ANALYSIS:
Path 0:
```

```

Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)
=====
ATTACKS: Mean attacks/frame: 0.000
=====

```

### EXPERIMENT 3 COMPLETED

Execution Time: 133.4s

#### Results:

```

Oracle: 7868.91 (theoretical maximum)
EXPNeuralUCB: 7520.98 (4.4% gap) [WINNER]
GNeuralUCB: 7266.99 (7.6% gap)
EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB

```

```

=====
QUANTUM EVALUATION COMPLETED!
=====
Total experiments: 3
Total time: 241.1s
Average time per experiment: 80.4s
=====

```

### DETAILED EXPERIMENT ANALYSIS

#### Attack Configuration:

```

Strategy: NONE
Intensity: 1.0

```

#### Per-Experiment Results:

#### EXPERIMENT 1 (4,000 frames, 52.0s)

```

|
+-- Oracle: 3307.29 (theoretical maximum)
+-- EXPNeuralUCB: 2854.95 (13.7% gap)
+-- GNeuralUCB: 3235.40 (2.2% gap)
+-- EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB

```

#### EXPERIMENT 2 (6,000 frames, 54.6s)

```

|
+-- Oracle: 5676.68 (theoretical maximum)
+-- EXPNeuralUCB: 5279.10 (7.0% gap)

```

--- GNeuralUCB: 5179.98 ( 8.7% gap)  
--- EXPUCB: 4552.40 ( 19.8% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 133.4s)

|  
--- Oracle: 7868.91 (theoretical maximum)  
--- EXPNeuralUCB: 7520.98 ( 4.4% gap)  
--- GNeuralUCB: 7266.99 ( 7.6% gap)  
--- EXPUCB: 6307.33 ( 19.8% gap)  
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

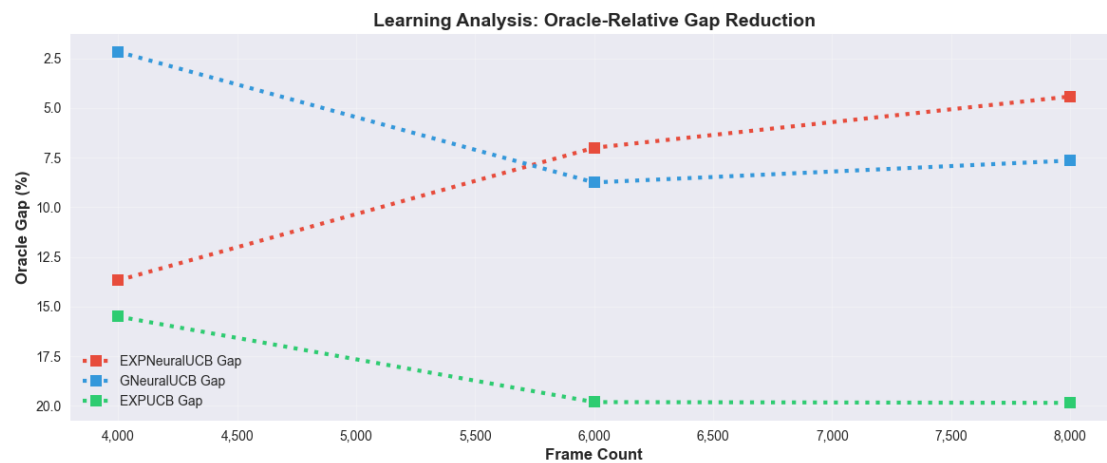
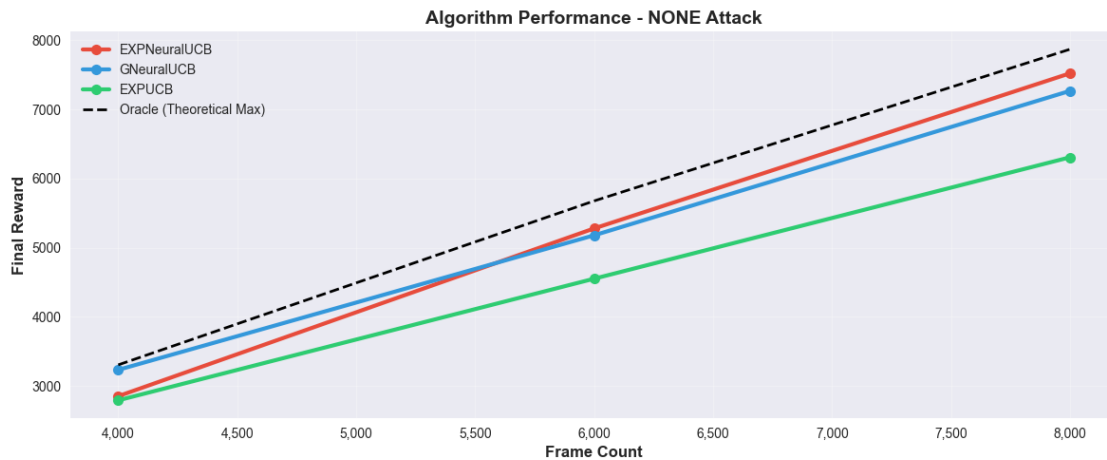
-----  
LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: EXPNeuralUCB -> +9.3pp improvement (gap closing)  
Gap: 13.7% -> 4.4%  
#2: EXPUCB -> +4.3pp decline (gap widening)  
Gap: 15.5% -> 19.8%  
#3: GNeuralUCB -> +5.5pp decline (gap widening)  
Gap: 2.2% -> 7.6%

Interpretation:

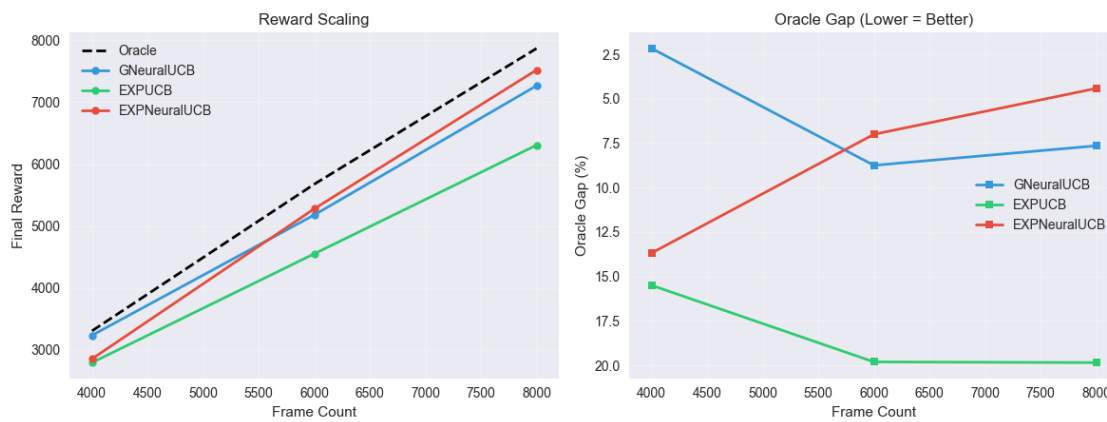
- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...



Visualization complete!

### None Environment Results



```
=====
TESTING RANDOM
=====
```

Running test for random environment...

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: RANDOM
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

```
=====
PROGRESS: Experiment 1/3
```

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: random (intensity=1.0)

EXPERIMENT 1: 4000 frames

```
=====
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

```
=====
ADVERSARIAL CONFIGURATION:
=====
```

```
| Attack Strategy: | RandomAttack |
=====
```

```
| Attack Rate: | 0.25 |
=====
Environment: random (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3192.62
 COMPLETED: in 0.00s, final reward: 3192.62
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...
```

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====
```

- NEURAL Progress: 100%| | 4000/4000 [00:09<00:00, 401.24it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100% | 4000/4000 [00:02<00:00, 1545.75it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

=====

- HYBRID Progress: 100% | 4000/4000 [00:13<00:00, 294.83it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 26.19s):

=====

Oracle (step-wise) : 3192.62  
GNeuralUCB (batch) : 2420.84  
EXPUCB (batch) : 2024.79  
EXPNeuralUCB (batch) : 2097.30  
WINNER: GNeuralUCB  
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 65.7%

#### ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	RandomAttack
Total Frames:	4,000
Total Attacks:	4,024
Attack Intensity:	0.252
=====	
PATH-SPECIFIC ANALYSIS:	
=====	
Path 0:	986 attacks (24.6% rate)
Path 1:	1041 attacks (26.0% rate)
Path 2:	1003 attacks (25.1% rate)
Path 3:	994 attacks (24.9% rate)
=====
ATTACKS: Mean attacks/frame: 1.006
=====
```

#### EXPERIMENT 1 COMPLETED

Execution Time: 26.2s

##### Results:

Oracle: 3192.62 (theoretical maximum)  
EXPNeuralUCB: 2097.30 ( 34.3% gap)  
GNeuralUCB: 2420.84 ( 24.2% gap) [WINNER]  
EXPUCB: 2024.79 ( 36.6% gap)  
Winner: GNeuralUCB

-----  
PROGRESS: Experiment 2/3

#### EXPERIMENT 2

=====
Frame Count: 6,000
Attack: random (intensity=1.0)
-----

EXPERIMENT 2: 6000 frames

#### QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths: | 4 |
```

```
| Qubit Config: | (8, 10, 8, 9) |
| Frame Length: | 6,000 |
=====
```

#### ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack |
| Attack Rate: | 0.25 |
=====
```

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5567.05

COMPLETED: in 0.01s, final reward: 5567.05

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====
```

- NEURAL Progress: 100%| 6000/6000 [00:15<00:00, 379.58it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

#### ORACLE ANALYSIS:

```
=====
```

```

| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====

```

EXPNeuralUCB initialized in 'exp3' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Linear UCB	Linear
		Learning
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB
=====

```

- EXP3 Progress: 100%| 6000/6000 [00:05<00:00, 1039.33it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====

```

EXPNeuralUCB initialized in 'hybrid' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

```

```
=====
- HYBRID Progress: 100%| | 6000/6000 [00:21<00:00, 278.63it/s]
```

```
 COMPLETED: EXPNeuralUCB (batch) completed successfully
```

```
RESULTS (completed in 43.19s):
```

```
=====
Oracle (step-wise) : 5567.05
GNeuralUCB (batch) : 3840.18
EXPUCB (batch) : 3310.90
EXPNeuralUCB (batch) : 3910.27
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
```

```
ATTACK PATTERN ANALYSIS:
```

```
=====
Strategy:	RandomAttack
Total Frames:	6,000
Total Attacks:	6,001
Attack Intensity:	0.250
=====
```

```
| PATH-SPECIFIC ANALYSIS: |
```

```

Path 0:	1490 attacks (24.8% rate)
Path 1:	1511 attacks (25.2% rate)
Path 2:	1476 attacks (24.6% rate)
Path 3:	1524 attacks (25.4% rate)
=====
```

```
 ATTACKS: Mean attacks/frame: 1.000
=====
```

```
EXPERIMENT 2 COMPLETED
```

```
Execution Time: 43.2s
```

```
Results:
```

```
Oracle: 5567.05 (theoretical maximum)
EXPNeuralUCB: 3910.27 (29.8% gap) [WINNER]
GNeuralUCB: 3840.18 (31.0% gap)
EXPUCB: 3310.90 (40.5% gap)
Winner: EXPNeuralUCB

```

```
PROGRESS: Experiment 3/3
```

```
=====
EXPERIMENT 3
=====
```

```
Frame Count: 8,000
```

```
Attack: random (intensity=1.0)
```

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack |
| Attack Rate: | 0.25 |
=====
```

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7777.26

COMPLETED: in 0.02s, final reward: 7777.26

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

=====

- NEURAL Progress: 100%| | 8000/8000 [01:12<00:00, 110.92it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====
ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Linear UCB	Linear
		Learning
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB
=====
```

- EXP3 Progress: 100%| | 8000/8000 [00:10<00:00, 767.18it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====
ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
```

|                  |                        |               |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100% | 8000/8000 [00:31<00:00, 256.93it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 113.80s):

```

=====
Oracle (step-wise) : 7777.26
GNeuralUCB (batch) : 5417.64
EXPUCB (batch) : 4675.52
EXPNeuralUCB (batch) : 5584.49
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.8%

```

ATTACK PATTERN ANALYSIS:

|                   |              |
|-------------------|--------------|
| Strategy:         | RandomAttack |
| Total Frames:     | 8,000        |
| Total Attacks:    | 8,063        |
| Attack Intensity: | 0.252        |

PATH-SPECIFIC ANALYSIS:

|         |                           |
|---------|---------------------------|
| Path 0: | 2018 attacks (25.2% rate) |
| Path 1: | 2041 attacks (25.5% rate) |
| Path 2: | 1996 attacks (24.9% rate) |
| Path 3: | 2008 attacks (25.1% rate) |

ATTACKS: Mean attacks/frame: 1.008

EXPERIMENT 3 COMPLETED

Execution Time: 113.8s

Results:

```

Oracle: 7777.26 (theoretical maximum)
EXPNeuralUCB: 5584.49 (28.2% gap) [WINNER]
GNeuralUCB: 5417.64 (30.3% gap)
EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB

```

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 184.0s  
Average time per experiment: 61.3s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: RANDOM  
Intensity: 1.0

Per-Experiment Results:

-----

EXPERIMENT 1 (4,000 frames, 26.2s)

|  
+-- Oracle: 3192.62 (theoretical maximum)  
+-- EXPNeuralUCB: 2097.30 ( 34.3% gap)  
+-- GNeuralUCB: 2420.84 ( 24.2% gap)  
+-- EXPUCB: 2024.79 ( 36.6% gap)  
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 43.2s)

|  
+-- Oracle: 5567.05 (theoretical maximum)  
+-- EXPNeuralUCB: 3910.27 ( 29.8% gap)  
+-- GNeuralUCB: 3840.18 ( 31.0% gap)  
+-- EXPUCB: 3310.90 ( 40.5% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 113.8s)

|  
+-- Oracle: 7777.26 (theoretical maximum)  
+-- EXPNeuralUCB: 5584.49 ( 28.2% gap)  
+-- GNeuralUCB: 5417.64 ( 30.3% gap)  
+-- EXPUCB: 4675.52 ( 39.9% gap)  
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

-----

LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: EXPNeuralUCB -> +6.1pp improvement (gap closing)

Gap: 34.3% -> 28.2%

#2: EXPUCB -> +3.3pp decline (gap widening)

Gap: 36.6% -> 39.9%

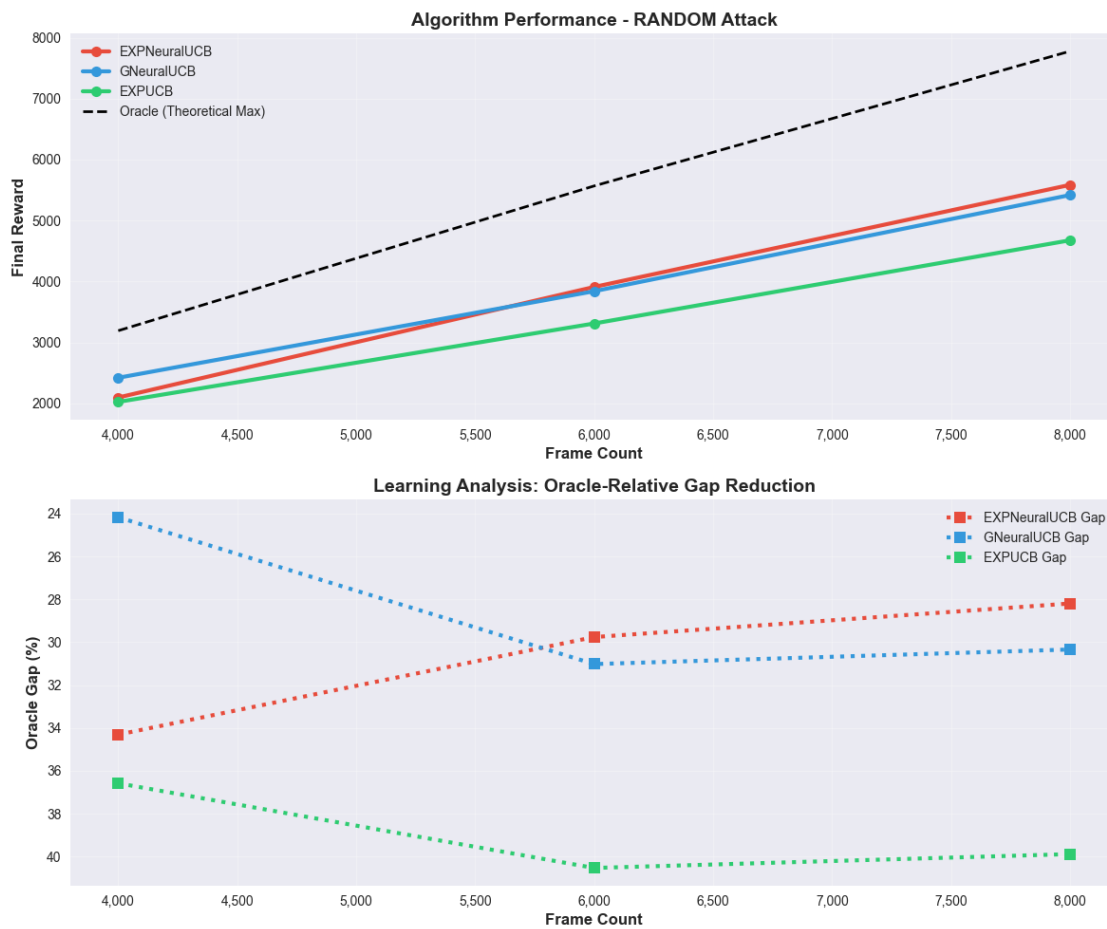
#3: GNeuralUCB -> +6.2pp decline (gap widening)

Gap: 24.2% -> 30.3%

Interpretation:

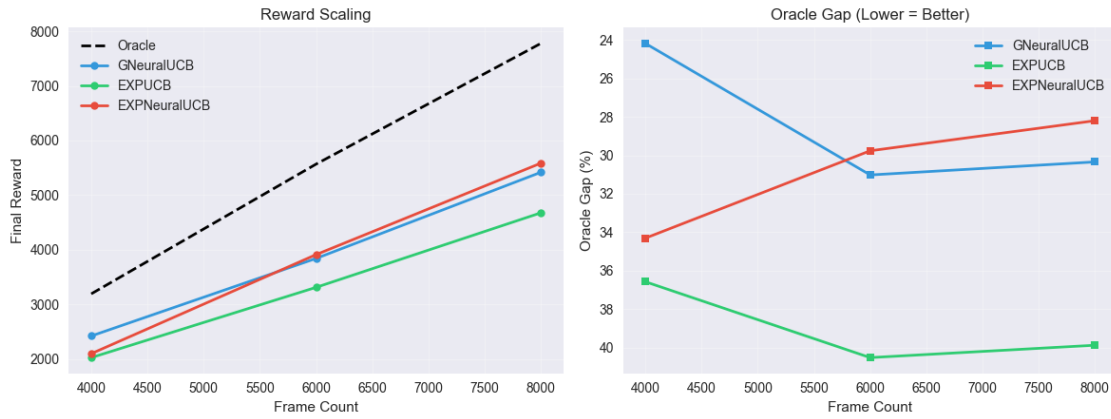
- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

# Random Environment Results



## TESTING MARKOV

Running test for markov environment...

## QUANTUM MULTIRUN EVALUATOR INITIALIZED

### Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: MARKOV
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

## STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

## EXPERIMENT 1

```
=====
Frame Count: 4,000
Attack: markov (intensity=1.0)

```

EXPERIMENT 1: 4000 frames

### QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

### ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3051.08

COMPLETED: in 0.00s, final reward: 3051.08

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
```

|                  |            |                |  |
|------------------|------------|----------------|--|
|                  |            | Robust         |  |
| Action Selection | Neural UCB | Nonlinear      |  |
|                  |            | Learning       |  |
| Parameters       | beta=0.2   | UCB Confidence |  |

=====

- NEURAL Progress: 100%| 4000/4000 [00:08<00:00, 479.42it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 4000/4000 [00:02<00:00, 1548.16it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

```

=====
ALGORITHM CONFIGURATION
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

- HYBRID Progress: 100%| | 4000/4000 [00:12<00:00, 316.21it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 23.76s):
=====
 Oracle (step-wise) : 3051.08
 GNeuralUCB (batch) : 1137.52
 EXPUCB (batch) : 1453.00
 EXPNeuralUCB (batch) : 1795.46
 WINNER: EXPNeuralUCB
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%

ATTACK PATTERN ANALYSIS:
=====
Strategy:	MarkovAttack
Total Frames:	4,000
Total Attacks:	8,000
| Attack Intensity: | 0.500 |
|-----|-----|-----|
| PATH-SPECIFIC ANALYSIS: |
|-----|-----|-----|
| Path 0: | 2543 attacks (63.6% rate) |
| Path 1: | 1474 attacks (36.9% rate) |
| Path 2: | 2482 attacks (62.1% rate) |
| Path 3: | 1501 attacks (37.5% rate) |
|-----|-----|-----|
 ATTACKS: Mean attacks/frame: 2.000
=====

EXPERIMENT 1 COMPLETED
Execution Time: 23.8s
Results:
 Oracle: 3051.08 (theoretical maximum)
 EXPNeuralUCB: 1795.46 (41.2% gap) [WINNER]

```

```
GNeuralUCB: 1137.52 (62.7% gap)
EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB
```

-----

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: markov (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 6,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |              |  |
|------------------|--------------|--|
| Attack Strategy: | MarkovAttack |  |
| Attack Rate:     | 1.00         |  |
| Simultaneous:    | 2            |  |

=====

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5474.92

COMPLETED: in 0.01s, final reward: 5474.92

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                  |  |
|-------------------|----------------------------------|--|
| Optimal Path:     | 0                                |  |
| Optimal Action:   | 4                                |  |
| Path Performance: | [np.float64(0.9461134921196183), |  |

np.float64(0.902918807873247), np.float64(0.8603089615724164),  
np.float64(0.8116651900474303)]

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

- NEURAL Progress: 100%| 6000/6000 [00:14<00:00, 408.33it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                     |
| Optimal Action:   | 4                                                                                                                                     |
| Path Performance: | [np.float64(0.9461134921196183),<br>np.float64(0.902918807873247), np.float64(0.8603089615724164),<br>np.float64(0.8116651900474303)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

- EXP3 Progress: 100%| 6000/6000 [00:05<00:00, 1049.22it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

```

| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====

```

EXPNeuralUCB initialized in 'hybrid' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

```

- HYBRID Progress: 100%| 6000/6000 [00:19<00:00, 300.72it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 40.62s):

```

=====
Oracle (step-wise) : 5474.92
GNeuralUCB (batch) : 3211.32
EXPUCB (batch) : 2604.25
EXPNeuralUCB (batch) : 3260.87
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%

```

#### ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	MarkovAttack
Total Frames:	6,000
Total Attacks:	12,000
Attack Intensity:	0.500
PATH-SPECIFIC ANALYSIS:	
-----	-----
Path 0:	3798 attacks (63.3% rate)
Path 1:	2217 attacks (37.0% rate)
Path 2:	3789 attacks (63.1% rate)
Path 3:	2196 attacks (36.6% rate)
-----	-----
=====

```

ATTACKS: Mean attacks/frame: 2.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 40.6s

Results:

Oracle: 5474.92 (theoretical maximum)  
EXPNeuralUCB: 3260.87 ( 40.4% gap) [WINNER]  
GNeuralUCB: 3211.32 ( 41.3% gap)  
EXPUCB: 2604.25 ( 52.4% gap)  
Winner: EXPNeuralUCB

-----

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: markov (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |              |  |
|------------------|--------------|--|
| Attack Strategy: | MarkovAttack |  |
| Attack Rate:     | 1.00         |  |
| Simultaneous:    | 2            |  |

=====

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7744.38

COMPLETED: in 0.01s, final reward: 7744.38

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

```
=====
```

- NEURAL Progress: 100%| 8000/8000 [01:07<00:00, 117.77it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

```
=====
- EXP3 Progress: 100%| | 8000/8000 [00:10<00:00, 792.58it/s]
```

```
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...
```

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

#### ===== ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====
```

```
- HYBRID Progress: 100%| | 8000/8000 [00:28<00:00, 280.10it/s]
```

```
 COMPLETED: EXPNeuralUCB (batch) completed successfully
```

#### RESULTS (completed in 106.93s):

```
=====
Oracle (step-wise) : 7744.38
GNeuralUCB (batch) : 4552.69
EXPUCB (batch) : 3605.14
EXPNeuralUCB (batch) : 4738.67
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%
```

#### ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	MarkovAttack
Total Frames:	8,000
Total Attacks:	16,000
Attack Intensity:	0.500
```

```

PATH-SPECIFIC ANALYSIS:
Path 0:
Path 1:
Path 2:
Path 3:
=====
ATTACKS: Mean attacks/frame: 2.000
=====

```

### EXPERIMENT 3 COMPLETED

Execution Time: 106.9s

#### Results:

```

Oracle: 7744.38 (theoretical maximum)
EXPNeuralUCB: 4738.67 (38.8% gap) [WINNER]
GNeuralUCB: 4552.69 (41.2% gap)
EXPUCB: 3605.14 (53.4% gap)
Winner: EXPNeuralUCB

```

### QUANTUM EVALUATION COMPLETED!

```

=====
Total experiments: 3
Total time: 172.3s
Average time per experiment: 57.4s
=====

```

### DETAILED EXPERIMENT ANALYSIS

#### Attack Configuration:

```

Strategy: MARKOV
Intensity: 1.0

```

#### Per-Experiment Results:

#### EXPERIMENT 1 (4,000 frames, 23.8s)

```

|
+-- Oracle: 3051.08 (theoretical maximum)
+-- EXPNeuralUCB: 1795.46 (41.2% gap)
+-- GNeuralUCB: 1137.52 (62.7% gap)
+-- EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

```

EXPERIMENT 2 (6,000 frames, 40.6s)

```
|
+-- Oracle: 5474.92 (theoretical maximum)
+-- EXPNeuralUCB: 3260.87 (40.4% gap)
+-- GNeuralUCB: 3211.32 (41.3% gap)
+-- EXPUCB: 2604.25 (52.4% gap)
Winner: EXPNeuralUCB
```

EXPERIMENT 3 (8,000 frames, 106.9s)

```
|
+-- Oracle: 7744.38 (theoretical maximum)
+-- EXPNeuralUCB: 4738.67 (38.8% gap)
+-- GNeuralUCB: 4552.69 (41.2% gap)
+-- EXPUCB: 3605.14 (53.4% gap)
Winner: EXPNeuralUCB
```

ORACLE-RELATIVE GAP ANALYSIS:

-----

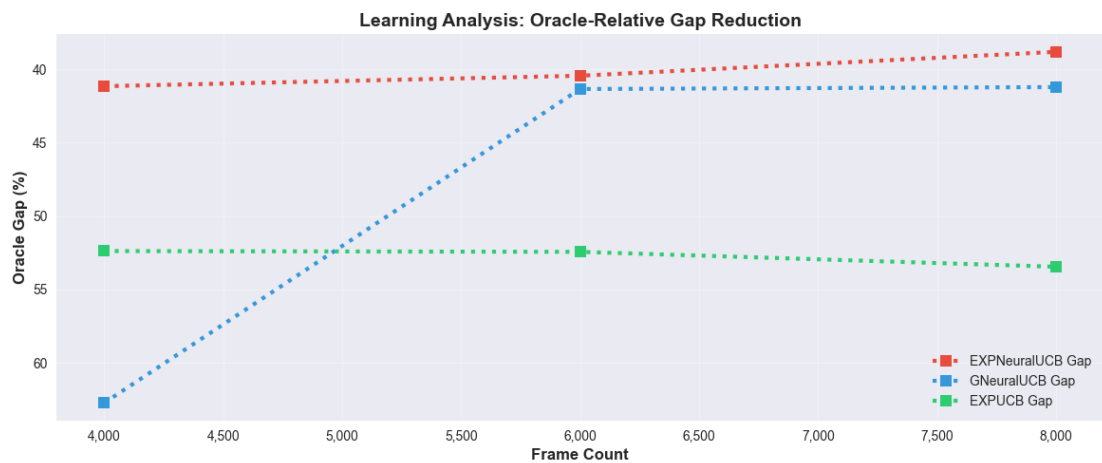
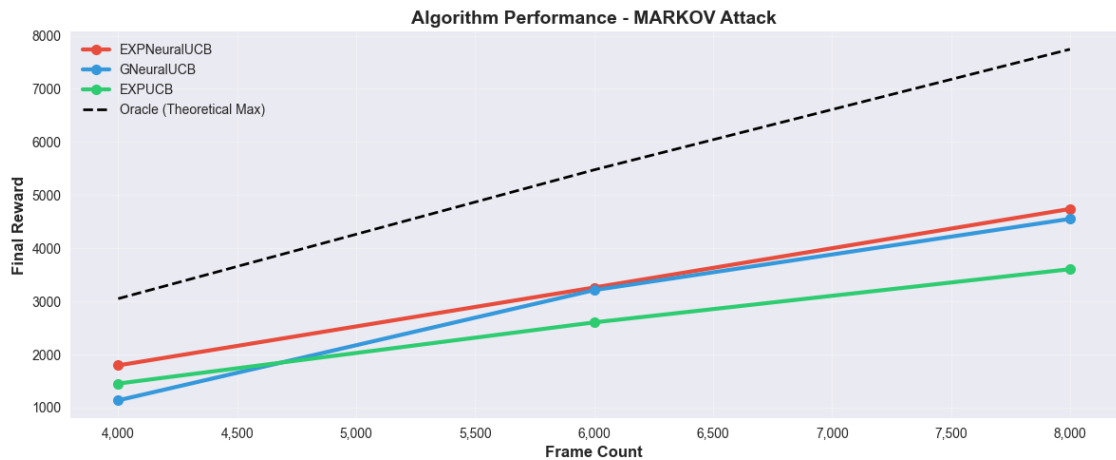
LEARNING PERFORMANCE (Gap Reduction to Oracle):

```
#1: GNeuralUCB -> +21.5pp improvement (gap closing)
 Gap: 62.7% -> 41.2%
#2: EXPNeuralUCB -> +2.3pp improvement (gap closing)
 Gap: 41.2% -> 38.8%
#3: EXPUCB -> +1.1pp decline (gap widening)
 Gap: 52.4% -> 53.4%
```

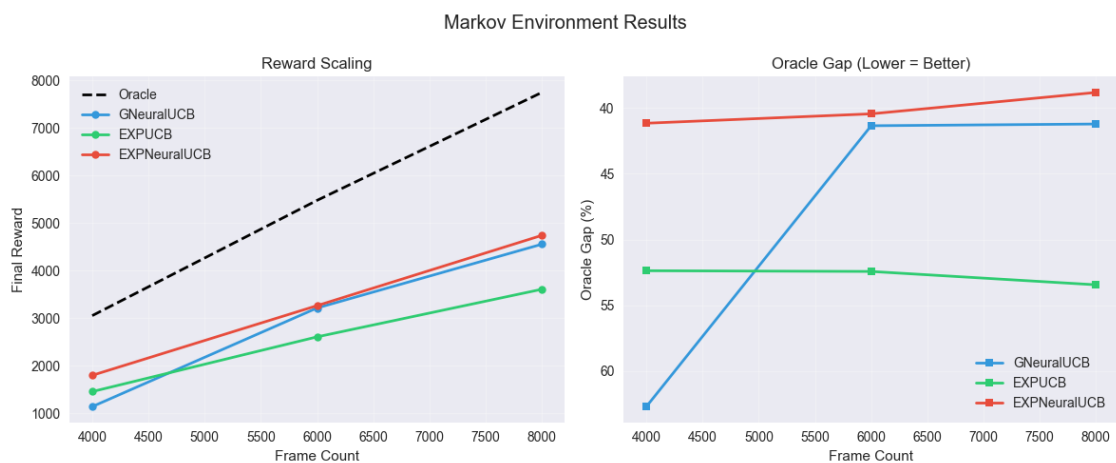
Interpretation:

- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...



Visualization complete!



```
=====
TESTING ADAPTIVE
=====
```

Running test for adaptive environment...

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: ADAPTIVE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: adaptive (intensity=1.0)

EXPERIMENT 1: 4000 frames

```
=====
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

```
=====
ADVERSARIAL CONFIGURATION:
=====
```

```
| Attack Strategy: | AdaptiveAttack |
```

```

| Attack Rate: | 1.00 |
| Memory Window: | 50 |
=====
Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3240.72
 COMPLETED: in 0.00s, final reward: 3240.72
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'neural' mode

#### =====

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====

```

```

- NEURAL Progress: 100%| | 4000/4000 [00:10<00:00, 367.61it/s]
 COMPLETED: GNeuralUCB (batch) completed successfully
 Running EXPUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

- EXP3 Progress: 100%| 4000/4000 [00:02<00:00, 1536.88it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

- HYBRID Progress: 100%| 4000/4000 [00:13<00:00, 303.10it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 26.76s):

```

=====
Oracle (step-wise) : 3240.72
GNeuralUCB (batch) : 2542.22
EXPUCB (batch) : 2093.89
EXPNeuralUCB (batch) : 2016.44
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 62.2%

```

#### ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	AdaptiveAttack
Total Frames:	4,000
Total Attacks:	4,000
Attack Intensity:	0.250

PATH-SPECIFIC ANALYSIS:
Path 0:
Path 1:
Path 2:
Path 3:
=====
ATTACKS: Mean attacks/frame: 1.000
=====

```

#### EXPERIMENT 1 COMPLETED

Execution Time: 26.8s

#### Results:

```

Oracle: 3240.72 (theoretical maximum)
EXPNeuralUCB: 2016.44 (37.8% gap)
GNeuralUCB: 2542.22 (21.6% gap) [WINNER]
EXPUCB: 2093.89 (35.4% gap)
Winner: GNeuralUCB

```

PROGRESS: Experiment 2/3

```

=====
EXPERIMENT 2
=====
Frame Count: 6,000
Attack: adaptive (intensity=1.0)

```

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000
=====
```

#### ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	AdaptiveAttack
Attack Rate:	1.00
Memory Window:	50
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5584.16

COMPLETED: in 0.01s, final reward: 5584.16

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====
```

- NEURAL Progress: 100%| 6000/6000 [00:15<00:00, 375.14it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

=====

- EXP3 Progress: 100%| 6000/6000 [00:05<00:00, 1045.03it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties    |
|------------------|------------|---------------|
| Group Selection  | EXP3       | Adversarially |
|                  |            | Robust        |
| Action Selection | Neural UCB | Nonlinear     |
|                  |            | Learning      |

|            |                        |             |
|------------|------------------------|-------------|
| Parameters | gamma=0.010, eta=0.050 | EXP3 Config |
|            | beta=0.2               | Neural UCB  |

- HYBRID Progress: 100% | 6000/6000 [00:23<00:00, 252.92it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 45.56s):

```
=====
Oracle (step-wise) : 5584.16
GNeuralUCB (batch) : 3903.85
EXPUCB (batch) : 3318.91
EXPNeuralUCB (batch) : 3980.39
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.3%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	AdaptiveAttack
Total Frames:	6,000
Total Attacks:	6,000
Attack Intensity:	0.250
```

-----

| PATH-SPECIFIC ANALYSIS: |

-----

```
Path 0:	2142 attacks (35.7% rate)
Path 1:	1439 attacks (24.0% rate)
Path 2:	947 attacks (15.8% rate)
Path 3:	1472 attacks (24.5% rate)
```

-----

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 45.6s

Results:

```
Oracle: 5584.16 (theoretical maximum)
EXPNeuralUCB: 3980.39 (28.7% gap) [WINNER]
GNeuralUCB: 3903.85 (30.1% gap)
EXPUCB: 3318.91 (40.6% gap)
Winner: EXPNeuralUCB
```

-----

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000  
Attack: adaptive (intensity=1.0)  
-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	AdaptiveAttack
Attack Rate:	1.00
Memory Window:	50
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7832.21

COMPLETED: in 0.01s, final reward: 7832.21

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |

| Parameters | beta=0.2 | Learning UCB Confidence |
|------------|----------|-------------------------|
| =====      |          |                         |

- NEURAL Progress: 100%| 8000/8000 [01:11<00:00, 112.07it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

=====

- EXP3 Progress: 100%| 8000/8000 [00:10<00:00, 779.08it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

# ALGORITHM CONFIGURATION

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100% | 8000/8000 [00:30<00:00, 265.54it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 111.91s):

```

Oracle (step-wise) : 7832.21
GNeuralUCB (batch) : 5597.04
EXPUCB (batch) : 4744.51
EXPNeuralUCB (batch) : 5498.66
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%

```

## ATTACK PATTERN ANALYSIS:

|                                    |                           |
|------------------------------------|---------------------------|
| Strategy:                          | AdaptiveAttack            |
| Total Frames:                      | 8,000                     |
| Total Attacks:                     | 8,000                     |
| Attack Intensity:                  | 0.250                     |
| PATH-SPECIFIC ANALYSIS:            |                           |
| Path 0:                            | 1844 attacks (23.1% rate) |
| Path 1:                            | 2126 attacks (26.6% rate) |
| Path 2:                            | 2217 attacks (27.7% rate) |
| Path 3:                            | 1813 attacks (22.7% rate) |
| ATTACKS: Mean attacks/frame: 1.000 |                           |

## EXPERIMENT 3 COMPLETED

Execution Time: 111.9s

Results:

```

Oracle: 7832.21 (theoretical maximum)
EXPNeuralUCB: 5498.66 (29.8% gap)
GNeuralUCB: 5597.04 (28.5% gap) [WINNER]
EXPUCB: 4744.51 (39.4% gap)

```

Winner: GNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 185.5s  
Average time per experiment: 61.8s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: ADAPTIVE

Intensity: 1.0

Per-Experiment Results:

=====

EXPERIMENT 1 (4,000 frames, 26.8s)

|  
+-- Oracle: 3240.72 (theoretical maximum)  
+-- EXPNeuralUCB: 2016.44 ( 37.8% gap)  
+-- GNeuralUCB: 2542.22 ( 21.6% gap)  
+-- EXPUCB: 2093.89 ( 35.4% gap)  
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 45.6s)

|  
+-- Oracle: 5584.16 (theoretical maximum)  
+-- EXPNeuralUCB: 3980.39 ( 28.7% gap)  
+-- GNeuralUCB: 3903.85 ( 30.1% gap)  
+-- EXPUCB: 3318.91 ( 40.6% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 111.9s)

|  
+-- Oracle: 7832.21 (theoretical maximum)  
+-- EXPNeuralUCB: 5498.66 ( 29.8% gap)  
+-- GNeuralUCB: 5597.04 ( 28.5% gap)  
+-- EXPUCB: 4744.51 ( 39.4% gap)  
Winner: GNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

=====

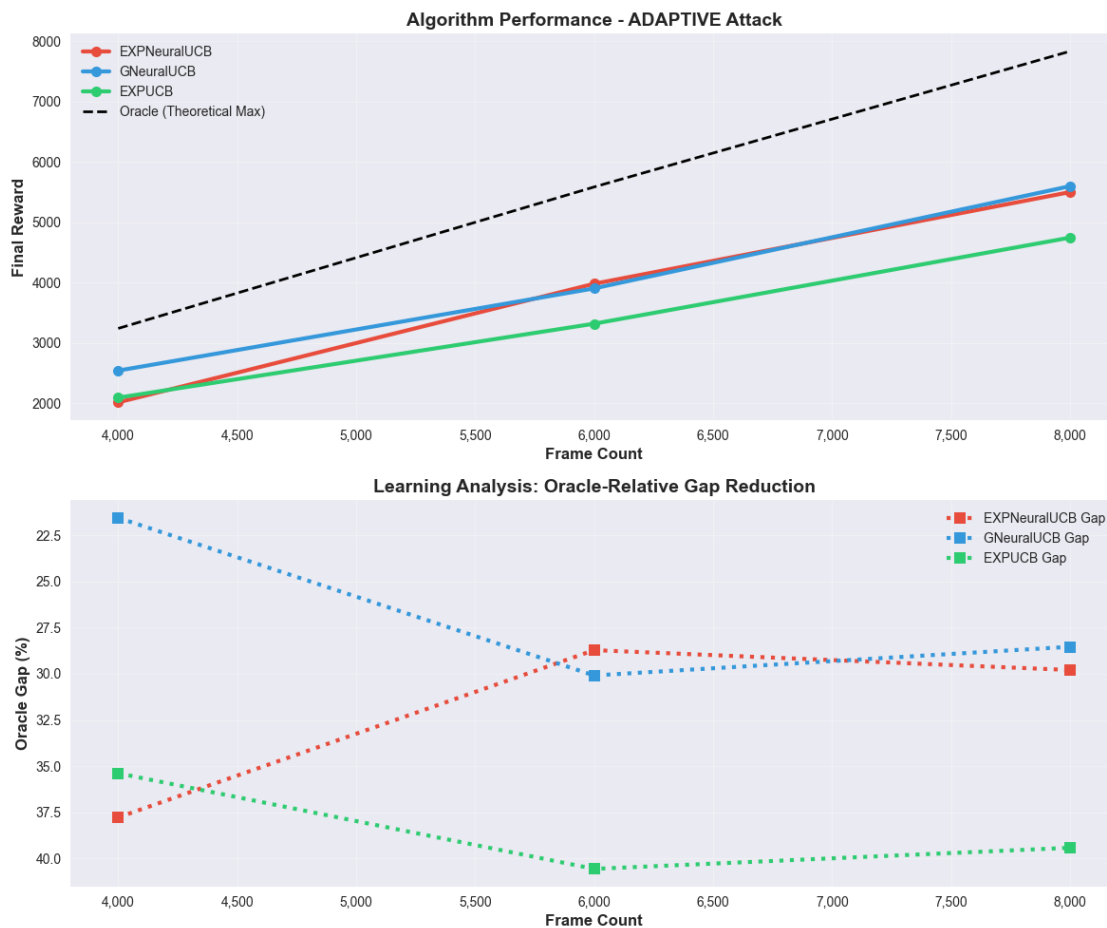
### LEARNING PERFORMANCE (Gap Reduction to Oracle):

- #1: EXPNeuralUCB -> +8.0pp improvement (gap closing)  
Gap: 37.8% -> 29.8%
- #2: EXPUCB -> +4.0pp decline (gap widening)  
Gap: 35.4% -> 39.4%
- #3: GNeuralUCB -> +7.0pp decline (gap widening)  
Gap: 21.6% -> 28.5%

### Interpretation:

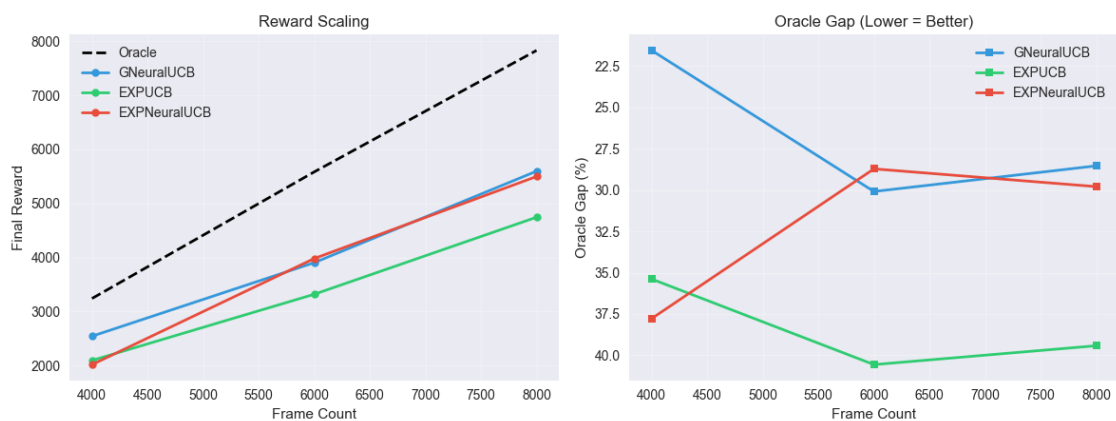
- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...

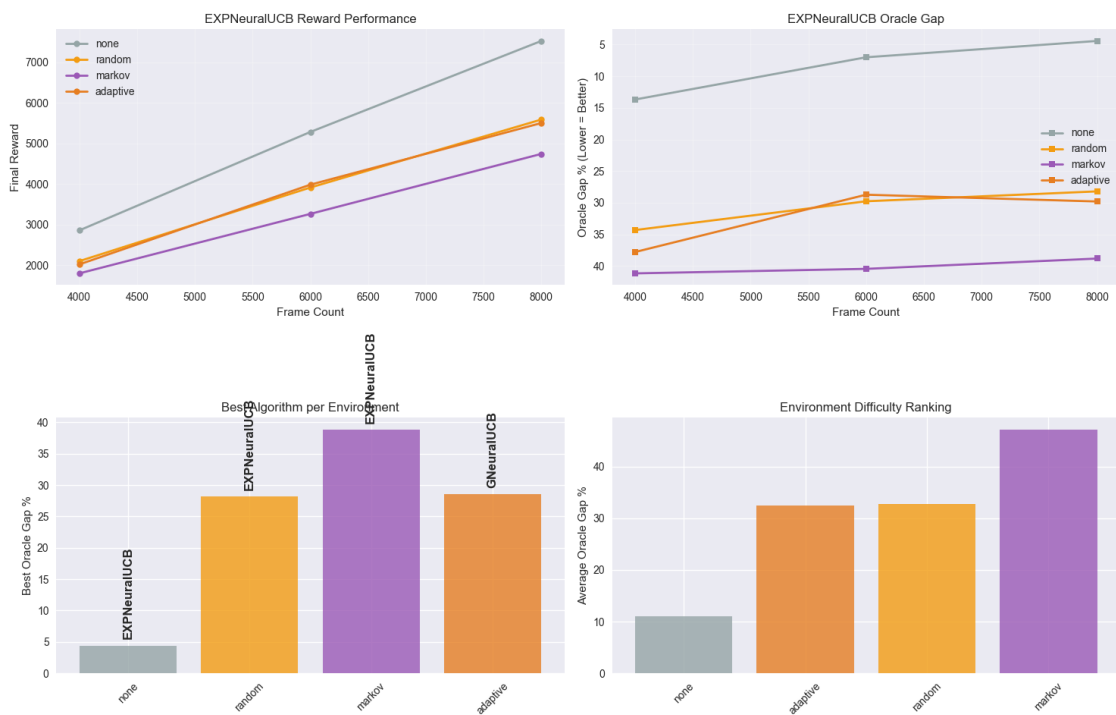


Visualization complete!

## Adaptive Environment Results



## ENVIRONMENT COMPARISON



[ ]:

[ ]:

```
@title
quantum_visualizer_enhanced.py

import matplotlib.pyplot as plt
import pandas as pd
```

```

import numpy as np
import seaborn as sns

class EnhancedQuantumVisualizer(QuantumVisualizer):
 """Enhanced visualization extending the base QuantumVisualizer with
 ↪advanced plots"""

 def __init__(self):
 super().__init__()
 self._setup_style()

 def _setup_style(self):
 plt.style.use('seaborn-v0_8')
 # Keep existing colors from base class
 self.env_colors['onlineadaptive'] = '#16a085' # Add missing color

 def _extract_data(self, attack_type):
 """Extract series for plotting from env_experiments structure (with
 ↪fallbacks)."""
 env_data = self._get_env_data(attack_type)

 # canonical order; include only those present
 first_exp = next(iter(env_data.values()))
 available_algs = set(first_exp.get('results', {}).keys())
 algorithms = [a for a in ("EXPNeuralUCB", "GNeuralUCB", "EXPUCB") if a
 ↪in available_algs]

 frame_counts = []
 oracle_rewards = []
 alg_rewards = {alg: [] for alg in algorithms}
 alg_gaps = {alg: [] for alg in algorithms}

 for exp_id in sorted(env_data.keys()):
 exp = env_data[exp_id]
 res = exp.get('results', {})
 gaps = exp.get('gaps', {})

 frame_counts.append(int(exp.get('frame_count', 0)))
 oracle_rewards.append(float(res.get('Oracle', {})).
 ↪get('final_reward', 0.0)))

 for alg in algorithms:
 alg_rewards[alg].append(float(res.get(alg, {})).
 ↪get('final_reward', 0.0)))
 alg_gaps[alg].append(float(gaps.get(alg, np.nan)))

 return frame_counts, oracle_rewards, alg_rewards, alg_gaps, algorithms

```

```

def plot_performance_dashboard(self, attack_types):
 """Enhanced dashboard visualization (cross-environment)."""
 missing = [env for env in attack_types if env not in self.evaluators]
 if missing:
 print(f"Missing environments: {missing}")
 return

 fig, axes = plt.subplots(2, 3, figsize=(18, 12))
 fig.suptitle(
 'QUANTUM ROUTING ALGORITHM PERFORMANCE DASHBOARD\n'
 'Lower Oracle Gap = Better Performance | Higher Reward = Better_
↳Performance',
 fontsize=16, fontweight='bold'
)

 # Collect all data
 all_gaps = []
 all_rewards = []

 for env in attack_types:
 try:
 frame_counts, oracle_rewards, alg_rewards, alg_gaps, algorithms_
↳ self._extract_data(env)
 except Exception:
 continue

 for alg in algorithms:
 for frame, gap, reward in zip(frame_counts, alg_gaps[alg],_
↳alg_rewards[alg]):
 all_gaps.append({
 'environment': env,
 'algorithm': alg,
 'frame_count': frame,
 'gap_percent': gap,
 'final_reward': reward
 })
 all_rewards.append({
 'environment': env,
 'algorithm': alg,
 'frame_count': frame,
 'final_reward': reward
 })

 gap_df = pd.DataFrame(all_gaps)
 reward_df = pd.DataFrame(all_rewards)

```

```

1. Oracle Gap Heatmap
if not gap_df.empty:
 gap_matrix = gap_df.pivot_table(index='environment',
 ↪columns='algorithm',
 values='gap_percent',
 ↪aggfunc='mean')
 sns.heatmap(gap_matrix, annot=True, fmt='.1f', cmap='RdYlGn_r',
 ax=axes[0,0], cbar_kws={'label': 'Oracle Gap %',
 ↪(Lower=Better)})
 axes[0,0].set_title('Oracle Gap by Environment\n(GREEN = Better)')
else:
 axes[0,0].set_visible(False)

2. Final Reward Heatmap
if not reward_df.empty:
 reward_matrix = reward_df.pivot_table(index='environment',
 ↪columns='algorithm',
 values='final_reward',
 ↪aggfunc='mean')
 sns.heatmap(reward_matrix, annot=True, fmt='.1f', cmap='RdYlGn',
 ax=axes[0,1], cbar_kws={'label': 'Final Reward',
 ↪(Higher=Better)})
 axes[0,1].set_title('Final Reward by Environment\n(GREEN = Better)')
else:
 axes[0,1].set_visible(False)

3. Algorithm Ranking (by avg gap)
if not gap_df.empty:
 avg_gaps = gap_df.groupby('algorithm')['gap_percent'].mean().
 ↪sort_values()
 colors = ['gold', 'silver', '#CD7F32', 'gray'][:len(avg_gaps)]
 axes[0,2].bar(range(len(avg_gaps)), avg_gaps.values, color=colors)
 axes[0,2].set_xticks(range(len(avg_gaps)))
 axes[0,2].set_xticklabels(avg_gaps.index, rotation=45)
 axes[0,2].set_title('Overall Algorithm Ranking\n(Lower = Better)')
 axes[0,2].set_ylabel('Average Oracle Gap %')
else:
 axes[0,2].set_visible(False)

4. Performance vs Frame Count
for env in attack_types:
 try:
 frame_counts, _, alg_rewards, alg_gaps, algorithms = self.
 ↪extract_data(env)
 except Exception:
 continue

```

```

 for alg in algorithms:
 axes[1,0].plot(frame_counts, alg_rewards[alg], 'o-',
 label=f"{alg} ({env})",
 color=self.colors.get(alg, 'gray'), alpha=0.7)

 axes[1,0].set_title('Reward Scaling with Frame Count')
 axes[1,0].set_xlabel('Frame Count')
 axes[1,0].set_ylabel('Final Reward')
 axes[1,0].legend(bbox_to_anchor=(1.05, 1), loc='upper left')
 axes[1,0].grid(True, alpha=0.3)

 # 5. Gap vs Frame Count
 for env in attack_types:
 try:
 frame_counts, _, alg_rewards, alg_gaps, algorithms = self.
↪_extract_data(env)
 except Exception:
 continue
 for alg in algorithms:
 axes[1,1].plot(frame_counts, alg_gaps[alg], 's-',
 label=f"{alg} ({env})",
 color=self.colors.get(alg, 'gray'), alpha=0.7)

 axes[1,1].set_title('Oracle Gap vs Frame Count')
 axes[1,1].set_xlabel('Frame Count')
 axes[1,1].set_ylabel('Oracle Gap %')
 axes[1,1].legend(bbox_to_anchor=(1.05, 1), loc='upper left')
 axes[1,1].grid(True, alpha=0.3)
 axes[1,1].invert_yaxis()

 # 6. Environment Difficulty
 if not gap_df.empty:
 env_difficulty = gap_df.groupby('environment')['gap_percent'].
↪mean().sort_values()
 colors = [self.env_colors.get(env, 'gray') for env in
↪env_difficulty.index]
 axes[1,2].bar(range(len(env_difficulty)), env_difficulty.values,
↪color=colors)
 axes[1,2].set_xticks(range(len(env_difficulty)))
 axes[1,2].set_xticklabels(env_difficulty.index, rotation=45)
 axes[1,2].set_title('Environment Difficulty\n(Lower = Easier)')
 axes[1,2].set_ylabel('Average Oracle Gap %')
 else:
 axes[1,2].set_visible(False)

 plt.tight_layout()
 plt.savefig('performance_dashboard.png', dpi=300, bbox_inches='tight')

```

```

plt.show()

def plot_algorithm_comparison(self, attack_types):
 """Algorithm comparison across environments."""
 fig, axes = plt.subplots(2, 2, figsize=(15, 12))
 fig.suptitle('ALGORITHM COMPARISON ANALYSIS', fontsize=16,
 ↪fontweight='bold')

 all_data = []
 for env in attack_types:
 try:
 frame_counts, _, alg_rewards, alg_gaps, algorithms = self.
 ↪_extract_data(env)
 except Exception:
 continue
 for alg in algorithms:
 for frame, gap, reward in zip(frame_counts, alg_gaps[alg],
 ↪alg_rewards[alg]):
 all_data.append({
 'environment': env,
 'algorithm': alg,
 'frame_count': frame,
 'gap_percent': gap,
 'final_reward': reward
 })

 df = pd.DataFrame(all_data)
 if df.empty:
 plt.close(fig)
 print("No data to compare.")
 return

 sns.boxplot(data=df, x='algorithm', y='gap_percent', ax=axes[0,0])
 axes[0,0].set_title('Performance Distribution (Lower = Better)')
 axes[0,0].set_ylabel('Oracle Gap %')
 axes[0,0].tick_params(axis='x', rotation=45)

 sns.boxplot(data=df, x='environment', y='gap_percent', ax=axes[0,1])
 axes[0,1].set_title('Environment Difficulty')
 axes[0,1].set_ylabel('Oracle Gap %')
 axes[0,1].tick_params(axis='x', rotation=45)

 pivot = df.pivot_table(index='algorithm', columns='environment',
 values='gap_percent', aggfunc='mean')
 sns.heatmap(pivot, annot=True, fmt='.1f', cmap='RdYlGn_r', ax=axes[1,0])
 axes[1,0].set_title('Algorithm vs Environment\n(Green = Better)')

```

```

Scaling improvement
scaling = []
for alg in df['algorithm'].unique():
 for env in df['environment'].unique():
 sub = df[(df['algorithm'] == alg) & (df['environment'] == env)]
 if sub['frame_count'].nunique() >= 2:
 min_f = sub['frame_count'].min()
 max_f = sub['frame_count'].max()
 min_gap = sub[sub['frame_count'] == min_f]['gap_percent'].
↳mean()
 max_gap = sub[sub['frame_count'] == max_f]['gap_percent'].
↳mean()
 scaling.append({'algorithm': alg, 'environment': env,
↳'improvement': (min_gap - max_gap)})

 if scaling:
 sc_df = pd.DataFrame(scaling)
 sns.barplot(data=sc_df, x='algorithm', y='improvement',
↳hue='environment', ax=axes[1,1])
 axes[1,1].set_title('Scaling Improvement (Higher = Better
↳Learning)')
 axes[1,1].set_ylabel('Gap Improvement (pp)')
 axes[1,1].tick_params(axis='x', rotation=45)
 else:
 axes[1,1].set_visible(False)

plt.tight_layout()
plt.savefig('algorithm_comparison.png', dpi=300, bbox_inches='tight')
plt.show()

Test functions using the enhanced class
def test_enhanced_visualizations():
 viz = EnhancedQuantumVisualizer()
 environments = ["none", "random", "markov", "adaptive"]

 for env in environments:
 print(f"\n{'='*50}\nTESTING {env.upper()}\n{'='*50}")
 viz.run_test(env)
 viz.plot_results(env) # Inherited from base class

 print("\n" + "="*60 + "\nCREATING ENHANCED VISUALIZATIONS\n" + "="*60)
 viz.plot_performance_dashboard(environments)
 viz.plot_algorithm_comparison(environments)
 viz.compare_environments(environments) # Inherited from base class
 return viz

if __name__ == "__main__":

```

```
print("TESTING ENHANCED QUANTUM VISUALIZER:")
viz = test_enhanced_visualizations()
```

TESTING ENHANCED QUANTUM VISUALIZER:

=====

TESTING NONE

=====

Running test for none environment...

=====

QUANTUM MULTIRUN EVALUATOR INITIALIZED

=====

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: NONE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

=====

=====

STARTING QUANTUM MULTIRUN EVALUATION

=====

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

=====

PROGRESS: Experiment 1/3

=====

EXPERIMENT 1

=====

Frame Count: 4,000

Attack: none (intensity=1.0)

-----

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |

```
| Frame Length: | 4,000 |
=====
```

#### ADVERSARIAL CONFIGURATION:

```
=====
```

```
| Attack Strategy: | NoAttack |
=====
```

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3307.29

COMPLETED: in 0.00s, final reward: 3307.29

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

#### ORACLE ANALYSIS:

```
=====
```

```
| Optimal Path: | 0 |
```

```
| Optimal Action: | 4 |
```

```
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
```

#### ALGORITHM CONFIGURATION

```
=====
```

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

```
=====
```

- NEURAL Progress: 100%| 4000/4000 [00:13<00:00, 300.33it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

#### ORACLE ANALYSIS:

```
=====
```

```
| Optimal Path: | 0 |
```

```
| Optimal Action: | 4 |
```

```
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

```
=====
```

- EXP3 Progress: 100%| 4000/4000 [00:02<00:00, 1477.43it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

```
=====
```

- HYBRID Progress: 100%| | 4000/4000 [00:14<00:00, 267.67it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 31.06s):

```
=====
Oracle (step-wise) : 3307.29
GNeuralUCB (batch) : 3235.40
EXPUCB (batch) : 2794.69
EXPNeuralUCB (batch) : 2854.95
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	NoAttack
Total Frames:	4,000
Total Attacks:	0
Attack Intensity:	0.000
```

-----

```
| PATH-SPECIFIC ANALYSIS: |
```

-----

```
Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)
```

-----

```
ATTACKS: Mean attacks/frame: 0.000
=====
```

EXPERIMENT 1 COMPLETED

Execution Time: 31.1s

Results:

```
Oracle: 3307.29 (theoretical maximum)
EXPNeuralUCB: 2854.95 (13.7% gap)
GNeuralUCB: 3235.40 (2.2% gap) [WINNER]
EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB
```

-----

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: none (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | NoAttack |
=====
```

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5676.68

COMPLETED: in 0.01s, final reward: 5676.68

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

- NEURAL Progress: 100% | 6000/6000 [00:22<00:00, 269.66it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100% | 6000/6000 [00:05<00:00, 1050.10it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component       | Method | Properties    |
|-----------------|--------|---------------|
| Group Selection | EXP3   | Adversarially |

|                  |                        |             |  |
|------------------|------------------------|-------------|--|
|                  |                        | Robust      |  |
| Action Selection | Neural UCB             | Nonlinear   |  |
|                  |                        | Learning    |  |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config |  |
|                  | beta=0.2               | Neural UCB  |  |

=====

- HYBRID Progress: 100% | 6000/6000 [00:17<00:00, 336.93it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 45.87s):

=====

|              |             |           |
|--------------|-------------|-----------|
| Oracle       | (step-wise) | : 5676.68 |
| GNeuralUCB   | (batch)     | : 5179.98 |
| EXPUCB       | (batch)     | : 4552.40 |
| EXPNeuralUCB | (batch)     | : 5279.10 |

WINNER: EXPNeuralUCB

EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%

ATTACK PATTERN ANALYSIS:

=====

|                   |          |  |
|-------------------|----------|--|
| Strategy:         | NoAttack |  |
| Total Frames:     | 6,000    |  |
| Total Attacks:    | 0        |  |
| Attack Intensity: | 0.000    |  |

-----

| PATH-SPECIFIC ANALYSIS: |

-----

|         |                        |
|---------|------------------------|
| Path 0: | 0 attacks ( 0.0% rate) |
| Path 1: | 0 attacks ( 0.0% rate) |
| Path 2: | 0 attacks ( 0.0% rate) |
| Path 3: | 0 attacks ( 0.0% rate) |

=====

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 45.9s

Results:

|               |                               |
|---------------|-------------------------------|
| Oracle:       | 5676.68 (theoretical maximum) |
| EXPNeuralUCB: | 5279.10 ( 7.0% gap) [WINNER]  |
| GNeuralUCB:   | 5179.98 ( 8.7% gap)           |
| EXPUCB:       | 4552.40 ( 19.8% gap)          |

Winner: EXPNeuralUCB

-----

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: none (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |          |  |
|------------------|----------|--|
| Attack Strategy: | NoAttack |  |
|------------------|----------|--|

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7868.91

COMPLETED: in 0.00s, final reward: 7868.91

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                  |  |
|-------------------|----------------------------------|--|
| Optimal Path:     | 0                                |  |
| Optimal Action:   | 4                                |  |
| Path Performance: | [np.float64(0.9836141108302526), |  |

np.float64(0.9637113766945926), np.float64(0.9435345403529187),  
np.float64(0.9202691292754374)]

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

|                 |            |                 |  |
|-----------------|------------|-----------------|--|
| Component       | Method     | Properties      |  |
| -----           | -----      | -----           |  |
| Group Selection | Simple UCB | NOT Adversarial |  |
|                 |            | Robust          |  |

|                  |            |                |
|------------------|------------|----------------|
| Action Selection | Neural UCB | Nonlinear      |
|                  |            | Learning       |
| Parameters       | beta=0.2   | UCB Confidence |

=====

- NEURAL Progress: 100%| 8000/8000 [00:53<00:00, 148.29it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 8000/8000 [00:07<00:00, 1133.72it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.9836141108302526),<br>np.float64(0.9637113766945926), np.float64(0.9435345403529187),<br>np.float64(0.9202691292754374)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

=====

- HYBRID Progress: 100%| 8000/8000 [00:23<00:00, 344.68it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

## RESULTS (completed in 84.30s): =====

Oracle (step-wise) : 7868.91  
 GNeuralUCB (batch) : 7266.99  
 EXPUCB (batch) : 6307.33  
 EXPNeuralUCB (batch) : 7520.98  
 WINNER: EXPNeuralUCB  
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%

## ATTACK PATTERN ANALYSIS: =====

|                   |          |
|-------------------|----------|
| Strategy:         | NoAttack |
| Total Frames:     | 8,000    |
| Total Attacks:    | 0        |
| Attack Intensity: | 0.000    |

-----

## PATH-SPECIFIC ANALYSIS: -----

|         |                        |
|---------|------------------------|
| Path 0: | 0 attacks ( 0.0% rate) |
| Path 1: | 0 attacks ( 0.0% rate) |
| Path 2: | 0 attacks ( 0.0% rate) |
| Path 3: | 0 attacks ( 0.0% rate) |

=====

ATTACKS: Mean attacks/frame: 0.000

=====

## EXPERIMENT 3 COMPLETED

Execution Time: 84.3s

### Results:

Oracle: 7868.91 (theoretical maximum)  
 EXPNeuralUCB: 7520.98 ( 4.4% gap) [WINNER]  
 GNeuralUCB: 7266.99 ( 7.6% gap)

EXPUCB: 6307.33 ( 19.8% gap)  
Winner: EXPNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 162.3s  
Average time per experiment: 54.1s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:  
Strategy: NONE  
Intensity: 1.0

Per-Experiment Results:

-----

EXPERIMENT 1 (4,000 frames, 31.1s)

|  
+-- Oracle: 3307.29 (theoretical maximum)  
+-- EXPNeuralUCB: 2854.95 ( 13.7% gap)  
+-- GNeuralUCB: 3235.40 ( 2.2% gap)  
+-- EXPUCB: 2794.69 ( 15.5% gap)  
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 45.9s)

|  
+-- Oracle: 5676.68 (theoretical maximum)  
+-- EXPNeuralUCB: 5279.10 ( 7.0% gap)  
+-- GNeuralUCB: 5179.98 ( 8.7% gap)  
+-- EXPUCB: 4552.40 ( 19.8% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 84.3s)

|  
+-- Oracle: 7868.91 (theoretical maximum)  
+-- EXPNeuralUCB: 7520.98 ( 4.4% gap)  
+-- GNeuralUCB: 7266.99 ( 7.6% gap)  
+-- EXPUCB: 6307.33 ( 19.8% gap)  
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

---

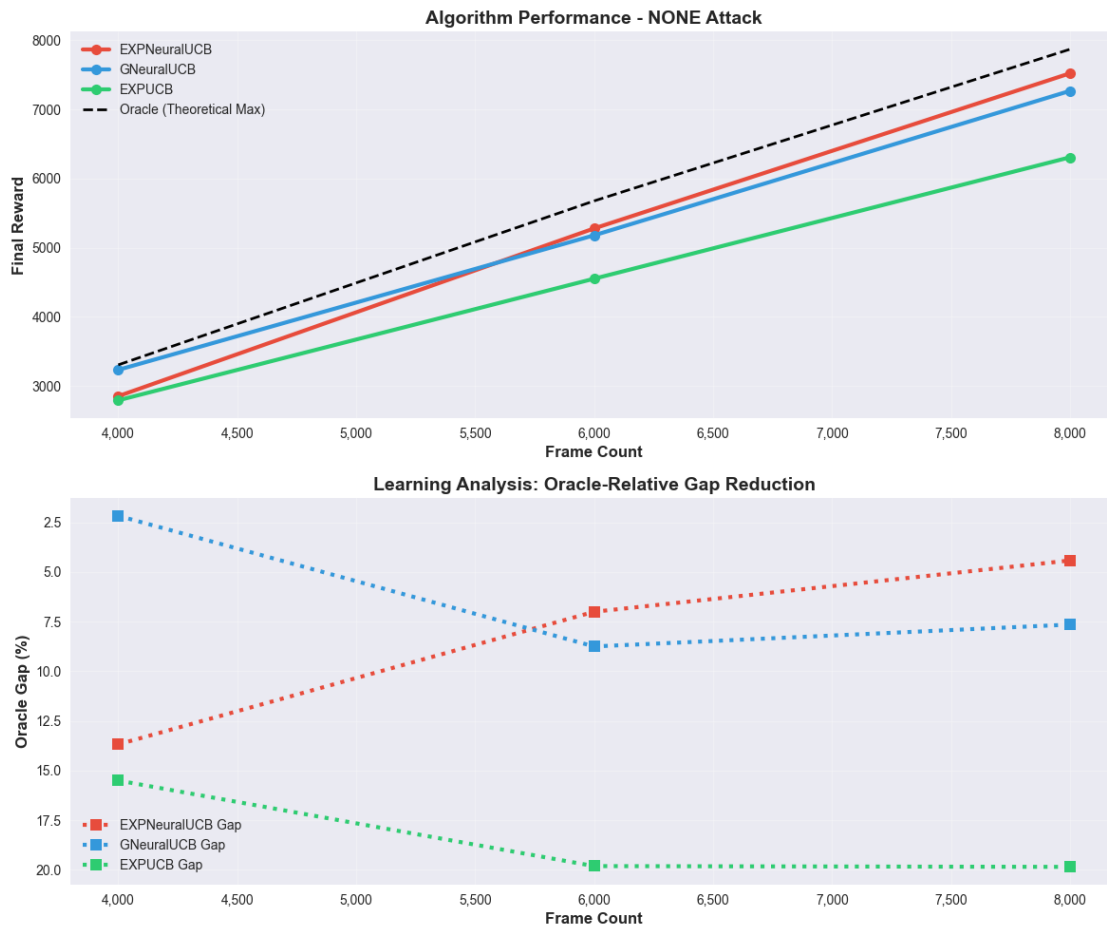
### LEARNING PERFORMANCE (Gap Reduction to Oracle):

- #1: EXPNeuralUCB -> +9.3pp improvement (gap closing)  
Gap: 13.7% -> 4.4%
- #2: EXPUCB -> +4.3pp decline (gap widening)  
Gap: 15.5% -> 19.8%
- #3: GNeuralUCB -> +5.5pp decline (gap widening)  
Gap: 2.2% -> 7.6%

### Interpretation:

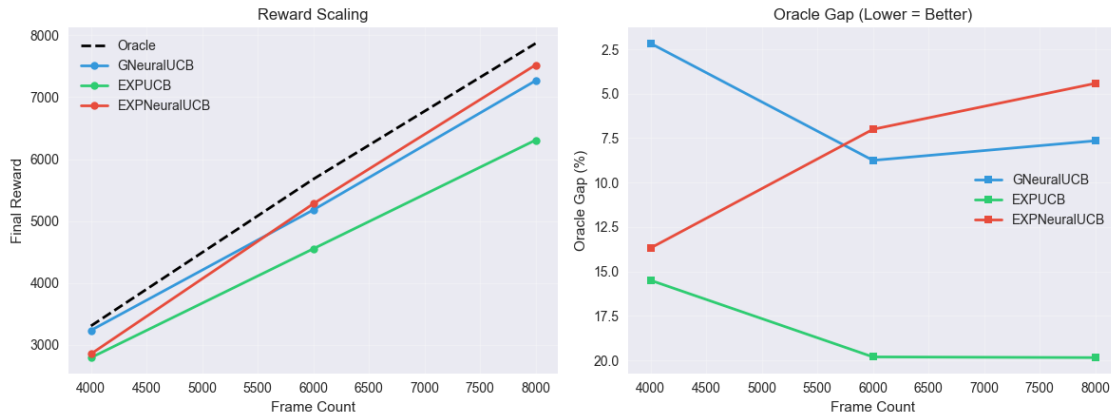
- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...



Visualization complete!

# None Environment Results



TESTING RANDOM

Running test for random environment...

QUANTUM MULTIRUN EVALUATOR INITIALIZED

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: RANDOM
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

## EXPERIMENT 1

```
=====
Frame Count: 4,000
Attack: random (intensity=1.0)

```

EXPERIMENT 1: 4000 frames

### QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

### ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack |
| Attack Rate: | 0.25 |
=====
```

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3192.62

COMPLETED: in 0.00s, final reward: 3192.62

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
```

|                  |            |                |
|------------------|------------|----------------|
| Action Selection | Neural UCB | Nonlinear      |
|                  |            | Learning       |
| Parameters       | beta=0.2   | UCB Confidence |

=====

- NEURAL Progress: 100%| 4000/4000 [00:06<00:00, 576.69it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

|                  |                                    |                           |
|------------------|------------------------------------|---------------------------|
| Component        | Method                             | Properties                |
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 4000/4000 [00:01<00:00, 2188.33it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100%| 4000/4000 [00:09<00:00, 433.70it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

## RESULTS (completed in 18.03s): =====

Oracle (step-wise) : 3192.62

GNeuralUCB (batch) : 2420.84

EXPUCB (batch) : 2024.79

EXPNeuralUCB (batch) : 2097.30

WINNER: GNeuralUCB

EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 65.7%

## ATTACK PATTERN ANALYSIS: =====

|                   |              |
|-------------------|--------------|
| Strategy:         | RandomAttack |
| Total Frames:     | 4,000        |
| Total Attacks:    | 4,024        |
| Attack Intensity: | 0.252        |

## PATH-SPECIFIC ANALYSIS: =====

|         |                           |
|---------|---------------------------|
| Path 0: | 986 attacks (24.6% rate)  |
| Path 1: | 1041 attacks (26.0% rate) |
| Path 2: | 1003 attacks (25.1% rate) |
| Path 3: | 994 attacks (24.9% rate)  |

ATTACKS: Mean attacks/frame: 1.006

=====

## EXPERIMENT 1 COMPLETED

Execution Time: 18.0s

## Results:

Oracle: 3192.62 (theoretical maximum)

EXPNeuralUCB: 2097.30 ( 34.3% gap)

GNeuralUCB: 2420.84 ( 24.2% gap) [WINNER]

EXPUCB: 2024.79 ( 36.6% gap)  
Winner: GNeuralUCB

-----  
PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: random (intensity=1.0)  
-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 6,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |              |  |
|------------------|--------------|--|
| Attack Strategy: | RandomAttack |  |
| Attack Rate:     | 0.25         |  |

=====

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5567.05

COMPLETED: in 0.00s, final reward: 5567.05

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                |  |
|-------------------|----------------------------------------------------------------|--|
| Optimal Path:     | 0                                                              |  |
| Optimal Action:   | 4                                                              |  |
| Path Performance: | [np.float64(0.9461134921196183),                               |  |
|                   | np.float64(0.902918807873247), np.float64(0.8603089615724164), |  |
|                   | np.float64(0.8116651900474303)]                                |  |

=====

EXPNeuralUCB initialized in 'neural' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

- NEURAL Progress: 100%| 6000/6000 [00:10<00:00, 549.82it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |  
| Path Performance: | [np.float64(0.9461134921196183),  
np.float64(0.902918807873247), np.float64(0.8603089615724164),  
np.float64(0.8116651900474303)] |  
=====

EXPNeuralUCB initialized in 'exp3' mode

# ===== ALGORITHM CONFIGURATION =====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

- EXP3 Progress: 100%| 6000/6000 [00:04<00:00, 1471.45it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

## ORACLE ANALYSIS: =====

| Optimal Path: | 0 |  
| Optimal Action: | 4 |

```
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

```
=====
- HYBRID Progress: 100%| | 6000/6000 [00:14<00:00, 413.28it/s]
```

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 29.56s):

```
=====
Oracle (step-wise) : 5567.05
GNeuralUCB (batch) : 3840.18
EXPUCB (batch) : 3310.90
EXPNeuralUCB (batch) : 3910.27
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	RandomAttack
Total Frames:	6,000
Total Attacks:	6,001
Attack Intensity:	0.250
=====
```

```
| PATH-SPECIFIC ANALYSIS: |
=====
```

|         |                           |
|---------|---------------------------|
| Path 0: | 1490 attacks (24.8% rate) |
| Path 1: | 1511 attacks (25.2% rate) |
| Path 2: | 1476 attacks (24.6% rate) |
| Path 3: | 1524 attacks (25.4% rate) |

```
=====
ATTACKS: Mean attacks/frame: 1.000
=====
```

EXPERIMENT 2 COMPLETED

Execution Time: 29.6s

Results:

Oracle: 5567.05 (theoretical maximum)  
EXPNeuralUCB: 3910.27 ( 29.8% gap) [WINNER]  
GNeuralUCB: 3840.18 ( 31.0% gap)  
EXPUCB: 3310.90 ( 40.5% gap)  
Winner: EXPNeuralUCB

-----  
PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: random (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |              |  |
|------------------|--------------|--|
| Attack Strategy: | RandomAttack |  |
| Attack Rate:     | 0.25         |  |

=====

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7777.26

COMPLETED: in 0.01s, final reward: 7777.26

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                 |   |  |
|-----------------|---|--|
| Optimal Path:   | 0 |  |
| Optimal Action: | 4 |  |

```
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |
|                  |            | Learning        |
| Parameters       | beta=0.2   | UCB Confidence  |

```
=====
```

- NEURAL Progress: 100%| 8000/8000 [00:50<00:00, 157.45it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

```
=====
```

- EXP3 Progress: 100%| 8000/8000 [00:07<00:00, 1117.11it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====
ALGORITHM CONFIGURATION
=====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

=====
- HYBRID Progress: 100%| | 8000/8000 [00:21<00:00, 378.38it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 79.18s):

```
=====
Oracle (step-wise) : 7777.26
GNeuralUCB (batch) : 5417.64
EXPUCB (batch) : 4675.52
EXPNeuralUCB (batch) : 5584.49
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.8%
```

ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	RandomAttack
Total Frames:	8,000
Total Attacks:	8,063
Attack Intensity:	0.252

PATH-SPECIFIC ANALYSIS:
```

```
Path 0:	2018 attacks (25.2% rate)
Path 1:	2041 attacks (25.5% rate)
Path 2:	1996 attacks (24.9% rate)
Path 3:	2008 attacks (25.1% rate)
```

```
=====
ATTACKS: Mean attacks/frame: 1.008
=====
```

### EXPERIMENT 3 COMPLETED

Execution Time: 79.2s

#### Results:

```
Oracle: 7777.26 (theoretical maximum)
EXPNeuralUCB: 5584.49 (28.2% gap) [WINNER]
GNeuralUCB: 5417.64 (30.3% gap)
EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB
```

```

```

### QUANTUM EVALUATION COMPLETED!

```
=====
Total experiments: 3
Total time: 127.5s
Average time per experiment: 42.5s
=====
```

### DETAILED EXPERIMENT ANALYSIS

#### Attack Configuration:

```
Strategy: RANDOM
Intensity: 1.0
```

#### Per-Experiment Results:

```

```

#### EXPERIMENT 1 (4,000 frames, 18.0s)

```
|
+-- Oracle: 3192.62 (theoretical maximum)
+-- EXPNeuralUCB: 2097.30 (34.3% gap)
+-- GNeuralUCB: 2420.84 (24.2% gap)
+-- EXPUCB: 2024.79 (36.6% gap)
Winner: GNeuralUCB
```

#### EXPERIMENT 2 (6,000 frames, 29.6s)

```
|
+-- Oracle: 5567.05 (theoretical maximum)
```

```

+-- EXPNeuralUCB: 3910.27 (29.8% gap)
+-- GNeuralUCB: 3840.18 (31.0% gap)
+-- EXPUCB: 3310.90 (40.5% gap)
Winner: EXPNeuralUCB

```

#### EXPERIMENT 3 (8,000 frames, 79.2s)

```

|
+-- Oracle: 7777.26 (theoretical maximum)
+-- EXPNeuralUCB: 5584.49 (28.2% gap)
+-- GNeuralUCB: 5417.64 (30.3% gap)
+-- EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB

```

#### ORACLE-RELATIVE GAP ANALYSIS:

##### LEARNING PERFORMANCE (Gap Reduction to Oracle):

```

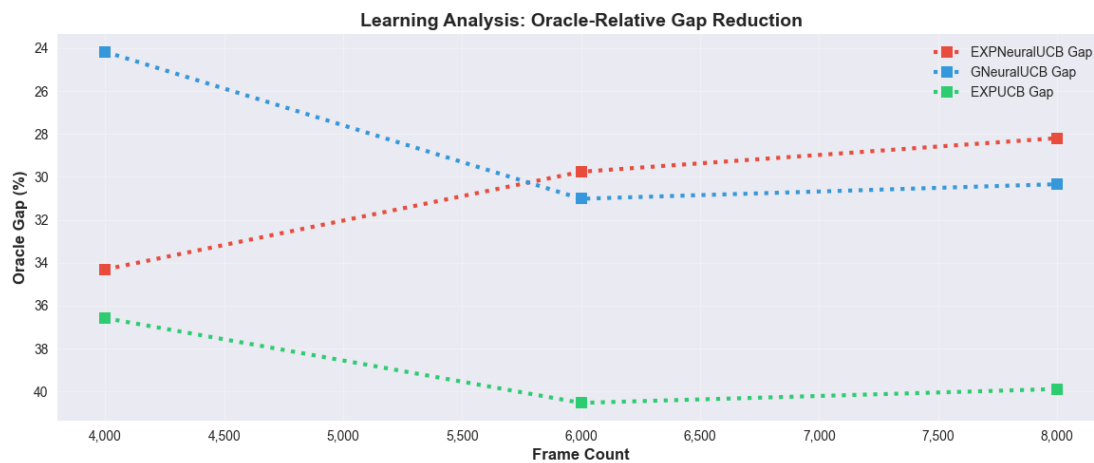
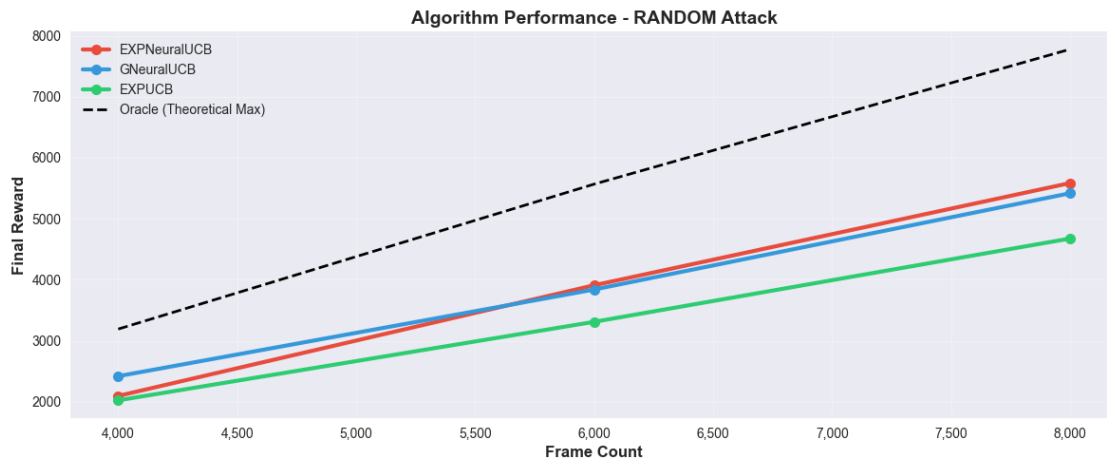
#1: EXPNeuralUCB -> +6.1pp improvement (gap closing)
 Gap: 34.3% -> 28.2%
#2: EXPUCB -> +3.3pp decline (gap widening)
 Gap: 36.6% -> 39.9%
#3: GNeuralUCB -> +6.2pp decline (gap widening)
 Gap: 24.2% -> 30.3%

```

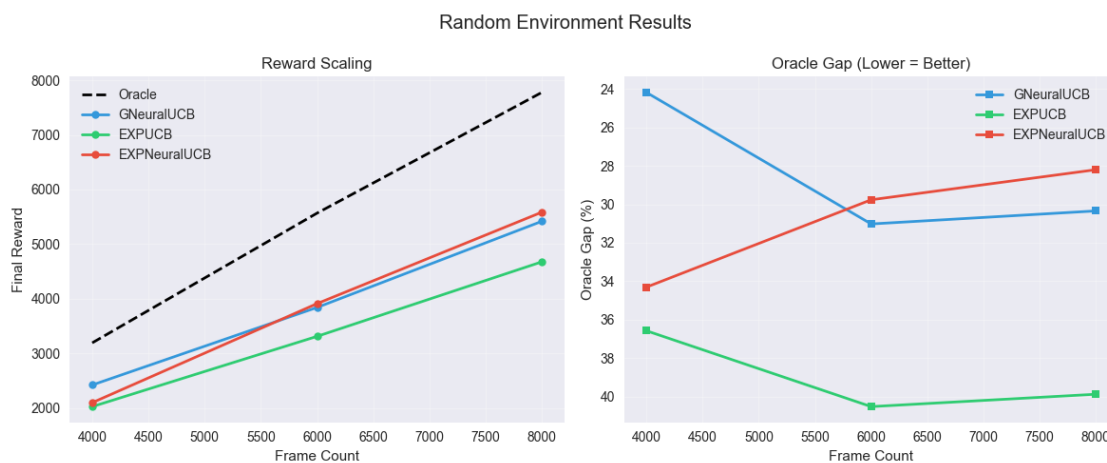
##### Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!



```
=====
TESTING MARKOV
=====
```

Running test for markov environment...

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: MARKOV
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

```
=====
PROGRESS: Experiment 1/3
```

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: markov (intensity=1.0)

-----

EXPERIMENT 1: 4000 frames

```
=====
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

```
=====
ADVERSARIAL CONFIGURATION:
=====
```

```
| Attack Strategy: | MarkovAttack |
```

```

| Attack Rate: | 1.00 |
| Simultaneous: | 2 |
=====
Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
 Running Oracle...
 Context strategy: context-free
 take_action params: 0, update params: 3
 COMPLETED: 4000 steps, final reward: 3051.08
 COMPLETED: in 0.00s, final reward: 3051.08
 COMPLETED: Oracle (step-wise) completed successfully
 Running GNeuralUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'neural' mode

#### =====

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====

```

```

- NEURAL Progress: 100%| | 4000/4000 [00:04<00:00, 822.58it/s]
 COMPLETED: GNeuralUCB (batch) completed successfully
 Running EXPUCB...

```

#### ORACLE ANALYSIS:

```

=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)] |
=====

```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

- EXP3 Progress: 100%| 4000/4000 [00:01<00:00, 2172.04it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Neural UCB                         | Nonlinear<br>Learning     |
| Parameters       | gamma=0.010, eta=0.050<br>beta=0.2 | EXP3 Config<br>Neural UCB |

- HYBRID Progress: 100%| 4000/4000 [00:08<00:00, 458.70it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 15.55s):

```

=====
Oracle (step-wise) : 3051.08
GNeuralUCB (batch) : 1137.52
EXPUCB (batch) : 1453.00
EXPNeuralUCB (batch) : 1795.46
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%

```

#### ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	MarkovAttack
Total Frames:	4,000
Total Attacks:	8,000
Attack Intensity:	0.500

PATH-SPECIFIC ANALYSIS:
Path 0:
Path 1:
Path 2:
Path 3:
=====
ATTACKS: Mean attacks/frame: 2.000
=====

```

#### EXPERIMENT 1 COMPLETED

Execution Time: 15.5s

#### Results:

```

Oracle: 3051.08 (theoretical maximum)
EXPNeuralUCB: 1795.46 (41.2% gap) [WINNER]
GNeuralUCB: 1137.52 (62.7% gap)
EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 2/3

```

=====
EXPERIMENT 2
=====
Frame Count: 6,000
Attack: markov (intensity=1.0)

```

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
```

```
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000
=====
```

#### ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5474.92

COMPLETED: in 0.00s, final reward: 5474.92

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	beta=0.2	UCB Confidence
=====
```

- NEURAL Progress: 100%| 6000/6000 [00:10<00:00, 586.52it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

- EXP3 Progress: 100% | 6000/6000 [00:04<00:00, 1450.28it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties    |
|------------------|------------|---------------|
| Group Selection  | EXP3       | Adversarially |
|                  |            | Robust        |
| Action Selection | Neural UCB | Nonlinear     |
|                  |            | Learning      |

|            |                        |             |
|------------|------------------------|-------------|
| Parameters | gamma=0.010, eta=0.050 | EXP3 Config |
|            | beta=0.2               | Neural UCB  |

- HYBRID Progress: 100% | 6000/6000 [00:13<00:00, 434.65it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 28.34s):

```

=====
Oracle (step-wise) : 5474.92
GNeuralUCB (batch) : 3211.32
EXPUCB (batch) : 2604.25
EXPNeuralUCB (batch) : 3260.87
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%

```

ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	MarkovAttack
Total Frames:	6,000
Total Attacks:	12,000
Attack Intensity:	0.500

```

PATH-SPECIFIC ANALYSIS:

```

Path 0:	3798 attacks (63.3% rate)
Path 1:	2217 attacks (37.0% rate)
Path 2:	3789 attacks (63.1% rate)
Path 3:	2196 attacks (36.6% rate)

```

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 2 COMPLETED

Execution Time: 28.3s

Results:

```

Oracle: 5474.92 (theoretical maximum)
EXPNeuralUCB: 3260.87 (40.4% gap) [WINNER]
GNeuralUCB: 3211.32 (41.3% gap)
EXPUCB: 2604.25 (52.4% gap)
Winner: EXPNeuralUCB

```

PROGRESS: Experiment 3/3

EXPERIMENT 3

Frame Count: 8,000  
Attack: markov (intensity=1.0)  
-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7744.38

COMPLETED: in 0.00s, final reward: 7744.38

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties      |
|------------------|------------|-----------------|
| Group Selection  | Simple UCB | NOT Adversarial |
|                  |            | Robust          |
| Action Selection | Neural UCB | Nonlinear       |

| Parameters | beta=0.2 | Learning UCB Confidence |
|------------|----------|-------------------------|
| =====      |          |                         |

- NEURAL Progress: 100%| 8000/8000 [00:48<00:00, 163.66it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Linear UCB             | Linear        |
|                  |                        | Learning      |
| Parameters       | gamma=0.100, eta=0.005 | EXP3 Config   |
|                  | beta=0.2               | Linear UCB    |

=====

- EXP3 Progress: 100%| 8000/8000 [00:06<00:00, 1151.01it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

# ALGORITHM CONFIGURATION

| Component        | Method                 | Properties    |
|------------------|------------------------|---------------|
| Group Selection  | EXP3                   | Adversarially |
|                  |                        | Robust        |
| Action Selection | Neural UCB             | Nonlinear     |
|                  |                        | Learning      |
| Parameters       | gamma=0.010, eta=0.050 | EXP3 Config   |
|                  | beta=0.2               | Neural UCB    |

- HYBRID Progress: 100% | 8000/8000 [00:21<00:00, 373.39it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 77.48s):

```

Oracle (step-wise) : 7744.38
GNeuralUCB (batch) : 4552.69
EXPUCB (batch) : 3605.14
EXPNeuralUCB (batch) : 4738.67
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%

```

## ATTACK PATTERN ANALYSIS:

|                                    |                           |
|------------------------------------|---------------------------|
| Strategy:                          | MarkovAttack              |
| Total Frames:                      | 8,000                     |
| Total Attacks:                     | 16,000                    |
| Attack Intensity:                  | 0.500                     |
| PATH-SPECIFIC ANALYSIS:            |                           |
| Path 0:                            | 5076 attacks (63.4% rate) |
| Path 1:                            | 2935 attacks (36.7% rate) |
| Path 2:                            | 5036 attacks (62.9% rate) |
| Path 3:                            | 2953 attacks (36.9% rate) |
| ATTACKS: Mean attacks/frame: 2.000 |                           |

## EXPERIMENT 3 COMPLETED

Execution Time: 77.5s

Results:

```

Oracle: 7744.38 (theoretical maximum)
EXPNeuralUCB: 4738.67 (38.8% gap) [WINNER]
GNeuralUCB: 4552.69 (41.2% gap)
EXPUCB: 3605.14 (53.4% gap)

```

Winner: EXPNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3  
Total time: 122.2s  
Average time per experiment: 40.7s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: MARKOV

Intensity: 1.0

Per-Experiment Results:

=====

EXPERIMENT 1 (4,000 frames, 15.5s)

|  
+-- Oracle: 3051.08 (theoretical maximum)  
+-- EXPNeuralUCB: 1795.46 ( 41.2% gap)  
+-- GNeuralUCB: 1137.52 ( 62.7% gap)  
+-- EXPUCB: 1453.00 ( 52.4% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 28.3s)

|  
+-- Oracle: 5474.92 (theoretical maximum)  
+-- EXPNeuralUCB: 3260.87 ( 40.4% gap)  
+-- GNeuralUCB: 3211.32 ( 41.3% gap)  
+-- EXPUCB: 2604.25 ( 52.4% gap)  
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 77.5s)

|  
+-- Oracle: 7744.38 (theoretical maximum)  
+-- EXPNeuralUCB: 4738.67 ( 38.8% gap)  
+-- GNeuralUCB: 4552.69 ( 41.2% gap)  
+-- EXPUCB: 3605.14 ( 53.4% gap)  
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

=====

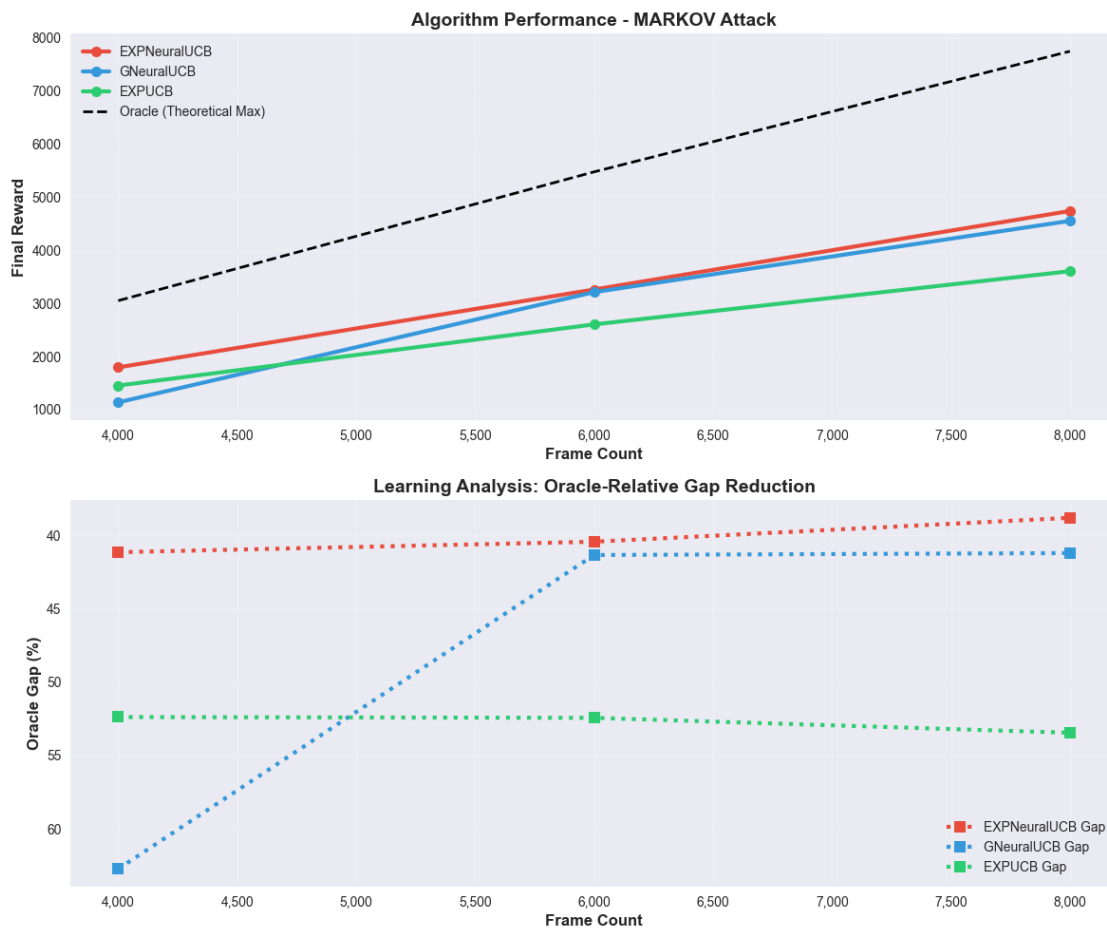
### LEARNING PERFORMANCE (Gap Reduction to Oracle):

- #1: GNeuralUCB -> +21.5pp improvement (gap closing)  
Gap: 62.7% -> 41.2%
- #2: EXPNeuralUCB -> +2.3pp improvement (gap closing)  
Gap: 41.2% -> 38.8%
- #3: EXPUCB -> +1.1pp decline (gap widening)  
Gap: 52.4% -> 53.4%

### Interpretation:

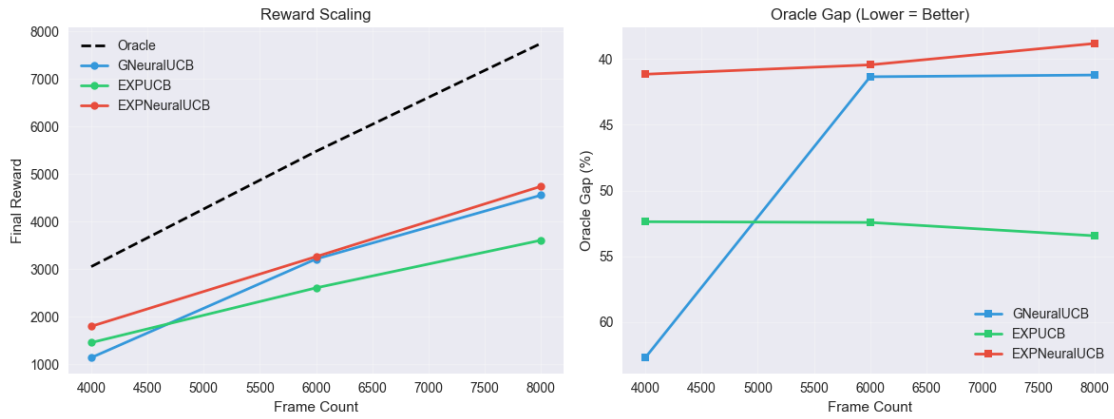
- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

### Generating performance visualization...



Visualization complete!

# Markov Environment Results



TESTING ADAPTIVE

Running test for adaptive environment...

QUANTUM MULTIRUN EVALUATOR INITIALIZED

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: ADAPTIVE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

## EXPERIMENT 1

```
=====
Frame Count: 4,000
Attack: adaptive (intensity=1.0)

```

EXPERIMENT 1: 4000 frames

### QUANTUM ENVIRONMENT INITIALIZED:

```
=====
Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000
=====
```

### ADVERSARIAL CONFIGURATION:

```
=====
Attack Strategy:	AdaptiveAttack
Attack Rate:	1.00
Memory Window:	50
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3240.72

COMPLETED: in 0.00s, final reward: 3240.72

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645),
np.float64(0.7476684769175549), np.float64(0.6599253551504429),
np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
```

|                  |            |                |  |
|------------------|------------|----------------|--|
|                  |            | Robust         |  |
| Action Selection | Neural UCB | Nonlinear      |  |
|                  |            | Learning       |  |
| Parameters       | beta=0.2   | UCB Confidence |  |

=====

- NEURAL Progress: 100%| 4000/4000 [00:07<00:00, 562.82it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 4000/4000 [00:01<00:00, 2177.65it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                      |
| Optimal Action:   | 4                                                                                                                                      |
| Path Performance: | [np.float64(0.8268235367879645),<br>np.float64(0.7476684769175549), np.float64(0.6599253551504429),<br>np.float64(0.5816045675472745)] |

=====

EXPNeuralUCB initialized in 'hybrid' mode

```

=====
ALGORITHM CONFIGURATION
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

- HYBRID Progress: 100%| | 4000/4000 [00:08<00:00, 449.15it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 17.91s):
=====
 Oracle (step-wise) : 3240.72
 GNeuralUCB (batch) : 2542.22
 EXPUCB (batch) : 2093.89
 EXPNeuralUCB (batch) : 2016.44
 WINNER: GNeuralUCB
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 62.2%

ATTACK PATTERN ANALYSIS:
=====
Strategy:	AdaptiveAttack
Total Frames:	4,000
Total Attacks:	4,000
| Attack Intensity: | 0.250 |
|-----|-----|-----|
| PATH-SPECIFIC ANALYSIS: |
|-----|-----|-----|
| Path 0: | 841 attacks (21.0% rate) |
| Path 1: | 1159 attacks (29.0% rate) |
| Path 2: | 896 attacks (22.4% rate) |
| Path 3: | 1104 attacks (27.6% rate) |
|-----|-----|-----|
 ATTACKS: Mean attacks/frame: 1.000
=====

EXPERIMENT 1 COMPLETED
Execution Time: 17.9s
Results:
 Oracle: 3240.72 (theoretical maximum)
 EXPNeuralUCB: 2016.44 (37.8% gap)

```

```
GNeuralUCB: 2542.22 (21.6% gap) [WINNER]
EXPUCB: 2093.89 (35.4% gap)
Winner: GNeuralUCB
```

-----

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000  
Attack: adaptive (intensity=1.0)

-----

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 6,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |                |  |
|------------------|----------------|--|
| Attack Strategy: | AdaptiveAttack |  |
| Attack Rate:     | 1.00           |  |
| Memory Window:   | 50             |  |

=====

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5584.16

COMPLETED: in 0.00s, final reward: 5584.16

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

|                   |                                  |  |
|-------------------|----------------------------------|--|
| Optimal Path:     | 0                                |  |
| Optimal Action:   | 4                                |  |
| Path Performance: | [np.float64(0.9461134921196183), |  |

np.float64(0.902918807873247), np.float64(0.8603089615724164),  
np.float64(0.8116651900474303)]

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

=====

- NEURAL Progress: 100%| 6000/6000 [00:11<00:00, 537.91it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

=====

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Optimal Path:     | 0                                                                                                                                     |
| Optimal Action:   | 4                                                                                                                                     |
| Path Performance: | [np.float64(0.9461134921196183),<br>np.float64(0.902918807873247), np.float64(0.8603089615724164),<br>np.float64(0.8116651900474303)] |

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

=====

- EXP3 Progress: 100%| 6000/6000 [00:04<00:00, 1477.07it/s]

COMPLETED: EXPUCB (batch) completed successfully  
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

```

| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183),
np.float64(0.902918807873247), np.float64(0.8603089615724164),
np.float64(0.8116651900474303)]
=====

```

EXPNeuralUCB initialized in 'hybrid' mode

#### ALGORITHM CONFIGURATION

```

=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====

```

- HYBRID Progress: 100%| 6000/6000 [00:14<00:00, 404.77it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 30.11s):

```

=====
Oracle (step-wise) : 5584.16
GNeuralUCB (batch) : 3903.85
EXPUCB (batch) : 3318.91
EXPNeuralUCB (batch) : 3980.39
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.3%

```

#### ATTACK PATTERN ANALYSIS:

```

=====
Strategy:	AdaptiveAttack
Total Frames:	6,000
Total Attacks:	6,000
Attack Intensity:	0.250
PATH-SPECIFIC ANALYSIS:	
-----	-----
Path 0:	2142 attacks (35.7% rate)
Path 1:	1439 attacks (24.0% rate)
Path 2:	947 attacks (15.8% rate)
Path 3:	1472 attacks (24.5% rate)
-----	-----
=====

```

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 30.1s

Results:

Oracle: 5584.16 (theoretical maximum)  
EXPNeuralUCB: 3980.39 ( 28.7% gap) [WINNER]  
GNeuralUCB: 3903.85 ( 30.1% gap)  
EXPUCB: 3318.91 ( 40.6% gap)  
Winner: EXPNeuralUCB

-----

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: adaptive (intensity=1.0)

-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

|                |               |  |
|----------------|---------------|--|
| Network Paths: | 4             |  |
| Qubit Config:  | (8, 10, 8, 9) |  |
| Frame Length:  | 8,000         |  |

=====

ADVERSARIAL CONFIGURATION:

=====

|                  |                |  |
|------------------|----------------|--|
| Attack Strategy: | AdaptiveAttack |  |
| Attack Rate:     | 1.00           |  |
| Memory Window:   | 50             |  |

=====

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take\_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7832.21

COMPLETED: in 0.00s, final reward: 7832.21

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method     | Properties                |
|------------------|------------|---------------------------|
| Group Selection  | Simple UCB | NOT Adversarial<br>Robust |
| Action Selection | Neural UCB | Nonlinear<br>Learning     |
| Parameters       | beta=0.2   | UCB Confidence            |

```
=====
```

- NEURAL Progress: 100%| 8000/8000 [00:51<00:00, 156.64it/s]

COMPLETED: GNeuralUCB (batch) completed successfully  
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
```

| Component        | Method                             | Properties                |
|------------------|------------------------------------|---------------------------|
| Group Selection  | EXP3                               | Adversarially<br>Robust   |
| Action Selection | Linear UCB                         | Linear<br>Learning        |
| Parameters       | gamma=0.100, eta=0.005<br>beta=0.2 | EXP3 Config<br>Linear UCB |

```
=====
- EXP3 Progress: 100%| | 8000/8000 [00:07<00:00, 1035.12it/s]
```

```
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...
```

#### ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526),
np.float64(0.9637113766945926), np.float64(0.9435345403529187),
np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

#### ALGORITHM CONFIGURATION

```
=====
Component	Method	Properties
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====
```

```
- HYBRID Progress: 100%| | 8000/8000 [00:21<00:00, 366.98it/s]
```

```
 COMPLETED: EXPNeuralUCB (batch) completed successfully
```

#### RESULTS (completed in 80.70s):

```
=====
Oracle (step-wise) : 7832.21
GNeuralUCB (batch) : 5597.04
EXPUCB (batch) : 4744.51
EXPNeuralUCB (batch) : 5498.66
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
```

#### ATTACK PATTERN ANALYSIS:

```
=====
Strategy:	AdaptiveAttack
Total Frames:	8,000
Total Attacks:	8,000
Attack Intensity:	0.250
```

```

PATH-SPECIFIC ANALYSIS:
Path 0:
Path 1:
Path 2:
Path 3:
=====
ATTACKS: Mean attacks/frame: 1.000
=====

```

### EXPERIMENT 3 COMPLETED

Execution Time: 80.7s

Results:

```

Oracle: 7832.21 (theoretical maximum)
EXPNeuralUCB: 5498.66 (29.8% gap)
GNeuralUCB: 5597.04 (28.5% gap) [WINNER]
EXPUCB: 4744.51 (39.4% gap)
Winner: GNeuralUCB

```

```

=====
QUANTUM EVALUATION COMPLETED!
=====
Total experiments: 3
Total time: 129.6s
Average time per experiment: 43.2s
=====

```

### DETAILED EXPERIMENT ANALYSIS

Attack Configuration:

```

Strategy: ADAPTIVE
Intensity: 1.0

```

Per-Experiment Results:

#### EXPERIMENT 1 (4,000 frames, 17.9s)

```

|
+-- Oracle: 3240.72 (theoretical maximum)
+-- EXPNeuralUCB: 2016.44 (37.8% gap)
+-- GNeuralUCB: 2542.22 (21.6% gap)
+-- EXPUCB: 2093.89 (35.4% gap)
Winner: GNeuralUCB

```

EXPERIMENT 2 (6,000 frames, 30.1s)

```
|
+-- Oracle: 5584.16 (theoretical maximum)
+-- EXPNeuralUCB: 3980.39 (28.7% gap)
+-- GNeuralUCB: 3903.85 (30.1% gap)
+-- EXPUCB: 3318.91 (40.6% gap)
Winner: EXPNeuralUCB
```

EXPERIMENT 3 (8,000 frames, 80.7s)

```
|
+-- Oracle: 7832.21 (theoretical maximum)
+-- EXPNeuralUCB: 5498.66 (29.8% gap)
+-- GNeuralUCB: 5597.04 (28.5% gap)
+-- EXPUCB: 4744.51 (39.4% gap)
Winner: GNeuralUCB
```

ORACLE-RELATIVE GAP ANALYSIS:

-----

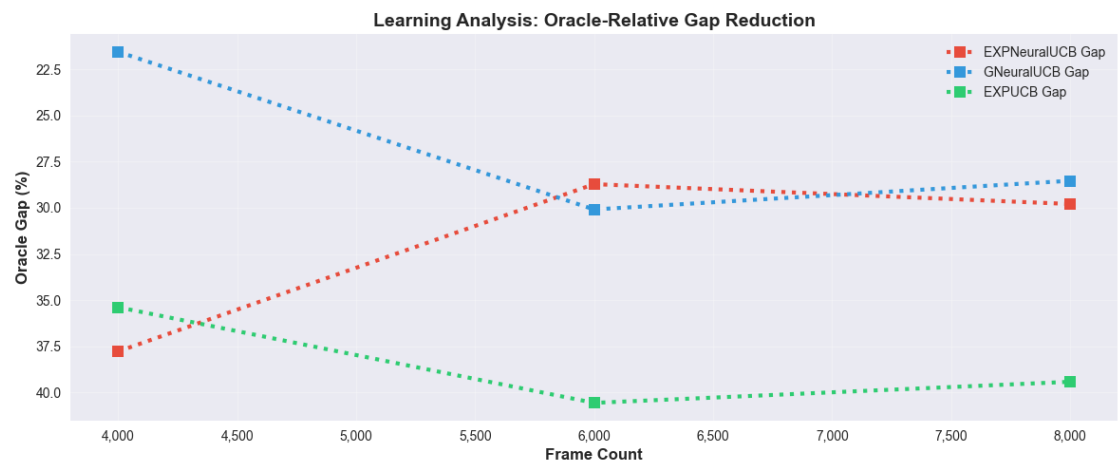
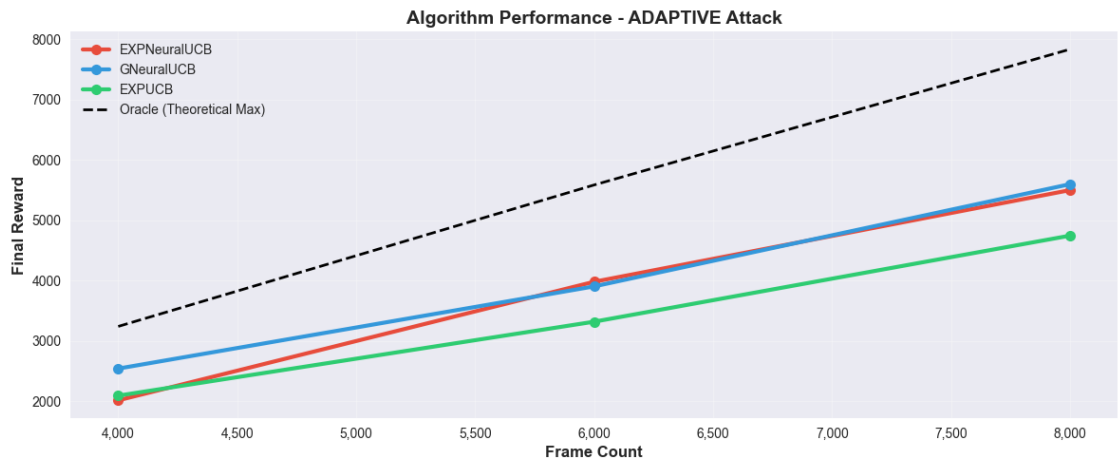
LEARNING PERFORMANCE (Gap Reduction to Oracle):

```
#1: EXPNeuralUCB -> +8.0pp improvement (gap closing)
 Gap: 37.8% -> 29.8%
#2: EXPUCB -> +4.0pp decline (gap widening)
 Gap: 35.4% -> 39.4%
#3: GNeuralUCB -> +7.0pp decline (gap widening)
 Gap: 21.6% -> 28.5%
```

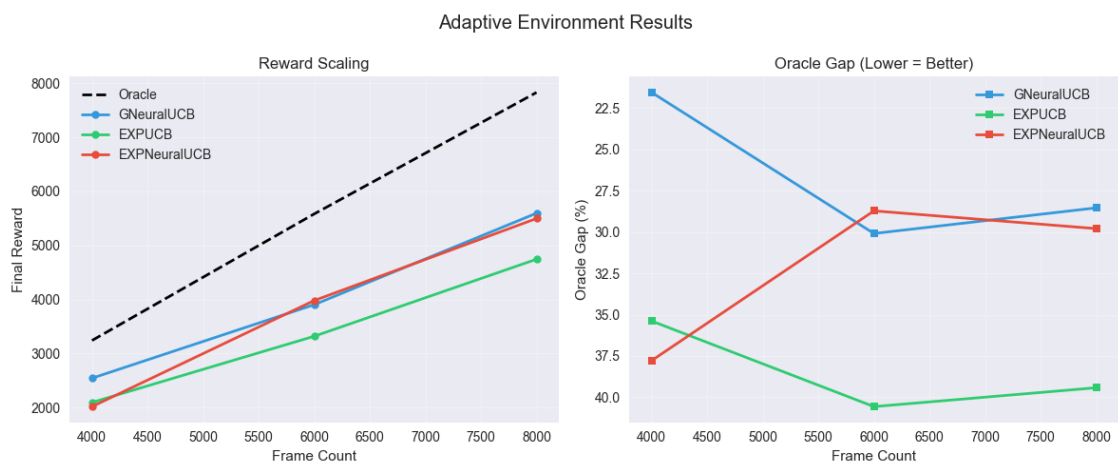
Interpretation:

- Lower gaps = Better performance relative to Oracle
  - Gap reduction = Algorithm learning and improving
  - Best learner closes the gap to Oracle over time
- 

Generating performance visualization...

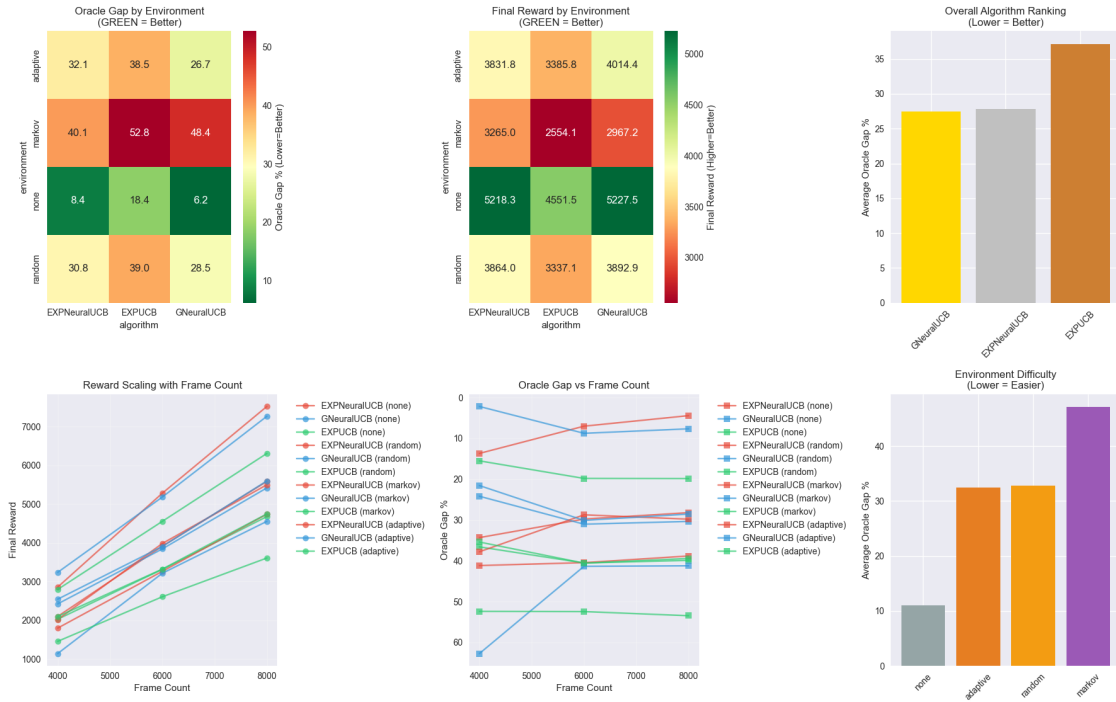


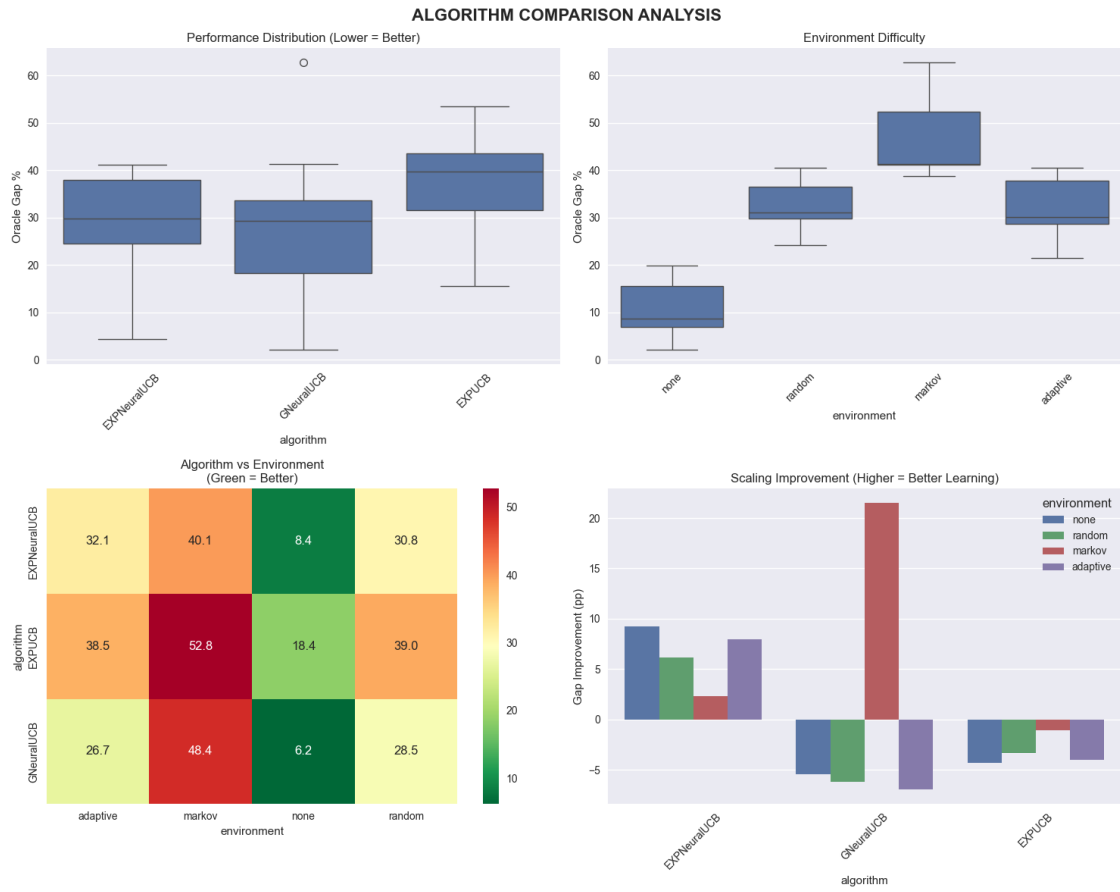
Visualization complete!



## CREATING ENHANCED VISUALIZATIONS

**QUANTUM ROUTING ALGORITHM PERFORMANCE DASHBOARD**  
Lower Oracle Gap = Better Performance | Higher Reward = Better Performance





ENVIRONMENT COMPARISON

